# EXPLICIT AND ASYMPTOTIC ESTIMATES FOR THE ZERO-COUNTING FUNCTION OF THE RIEMANN ZETA-FUNCTION

VÍCTOR CASAÑA-GINER

Abstract. Let $N(T)$ be the number of non-trivial zeros of the Riemann zeta-function with imaginary part in $(0,T]$, counted with multiplicity, and let $S(T)$ be the argument of $\zeta$ on the critical line, normalised so that $N(T)=\frac{1}{\pi}\vartheta(T)+1+S(T)$. Two estimates are proved by the method of Turing type that Amberger introduced for Dedekind zeta-functions and that Bellotti and Wong extended to Artin $L$-functions. The first is explicit: for every $T\geq 1$,

$$\left|N(T)-\frac{T}{2\pi}\log\frac{T}{2\pi e}\right|\leq 0.08382\log T+5.326,\qquad |S(T)|\leq 0.08382\log T+4.444,$$

and a piecewise linear bound, formed by six lines, lies below the explicit bounds of Amberger and of Bellotti and Wong for every $T\geq 1$. The kernels of the method are found by convex optimisation instead of by hand, and the constant coming from the Euler product is evaluated rigorously in ball arithmetic over all prime powers up to $10^8$. The second estimate is asymptotic: $\limsup_{T\to\infty}|S(T)|/\log T<0.0748596142711$. It comes from a hybrid of the kernel majorant with Jensen's formula on discs to the right of the critical line, where the growth of $\zeta$ is controlled by the subconvexity bounds of Bourgain, of Trudgian and Yang, of Tao, Trudgian and Yang, and of Heath-Brown. The constant is smaller than the value $0.0786252\ldots$ below which the method of Backlund cannot go, even under the Lindelöf hypothesis, for the family of discs centred near $s=1$ considered in the author's earlier comparison. Every numerical inequality involving the chosen kernels and constants is certified in ball arithmetic, and no numerical information on the zeros is used.

## Contents

1. Introduction 2
2. Results 4
3. The identity of Turing type 6
4. The explicit estimate 8
4.1. The decomposition 8
4.2. The Euler product 10
4.3. The elementary part 12
4.4. The kernel inequality 14
4.5. Proof of Theorem 2.1 and of Corollary 2.2 15
5. Jensen discs and admissible profiles 16
6. The asymptotic estimate 18
7. Methods and remarks 20
7.1. Where the kernel inequality binds 20
7.2. Optimisation 20
7.3. Certification 21
7.4. Comparison of the two methods 21

*Date*: 1 November 2026.

2020 *Mathematics Subject Classification.* Primary 11M06, 11M26; Secondary 11Y35, 65G30.

*Key words and phrases.* Riemann zeta-function, zero-counting function, argument of the zeta-function, explicit bounds, Turing's method, Jensen's formula, subconvexity, ball arithmetic.

ORCID 0000-0002-8252-2608.

7.5. An open question 22
Data availability 22
AI declaration 22
Mathematical software 22
References 22
Appendix A. Data tables 24
Appendix B. Programs 29
Appendix C. Outputs of the certification programs 45
Appendix D. Input data of the certificates 51

## 1. Introduction

Let $\zeta$ be the Riemann zeta-function and

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\tfrac{s}{2})\zeta(s) \tag{1.1}$$

Riemann's $\xi$-function, an entire function that satisfies $\xi(s) = \xi(1-s)$ [25, §2.1, (2.1.12)–(2.1.13)] and is real and positive on the real axis: on $s > 1$ every factor is positive, on $0 < s < 1$ one has $s(s-1) < 0$ and $\zeta(s) < 0$, $\xi(0) = \xi(1) = \frac{1}{2}$, and $s < 0$ reduces to $s > 1$ by the functional equation. By the Euler product, $\zeta$ has no zeros in $\operatorname{Re} s > 1$, by the theorem of Hadamard and de la Vallée Poussin none on $\operatorname{Re} s = 1$ [8, Ch. 13], and by the functional equation none in $\operatorname{Re} s \le 0$ other than the trivial zeros $-2, -4, \dots$, which are cancelled in $\xi$ by the poles of $\Gamma(s/2)$. The zeros of $\xi$ are therefore the non-trivial zeros of $\zeta$, all in the strip $0 < \operatorname{Re} s < 1$. Throughout, $\rho = \beta + i\gamma$ runs over these zeros, counted with multiplicity, and

$$N(T) = \#\{\rho :\ 0 < \gamma \le T\} \qquad (T > 0), \tag{1.2}$$

so that $N(T)$ is right-continuous and $N(T) = N(T^+)$ at an ordinate $T$. Since $\xi(\bar{s}) = \overline{\xi(s)}$, the zeros come in conjugate pairs, and $N(T)$ counts those with positive ordinate among the zeros with $|\gamma| \le T$. Riemann stated and von Mangoldt [30] proved that

$$N(T) = \mathcal{M}(T) + O(\log T), \qquad \mathcal{M}(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}. \tag{1.3}$$

The function

$$\vartheta(T) = \operatorname{Im}\operatorname{Log}\Gamma\left(\tfrac{1}{4} + \tfrac{iT}{2}\right) - \frac{T}{2}\log\pi \tag{1.4}$$

is the Riemann–Siegel theta function, with the branch of $\operatorname{Log}\Gamma$ that is real on $(0,\infty)$, and the argument of $\zeta$ on the critical line is the function $S$ defined for $T > 0$ by

$$N(T) = \frac{1}{\pi}\vartheta(T) + 1 + S(T) \tag{1.5}$$

[25, §9.3]; with the convention (1.2), $S(T) = S(T^+)$ at the ordinates. Since $\vartheta(T)/\pi + 1 = \mathcal{M}(T) + \frac{7}{8} + O(1/T)$ (Lemma 4.8 below), the estimate (1.3) is equivalent to $S(T) = O(\log T)$.

The problem of making (1.3) explicit, in the form

$$\left|N(T) - \mathcal{M}(T)\right| \le C_1\log T + C_2\log\log T + C_3 \qquad (T \ge T_0), \tag{1.6}$$

goes back to Backlund [3], after von Mangoldt [30] and Grossmann [12], and was taken up by Rosser [21], Trudgian [26, 27], Platt and Trudgian [18], Hasanalizade, Shen and Wong [13] and Bellotti and Wong [4], all by the method of Backlund: the variation of $\arg\zeta$ on the horizontal segment from $\frac{1}{2} + iT$ to the right is controlled by Jensen's formula on a disc centred at a point $c + iT$ with $c > 1$, and the growth of $\zeta$ on that disc by convexity or subconvexity bounds. Table 1 lists the constants obtained in this way, in the form (1.6) and with the first height $T_0$ from which each bound is proved.

The leading constant obtainable by this method is limited from below: for the family of discs used in [7], centred near $s = 1$ with radius near 1, the leading constant cannot be smaller than

$$\Lambda(0) = \frac{3\sqrt{3} - \pi}{12\pi \log 2} = 0.0786252\ldots \tag{1.7}$$

for any admissible growth profile of $\zeta$, even under the Lindelöf hypothesis [7, Remark 5.2]; whether other discs could do better is not settled there.

In 2025 Amberger [1] introduced a different method, inspired by Turing's [29, Lemma 1], for Dedekind zeta-functions, and Bellotti and Wong [5] extended it to Artin $L$-functions and sharpened it. Instead of a disc, it uses an exact identity: for $d > \frac{1}{2}$, $N(T)$ equals a combination of the arguments of $\xi$ at the two points $\frac{1}{2} + d + iT$ and $\frac{1}{2} + 2d + iT$, which lie in the zero-free half-plane $\operatorname{Re} s > 1$, plus a sum over all the zeros of an explicit function $f(\frac{1}{2} - \beta, T - \gamma)$ (Lemma 3.1). The sum over the zeros is then estimated from above and from below by sums of functions of the form $\operatorname{Re}(x + it)^{-j}$ and $\operatorname{Im}(x+it)^{-j}$, whose sums over the zeros are the derivatives of $\xi'/\xi$ at $\frac{1}{2}+d+iT$, again in the zero-free half-plane. For the Riemann zeta-function the method gives $|N(T) - \mathcal{M}(T)| \le 0.097 \log T + 4.807$ for all $T \ge 1$ in [1, Cor. 2], and $\min(0.0946 \log T + 5.746,\ 0.097 \log T + 4.954)$ in [5, Cor. 1.3]. Both articles choose the coefficients of the majorant by trial, with three or four of them, and Bellotti and Wong [5, §3] ask what the method can give when more coefficients are allowed and chosen by convex optimisation, and whether the infimum of the leading constant over all such majorants is zero.

This article contains two contributions to that method. The first is quantitative. The majorant is taken from a family with nineteen coefficients (ten real and nine imaginary parts of the kernels $(x+it)^{-j}$, $1 \le j \le 10$, in the notation of Section 3) and the coefficients are chosen by a second-order cone programme, with the inequality that defines the majorant imposed on a grid and refined by cutting planes. The constant that comes from the Euler product is computed rigorously in ball arithmetic over all prime powers up to $10^8$: for the primes $p \le 1000$ the maximum over the common phase of all the powers of $p$ (an observation of Amberger [1, Lemma 9]) is bounded rigorously by a phase grid with a derivative correction, and for the remaining prime powers the Euclidean majorant is used; the sum is bounded beyond $10^8$ by partial summation with the inequality $\psi(x) < 1.03883x$ of Rosser and Schoenfeld [22, Thm. 12]. The remaining part, which comes from $\Gamma$, from $s(s-1)$ and from the main term, is evaluated as a single function of $T$ whose supremum over $T \ge 1$ is certified. The result (Theorem 2.1) is a piecewise linear bound, valid for all $T \ge 1$, that lies below the bounds of [1] and [5] at every height; its last line has leading constant 0.08382.

The second contribution is methodological. For the kernel families considered here, the active constraints of the optimisation lie at hypothetical zeros with $\beta$ near 0 or near 1 (Section 7.1 explains why the kernel inequality is controlled by its boundary), and the method of Turing type, which uses no information on $\zeta$ inside the critical strip, cannot exclude such zeros. In the hybrid of Section 5 the majorant is allowed to fail near $\beta \in \{0, 1\}$, and the excess is paid by weights of the form $\log^+(R/|\rho^* - s_0|)$, where $s_0 = c + i(T + v)$ with $c > 1$ and $\rho^*$ is the one of $\rho$, $1 - \bar{\rho}$ with real part $\ge \frac{1}{2}$. The sum of such a weight over the zeros is bounded by Jensen's formula on the disc $|s - s_0| < R$, and the integral of $\log|\zeta|$ on its boundary by the subconvexity bounds for $\zeta$ in the strip. The coefficients of the majorant and the weights of the discs are chosen together by linear programming. This gives (Theorem 2.3)

$$\limsup_{T \to \infty} \frac{|S(T)|}{\log T} < 0.0748596142711,$$

below the value (1.7). The bound is asymptotic: the subconvexity estimates used have no explicit constants, and no height from which the inequality holds is computed.

The identity of Lemma 3.1, the family of kernel functions and the way their sums over the zeros are evaluated are due to Amberger [1] and to Bellotti and Wong [5], as is the maximum over

the phase in the Euler product; they are proved here again, with the normalisation used in this article, so that the article is self-contained. The choice of the kernel by convex optimisation, the certification of every inequality of the proofs in ball arithmetic [16], and the hybrid with Jensen's formula have, to the author's knowledge, not been used for this problem before. Of the sources used, [1, 5, 28, 23] are arXiv preprints, cited with the version numbers given in the bibliography, [7] is a submitted manuscript of the author (its results are used only for the comparison with Backlund's method, not in the proofs), and the remaining references are published. Backlund's method and the method of Turing type are compared in Section 7, where the computations are also described. All numerical statements in the proofs are certified in Arb ball arithmetic [16] through `python-flint`; the programs and their outputs are reproduced after the bibliography (Appendices B–D), and no computation of zeros of $\zeta$, in particular no verification of the Riemann hypothesis up to some height, is used anywhere.

## 2. Results

Write $L = \log T$. The bounds below are lower envelopes of finitely many lines $A_1 L + A_3$ in the variable $L$, each line valid on its own for all $T \geq 1$.

**Theorem 2.1** (Explicit estimate)**.** *For every $T \geq 1$, with $L = \log T$,*

$$\begin{aligned} |N(T) - \mathcal{M}(T)| \leq \min\{&0.20176L + 1.369,\ 0.10823L + 2.171,\ 0.09473L + 2.834,\\ &0.09349L + 2.987,\ 0.08399L + 5.202,\ 0.08382L + 5.326\}, \end{aligned} \tag{2.1}$$

$$\begin{aligned} |S(T)| \leq \min\{&0.20176L + 0.844,\ 0.10823L + 1.286,\ 0.09473L + 1.946,\\ &0.09349L + 2.099,\ 0.08399L + 4.318,\ 0.08382L + 4.444\}. \end{aligned} \tag{2.2}$$

The six lines are called $\ell_5, \ell_{20}, \ell_{60}, \ell_{150}, \ell_{400}, \ell_{6000}$, after the value of $L$ for which each was optimised. In (2.1) they are the smallest, respectively, on $[0, L_1]$, $[L_1, L_2]$, $[L_2, L_3]$, $[L_3, L_4]$, $[L_4, L_5]$ and $[L_5, \infty)$, where the breakpoints, at which two adjacent lines agree, are

$$L_1 = \tfrac{80200}{9353} = 8.574788\ldots, \qquad L_2 = \tfrac{442}{9} = 49.111111\ldots, \qquad L_3 = \tfrac{3825}{31} = 123.387096\ldots,$$
$$L_4 = \tfrac{4430}{19} = 233.157894\ldots, \qquad L_5 = \tfrac{12400}{17} = 729.411764\ldots$$

(These are the breakpoints of the envelope for $N(T) - \mathcal{M}(T)$; the envelope for $S(T)$ has the same slopes but different intercepts, hence slightly different breakpoints, which are not needed.) Each line is proved on its own for all $T \geq 1$; in particular

$$|N(T) - \mathcal{M}(T)| \leq 0.08382 \log T + 5.326, \qquad |S(T)| \leq 0.08382 \log T + 4.444 \qquad (T \geq 1). \tag{2.3}$$

**Corollary 2.2** (Comparison)**.** *For every $T \geq 1$ the right side of* (2.1) *is smaller than* $\min(0.0946L + 5.746,\ 0.097L + 4.954)$ [5, Cor. 1.3] *by at least* 2.231*, and smaller than the minimum of the seven bounds* $0.097L + 4.807$*,* $0.1L + 4.140$*,* $0.12L + 2.830$*,* $0.14L + 2.306$*,* $0.16L + 2.027$*,* $0.18L + 1.847$*,* $0.2L + 1.722$ *of* [1, Cor. 2 and Table 6] *by at least* 0.291*. The single line* $0.08399L + 5.202$ *is smaller than* $0.0946L + 5.746$ *for every $T \geq 1$, and smaller than* $0.097L + 4.954$ *for $L > (5.202 - 4.954)/(0.097 - 0.08399) = 19.062\ldots$; the single line* $0.08382L + 5.326$ *is smaller than* $0.0946L + 5.746$ *for every $T \geq 1$, and smaller than* $0.097L + 4.954$ *for $L > (5.326 - 4.954)/(0.097 - 0.08382) = 28.224\ldots$.*

In the form (1.6) with $C_2 = 0$ and $T_0 = 1$, the last line of Theorem 2.1 has $(C_1, C_3) = (0.08382, 5.326)$; Table 1 puts it next to the earlier results. For $S(T)$, Bellotti and Wong [4, Cor. 1.5] prove $|S(T)| \leq 0.10076L + 0.24460 \log L + 7.20844$ for $T \geq e$, and [5, Cor. 1.3] gives, through $|S| \leq |N - \mathcal{M}| + \frac{7}{8} + |\varpi|/\pi$ (Lemma 4.8), $|S(T)| \leq 0.0946L + 6.691$ for $T \geq e$.

For the asymptotic coefficient

$$\mathfrak{a} = \limsup_{T \to \infty} \frac{|S(T)|}{\log T}, \tag{2.4}$$

| | $C_1$ | $C_2$ | $C_3$ | $T_0$ |
|---|---|---|---|---|
| von Mangoldt [30] (1905) | 0.4320 | 1.9167 | 13.0788 | 28.5580 |
| Grossmann [12] (1913) | 0.2907 | 1.7862 | 7.0120 | 50 |
| Backlund [3] (1916) | 0.1370 | 0.4430 | 5.2250 | 200 |
| Rosser [21] (1941) | 0.1370 | 0.4430 | 2.4630 | 2 |
| Trudgian [26] (2012) | 0.1700 | 0 | 2.8730 | $e$ |
| Trudgian [27] (2014) | 0.1120 | 0.2780 | 3.3850 | $e$ |
| Platt and Trudgian [18] (2015) | 0.1100 | 0.2900 | 3.165 | $e$ |
| Hasanalizade, Shen and Wong [13] (2022) | 0.1038 | 0.2573 | 9.4925 | $e$ |
| Bellotti and Wong [4] (2025) | 0.10076 | 0.24460 | 8.08344 | $e$ |
| Amberger [1, Cor. 2] (2025, v5 of 2026) | 0.097 | 0 | 4.807 | 1 |
| Bellotti and Wong [5, Cor. 1.3] (2026) | 0.0946 | 0 | 5.746 | 1 |
| Theorem 2.1, line $\ell_{400}$ | 0.08399 | 0 | 5.202 | 1 |
| Theorem 2.1, line $\ell_{6000}$ | 0.08382 | 0 | 5.326 | 1 |

TABLE 1. Explicit bounds $|N(T)-\mathcal{M}(T)| \le C_1 \log T + C_2 \log\log T + C_3$ for $T \ge T_0$. The rows up to 2025 are taken from [5, Table 3].

the explicit bounds give $\mathfrak{a} \le 0.0946$ [5] and $\mathfrak{a} \le 0.08382$ (Theorem 2.1), and Backlund's method with the exponent pair of Bourgain gives $\mathfrak{a} \le 0.0968160\ldots$ [7, Cor. 1.2].

**Theorem 2.3** (Hybrid estimate)**.** $\mathfrak{a} \le C_{\mathrm{H}}$*, where* $C_{\mathrm{H}}$ *is the number* (6.1) *defined by the data of Section 6, and*

$$C_{\mathrm{H}} = 0.0748596142710907\ldots < 0.0748596142711. \tag{2.5}$$

*In particular* $\mathfrak{a} < \Lambda(0)$*, with* $\Lambda(0)$ *as in* (1.7)*.*

Theorem 2.3 is asymptotic: it states that for every $\varepsilon > 0$ there is $T_0(\varepsilon)$ with $|S(T)| \le (C_{\mathrm{H}} + \varepsilon)\log T$ for $T \ge T_0(\varepsilon)$, and $T_0(\varepsilon)$ is not computed, because the subconvexity bounds that enter the proof (Lemma 5.3) have no explicit constants. The explicit bounds of Theorem 2.1 are not improved by it at any finite height, and Theorem 2.1 itself uses no disc.

*Remark* 2.4 (Conventions). Amberger [1] and Bellotti and Wong [5] count a zero with imaginary part exactly $T$ with weight $\frac{1}{2}$, that is, they use $\frac{1}{2}(N(T^+)+N(T^-))$. Since the bounds of Theorem 2.1 are proved for every $T$ that is not an ordinate and $\mathcal{M}$ is continuous, they hold for $N(T^+)$ and for $N(T^-)$ at an ordinate, hence also for their mean. The comparisons of Corollary 2.2 are therefore between bounds for the same quantity.

*Remark* 2.5 (On the infimum of the leading constant). Bellotti and Wong [5, §3] ask whether the infimum of $C_1$ over all majorants of the type used in the method, with finitely many kernel functions, is 0. The program `kernel_only.py` of Appendix B (output in Appendix C) gives, for the family (3.13) with the constraint (K) imposed on a grid and refined by cutting planes, and $d$ optimised over $\{0.7, 0.8, 0.9\}$, the leading constants 0.08831, 0.08071 and 0.07983 for $m = 4, 10, 20$ (grid optima, not certified; only runs whose cutting planes converged, with a residual violation below $10^{-5}$ on the fine grid, are quoted, and for $m = 40$ the iteration did not converge, the linear programme being badly conditioned), which suggest a positive limit near 0.079. The same computations with the constraint imposed only at $\beta = \frac{1}{2}$, as would suffice under the Riemann hypothesis, give 0.04375 and 0.02002 for $m = 4, 10$ (for $m = 20$ the iteration did not converge). For the kernel families tested, the computations and the boundary principle of Section 7.1 indicate that the dominant obstruction comes from the constraints near $\beta \in \{0, 1\}$; these are the constraints that the hybrid of Theorem 2.3 relaxes, at the price of using subconvexity bounds for $\zeta$ inside the strip.

## 3. The identity of Turing type

On the closed half-plane $\operatorname{Re} s \geq 1$ the function $\xi$ has no zeros ($\zeta$ has none, $\Gamma$ has none, and $\xi(1) = \frac{1}{2}$), and the half-plane is simply connected, so $\arg \xi$ has a continuous branch there; it is normalised by $\arg \xi(\sigma) = 0$ for real $\sigma \geq 1$, where $\xi(\sigma) > 0$. For $d > \frac{1}{2}$ put

$$\sigma_1 = \tfrac{1}{2} + d, \qquad \sigma_2 = \tfrac{1}{2} + 2d, \qquad s_1 = \sigma_1 + iT, \qquad s_2 = \sigma_2 + iT. \tag{3.1}$$

For $b \in \mathbb{R}$ with $|b| < d$ and $t \neq 0$ put

$$f(b,t) = 2\arctan\frac{d+b}{t} + 2\arctan\frac{d-b}{t} - \arctan\frac{2d+b}{t} - \arctan\frac{2d-b}{t}, \tag{3.2}$$

where arctan is the principal branch, with values in $(-\frac{\pi}{2}, \frac{\pi}{2})$, here and throughout; and for a zero $\rho = \beta + i\gamma$ and a height $T$ write

$$b_\rho = \tfrac{1}{2} - \beta, \qquad t_\rho = T - \gamma. \tag{3.3}$$

Since $0 < \beta < 1$, one has $|b_\rho| < \frac{1}{2} < d$. The function $f$ is even in $b$ and odd in $t$.

**Lemma 3.1** (Identity of Turing type [29, 1, 5])**.** *Let $d > \frac{1}{2}$ and let $T > 0$ not be an ordinate. Then*

$$N(T) = \frac{1}{\pi}\Big(2\arg\xi(s_1) - \arg\xi(s_2)\Big) + \frac{1}{2\pi}\sum_\rho f(b_\rho, t_\rho), \tag{3.4}$$

*where the series converges absolutely.*

*Proof. The argument principle.* Let $\mathcal{R}$ be the rectangle with vertices $\sigma_2$, $\sigma_2 + iT$, $1-\sigma_2+iT$, $1-\sigma_2$. The function $\xi$ has no zeros on $\partial\mathcal{R}$: none on the vertical sides, which lie outside the critical strip; none on the top side, because $T$ is not an ordinate; and none on the real segment, because $\xi(\sigma) > 0$ for all real $\sigma$ (for $\sigma \geq 1$ this was noted above; for $0 < \sigma < 1$ one has $\sigma(\sigma-1) < 0$ and $\zeta(\sigma) < 0$, the latter because $(1-2^{1-\sigma})\zeta(\sigma) = \sum_{n\geq 1}(-1)^{n-1}n^{-\sigma} > 0$ and $1 - 2^{1-\sigma} < 0$; the case $\sigma \leq 0$ follows from $\xi(\sigma) = \xi(1-\sigma)$). Since $T$ is not an ordinate, the zeros of $\xi$ inside $\mathcal{R}$, namely the non-trivial zeros with $0 < \gamma < T$, are exactly the $N(T)$ zeros counted in (1.2), so by the argument principle the variation of $\arg\xi$ along $\partial\mathcal{R}$, counterclockwise, is $2\pi N(T)$.

Let $\Delta_1$ be the variation of $\arg\xi$ along the path $\pi_1$ from $\sigma_2$ to $\sigma_2 + iT$ and then to $\frac{1}{2} + iT$. The reflection $s \mapsto 1 - \bar{s}$ in the critical line maps $\pi_1$ onto the path $\pi_1'$ from $1-\sigma_2$ to $1-\sigma_2+iT$ to $\frac{1}{2}+iT$, and $\xi(1-\bar{s}) = \xi(\bar{s}) = \overline{\xi(s)}$, so the variation of $\arg\xi$ along $\pi_1'$ is $-\Delta_1$. The boundary $\partial\mathcal{R}$, counterclockwise, is $\pi_1$ followed by the reverse of $\pi_1'$ followed by the real segment from $1-\sigma_2$ to $\sigma_2$, on which $\arg\xi$ is constant. Hence $2\pi N(T) = \Delta_1 + \Delta_1$, that is, $\Delta_1 = \pi N(T)$.

Along the vertical part of $\pi_1$ the variation is $\arg\xi(s_2)$, the branch defined above. Along the horizontal part, $\xi$ has no zeros, and with

$$F(\sigma) = \operatorname{Im}\frac{\xi'}{\xi}(\sigma + iT) = \frac{\partial}{\partial\sigma}\arg\xi(\sigma + iT) \qquad (\tfrac{1}{2} \leq \sigma \leq \sigma_2), \tag{3.5}$$

a continuous function, the variation from $\sigma_2 + iT$ to $\frac{1}{2} + iT$ is $-\int_{1/2}^{\sigma_2} F(\sigma)\,d\sigma$. Therefore

$$\pi N(T) = \arg\xi(s_2) - \int_{1/2}^{\sigma_2} F(\sigma)\,d\sigma. \tag{3.6}$$

Since $\int_{\sigma_1}^{\sigma_2} F(\sigma)\,d\sigma = \arg\xi(s_2) - \arg\xi(s_1)$ (the segment lies in $\operatorname{Re} s \geq 1$, where the branch is the one fixed above) and $\int_{1/2}^{\sigma_1} F(\sigma + d)\,d\sigma = \int_{\sigma_1}^{\sigma_2} F$,

$$\pi N(T) = 2\arg\xi(s_1) - \arg\xi(s_2) - \int_{1/2}^{\sigma_1}\big(F(\sigma) - F(\sigma + d)\big)\,d\sigma. \tag{3.7}$$

*The sum over the zeros.* By the Hadamard product [8, Ch. 12],

$$\frac{\xi'}{\xi}(s) = B + \sum_{\rho}\Big(\frac{1}{s-\rho} + \frac{1}{\rho}\Big), \tag{3.8}$$

with a real constant $B$ and absolute convergence for every $s$ that is not a zero. Taking imaginary parts at $\sigma + iT$ and at $\sigma + d + iT$ and subtracting, the constants cancel and

$$F(\sigma) - F(\sigma + d) = \sum_{\rho} \operatorname{Im}\Big(\frac{1}{\sigma + iT - \rho} - \frac{1}{\sigma + d + iT - \rho}\Big), \tag{3.9}$$

an absolutely convergent series of differences (the two series (3.8), with their regularising terms $1/\rho$, are each absolutely convergent, and the terms $1/\rho$ cancel in the difference; nothing is asserted about $\sum_\rho (s-\rho)^{-1}$ alone). Since $\frac{1}{s-\rho} - \frac{1}{s+d-\rho} = \frac{d}{(s-\rho)(s+d-\rho)}$, for $\frac12 \le \sigma \le \sigma_1$ the $\rho$-th term has modulus at most $d/|t_\rho|^2$, because $|\sigma + iT - \rho| \ge |t_\rho|$ and $|\sigma + d + iT - \rho| \ge |t_\rho|$; since $T$ is not an ordinate, $t_\rho \ne 0$ for every $\rho$, only finitely many zeros have $|t_\rho| < 1$, and $\sum_{|t_\rho|\ge 1} |t_\rho|^{-2} < \infty$ [25, §9.4], the series (3.9) is dominated by a summable sequence independent of $\sigma$ and may be integrated term by term over $[\frac12, \sigma_1]$. With $\operatorname{Im}\frac{1}{x+it} = -\frac{t}{x^2+t^2}$ and $\int \frac{-t\,dx}{x^2+t^2} = -\arctan\frac{x}{t}$,

$$\int_{1/2}^{\sigma_1} \operatorname{Im}\Big(\frac{1}{\sigma + iT - \rho} - \frac{1}{\sigma + d + iT - \rho}\Big) d\sigma = g(b_\rho, t_\rho), \tag{3.10}$$

where

$$g(b,t) = -2\arctan\frac{b+d}{t} + \arctan\frac{b}{t} + \arctan\frac{b+2d}{t}, \tag{3.11}$$

since $\sigma - \beta$ runs from $b_\rho$ to $b_\rho + d$ in the first term and $\sigma + d - \beta$ from $b_\rho + d$ to $b_\rho + 2d$ in the second. For $|b| \le \frac12$ and $|t| \ge 1$, the expansion $\arctan(y/t) = y/t - y^3/(3t^3) + O(|t|^{-5})$ gives $g(b,t) = O(|t|^{-3})$, because $-2(b+d) + b + (b+2d) = 0$; hence $\sum_\rho g(b_\rho, t_\rho)$ converges absolutely.

The map $\rho \mapsto 1 - \bar\rho$ is a bijection of the set of zeros that preserves multiplicities and maps $(b_\rho, t_\rho)$ to $(-b_\rho, t_\rho)$. Hence

$$\sum_\rho g(b_\rho, t_\rho) = \frac12 \sum_\rho \big(g(b_\rho, t_\rho) + g(-b_\rho, t_\rho)\big) = -\frac12 \sum_\rho f(b_\rho, t_\rho),$$

because $g(b,t) + g(-b,t) = -f(b,t)$: the terms $\arctan\frac{b}{t}$ and $\arctan\frac{-b}{t}$ cancel, and the others give (3.2) with the opposite sign. Inserting this in (3.7) proves (3.4). □

For an index $\iota = (\mathrm{X}, j)$ with $\mathrm{X} \in \{\mathrm{R}, \mathrm{I}\}$ and $j \ge 1$, and for $x > 0$, $t \in \mathbb{R}$, put

$$F_{\mathrm{R},j}(x,t) = \operatorname{Re}(x+it)^{-j}, \quad F_{\mathrm{I},j}(x,t) = -\operatorname{Im}(x+it)^{-j}, \quad G_\iota(b,t) = F_\iota(d+b,t) + F_\iota(d-b,t) \ \ (|b| < d). \tag{3.12}$$

A *kernel* is a real linear combination

$$U = \sum_{\iota \in \mathcal{B}} c_\iota G_\iota, \qquad \mathcal{B} = \mathcal{B}_m = \{(\mathrm{R}, j) : 1 \le j \le m\} \cup \{(\mathrm{I}, j) : 2 \le j \le m\}. \tag{3.13}$$

Each $G_\iota$ is even in $b$; $G_{\mathrm{R},j}$ is even in $t$ and $G_{\mathrm{I},j}$ is odd in $t$, because $(x - it)^{-j} = \overline{(x+it)^{-j}}$. Write $\Phi_{\mathrm{R}}(z) = \operatorname{Re} z$, $\Phi_{\mathrm{I}}(z) = -\operatorname{Im} z$ (whenever $\iota = (\mathrm{X}, j)$ is given, X and $j$ in the formulas that follow are the two components of $\iota$, so that $G_\iota(b,t) = \Phi_{\mathrm{X}}\big((d+b+it)^{-j}\big) + \Phi_{\mathrm{X}}\big((d-b+it)^{-j}\big)$), and

$$D_1(s) = \frac{1}{s} + \frac{1}{s-1} - \frac12 \log\pi + \frac12 \psi_\Gamma\Big(\frac{s}{2}\Big), \qquad D_j(s) = \frac{1}{s^j} + \frac{1}{(s-1)^j} + \frac{(-1)^{j-1}}{(j-1)!\,2^j} \psi_\Gamma^{(j-1)}\Big(\frac{s}{2}\Big) \quad (j \ge 2), \tag{3.14}$$

where $\psi_\Gamma = \Gamma'/\Gamma$ is the digamma function (the letter $\psi$ alone denotes the Chebyshev function $\psi(x) = \sum_{n \le x} \Lambda(n)$ in Section 4.2).

**Lemma 3.2** (Sums of the kernel functions over the zeros)**.** *Let $d > \frac{1}{2}$, $T > 0$ and $s_1 = \sigma_1 + iT$. For every $\iota = (\mathrm{X}, j) \in \mathcal{B}$ the series $\sum_\rho G_\iota(b_\rho, t_\rho)$ converges absolutely and*

$$\sum_\rho G_\iota(b_\rho, t_\rho) = 2\,\Phi_{\mathrm{X}}\big(D_j(s_1)\big) - \frac{2}{(j-1)!}\sum_{n\geq 2}\Lambda(n)(\log n)^{j-1}\,\Phi_{\mathrm{X}}\big(n^{-s_1}\big), \tag{3.15}$$

*where $\Phi_{\mathrm{R}}(n^{-s_1}) = n^{-\sigma_1}\cos(T\log n)$ and $\Phi_{\mathrm{I}}(n^{-s_1}) = n^{-\sigma_1}\sin(T\log n)$.*

*Proof.* Since $s_1 - \rho = (d + b_\rho) + it_\rho$ and $s_1 - (1-\bar\rho) = (d - b_\rho) + it_\rho$, and $\rho \mapsto 1-\bar\rho$ permutes the zeros, $\sum_\rho G_\iota(b_\rho, t_\rho) = 2\sum_\rho F_\iota(s_1 - \rho)$, where $F_\iota(z)$ stands for $F_\iota(\operatorname{Re} z, \operatorname{Im} z)$, provided the series converge absolutely. For $j \geq 2$: only finitely many zeros have $|s_1 - \rho| < 1$ (the zeros are isolated and $s_1$ is not a zero), for the others $|s_1-\rho|^{-j} \leq |s_1-\rho|^{-2}$, and $\sum_\rho |s_1-\rho|^{-2} < \infty$ [25, §9.4]. For $j = 1$, $\operatorname{Re}\frac{1}{s_1-\rho} = \frac{\sigma_1-\beta}{|s_1-\rho|^2} > 0$ because $\sigma_1 = \frac{1}{2} + d > 1 > \beta$, and by (3.8) together with $B = -\sum_\rho \operatorname{Re}\frac{1}{\rho}$ [8, Ch. 12], taking real parts, $\operatorname{Re}\frac{\xi'}{\xi}(s) = \sum_\rho \operatorname{Re}\frac{1}{s-\rho}$ for $\operatorname{Re} s > 1$, where the series has positive terms; so $\sum_\rho \operatorname{Re}\frac{1}{s_1-\rho}$ converges to $\operatorname{Re}\frac{\xi'}{\xi}(s_1)$.

From (1.1), for $\operatorname{Re} s > 1$,

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi_\Gamma\Big(\frac{s}{2}\Big) + \frac{\zeta'}{\zeta}(s) = D_1(s) - \sum_{n\geq 2}\Lambda(n)n^{-s}, \tag{3.16}$$

which gives (3.15) for $j = 1$. For $j \geq 2$, differentiating (3.8) $j-1$ times term by term (the differentiated series converge absolutely and locally uniformly away from the zeros) gives $\sum_\rho (s-\rho)^{-j} = \frac{(-1)^{j-1}}{(j-1)!}\big(\frac{\xi'}{\xi}\big)^{(j-1)}(s)$, and differentiating (3.16) $j-1$ times gives

$$\frac{(-1)^{j-1}}{(j-1)!}\Big(\frac{\xi'}{\xi}\Big)^{(j-1)}(s) = D_j(s) - \frac{1}{(j-1)!}\sum_{n\geq 2}\Lambda(n)(\log n)^{j-1}n^{-s},$$

because $\frac{d^{j-1}}{ds^{j-1}}s^{-1} = (-1)^{j-1}(j-1)!\,s^{-j}$ and $\frac{d^{j-1}}{ds^{j-1}}\frac{1}{2}\psi_\Gamma(\frac{s}{2}) = 2^{-j}\psi_\Gamma^{(j-1)}(\frac{s}{2})$. Applying $\Phi_{\mathrm{X}}$ and multiplying by 2 proves (3.15). □

## 4. The explicit estimate

4.1. **The decomposition.** Fix $d > \frac{1}{2}$, $m \geq 2$ and a kernel $U = \sum_{\iota\in\mathcal{B}_m} c_\iota G_\iota$ such that

$$f(b,t) \leq U(b,t) \qquad \text{for } 0 \leq b \leq \tfrac{1}{2},\ t \neq 0. \tag{K}$$

Since $f$ and $U$ are even in $b$, (K) holds for $|b| \leq \frac{1}{2}$. Put $\varepsilon_{\mathrm{R}} = 1$, $\varepsilon_{\mathrm{I}} = -1$ and, for $L > 0$,

$$a(L) = -\frac{1}{\pi}\sum_{(\mathrm{R},j)\in\mathcal{B}} c_{\mathrm{R},j}\frac{L^{j-1}}{(j-1)!}, \qquad b(L) = \frac{1}{\pi}\Big(\frac{e^{-dL}-2}{L} - \sum_{(\mathrm{I},j)\in\mathcal{B}} c_{\mathrm{I},j}\frac{L^{j-1}}{(j-1)!}\Big). \tag{4.1}$$

For $s = \sigma + iT$ with $\sigma > 1$ and $T > 0$ put

$$\mathcal{A}(s) = \arg s + \arg(s-1) - \frac{T}{2}\log\pi + \operatorname{Im}\operatorname{Log}\Gamma\Big(\frac{s}{2}\Big), \qquad \arg s = \arctan\frac{T}{\sigma},\quad \arg(s-1) = \arctan\frac{T}{\sigma-1}, \tag{4.2}$$

and

$$\begin{gathered}\mathcal{E}(T) = \frac{1}{\pi}\big(2\mathcal{A}(s_1) - \mathcal{A}(s_2)\big) - \mathcal{M}(T),\\ \mathcal{E}_u(T) = \mathcal{E}(T) + \frac{1}{\pi}\sum_{\iota\in\mathcal{B}} c_\iota\Phi_{\mathrm{X}}\big(D_j(s_1)\big), \qquad \mathcal{E}_\ell(T) = \mathcal{E}(T) - \frac{1}{\pi}\sum_{\iota\in\mathcal{B}}\varepsilon_{\mathrm{X}}c_\iota\Phi_{\mathrm{X}}\big(D_j(s_1)\big).\end{gathered} \tag{4.3}$$

**Proposition 4.1** (Two-sided estimate)**.** *Under* (K)*, for every $T > 0$ that is not an ordinate,*

$$\begin{aligned} N(T) - \mathcal{M}(T) &\le \mathcal{E}_u(T) + \sum_{n\ge 2} \Lambda(n) n^{-\sigma_1} \big(a(\log n)\cos(T\log n) + b(\log n)\sin(T\log n)\big), \\ N(T) - \mathcal{M}(T) &\ge \mathcal{E}_\ell(T) + \sum_{n\ge 2} \Lambda(n) n^{-\sigma_1} \big(-a(\log n)\cos(T\log n) + b(\log n)\sin(T\log n)\big), \end{aligned} \tag{4.4}$$

*and both series converge absolutely.*

*Proof.* For $\operatorname{Re} s > 1$, $\operatorname{Log}\zeta(s) = \sum_{n\ge 2} \frac{\Lambda(n)}{\log n} n^{-s}$ is the branch of $\log\zeta$ that is real on $(1,\infty)$ and continuous on the half-plane $\operatorname{Re} s > 1$, and $\xi(s) = \frac{1}{2}\cdot s\cdot(s-1)\cdot \pi^{-s/2}\cdot\Gamma(\frac{s}{2})\cdot\zeta(s)$ is a product of non-vanishing factors each of which has a continuous argument on $\operatorname{Re} s \ge 1$, $T > 0$, vanishing on the real axis: $\arg s$ and $\arg(s-1)$ as in (4.2), $-\frac{T}{2}\log\pi$, $\operatorname{Im}\operatorname{Log}\Gamma(\frac{s}{2})$ and $\operatorname{Im}\operatorname{Log}\zeta(s)$. Their sum is a continuous argument of $\xi$ vanishing on the real axis, hence it is the branch of Section 3:

$$\arg\xi(s) = \mathcal{A}(s) - \sum_{n\ge 2} \frac{\Lambda(n)}{\log n} n^{-\sigma}\sin(T\log n) \qquad (s = \sigma + iT,\ \sigma > 1,\ T > 0). \tag{4.5}$$

Since $n^{-\sigma_2} = n^{-\sigma_1} e^{-d\log n}$,

$$\frac{1}{\pi}\big(2\arg\xi(s_1) - \arg\xi(s_2)\big) = \frac{1}{\pi}\big(2\mathcal{A}(s_1) - \mathcal{A}(s_2)\big) + \frac{1}{\pi}\sum_{n\ge 2}\Lambda(n) n^{-\sigma_1}\frac{e^{-d\log n} - 2}{\log n}\sin(T\log n). \tag{4.6}$$

*Upper bound.* By (K) at $(b_\rho, t_\rho)$, $f(b_\rho,t_\rho) \le U(b_\rho,t_\rho)$ for every zero, and the series $\sum_\rho U(b_\rho,t_\rho)$ converges absolutely by Lemma 3.2. Hence, by Lemma 3.1 and (3.15),

$$\begin{aligned} \frac{1}{2\pi}\sum_\rho f(b_\rho,t_\rho) &\le \frac{1}{2\pi}\sum_\iota c_\iota \sum_\rho G_\iota(b_\rho,t_\rho) \\ &= \frac{1}{\pi}\sum_\iota c_\iota \Phi_{\mathrm{X}}\big(D_j(s_1)\big) - \frac{1}{\pi}\sum_{n\ge 2}\Lambda(n) n^{-\sigma_1}\sum_\iota c_\iota \frac{(\log n)^{j-1}}{(j-1)!}\Phi_{\mathrm{X}}\big(e^{-iT\log n}\big), \end{aligned}$$

and $\Phi_{\mathrm{R}}(e^{-iT\log n}) = \cos(T\log n)$, $\Phi_{\mathrm{I}}(e^{-iT\log n}) = \sin(T\log n)$, so, with $Q(L) = \sum_{(\mathrm{I},j)} c_{\mathrm{I},j} L^{j-1}/(j-1)! = \frac{e^{-dL}-2}{L} - \pi b(L)$, the inner sum over $\iota$ is $-\pi a(\log n)\cos(T\log n) + Q(\log n)\sin(T\log n)$ and the right side equals

$$\begin{aligned} &\frac{1}{\pi}\sum_\iota c_\iota \Phi_{\mathrm{X}}(D_j(s_1)) + \sum_{n\ge 2}\Lambda(n) n^{-\sigma_1}\big(a(\log n)\cos(T\log n) + b(\log n)\sin(T\log n)\big) \\ &\qquad - \frac{1}{\pi}\sum_{n\ge 2}\Lambda(n) n^{-\sigma_1}\frac{e^{-d\log n}-2}{\log n}\sin(T\log n). \end{aligned}$$

Adding (4.6) cancels the last sum and gives the first line of (4.4).

*Lower bound.* Since $f(b,-t) = -f(b,t)$, (K) at $(b_\rho, -t_\rho)$ gives $f(b_\rho,t_\rho) \ge -U(b_\rho,-t_\rho)$, and $U(b,-t) = \sum_\iota \varepsilon_{\mathrm{X}} c_\iota G_\iota(b,t)$ by the parity of the $G_\iota$ in $t$. Hence

$$\begin{aligned} \frac{1}{2\pi}\sum_\rho f(b_\rho,t_\rho) &\ge -\frac{1}{\pi}\sum_\iota \varepsilon_{\mathrm{X}} c_\iota \Phi_{\mathrm{X}}\big(D_j(s_1)\big) \\ &\qquad + \frac{1}{\pi}\sum_{n\ge 2}\Lambda(n) n^{-\sigma_1}\sum_\iota \varepsilon_{\mathrm{X}} c_\iota \frac{(\log n)^{j-1}}{(j-1)!}\Phi_{\mathrm{X}}\big(e^{-iT\log n}\big). \end{aligned}$$

Here the inner sum over $\iota$ is $-\pi a(\log n)\cos(T\log n) - Q(\log n)\sin(T\log n)$, by (4.1) and $\varepsilon_{\mathrm{I}} = -1$, so the right side equals

$$-\frac{1}{\pi}\sum_{\iota}\varepsilon_{\mathrm{X}}c_\iota\Phi_{\mathrm{X}}(D_j(s_1)) + \sum_{n\geq 2}\Lambda(n)n^{-\sigma_1}\big(-a(\log n)\cos(T\log n) + b(\log n)\sin(T\log n)\big) \\ - \frac{1}{\pi}\sum_{n\geq 2}\Lambda(n)n^{-\sigma_1}\frac{e^{-d\log n}-2}{\log n}\sin(T\log n).$$

Adding (4.6) cancels the last sum and gives the second line of (4.4). All the series over $n$ converge absolutely, because $a(\log n)$ and $b(\log n)$ are bounded by polynomials in $\log n$ and $\sum_{n\geq 2}\Lambda(n)n^{-\sigma_1}(\log n)^r < \infty$ for every $r \geq 0$ when $\sigma_1 > 1$. □

For a prime $p$ write $\lambda_k = (\log p)\,p^{-k\sigma_1}$, $a_k = a(k\log p)$, $b_k = b(k\log p)$ and

$$f_p(x) = \sum_{k\geq 1}\lambda_k(a_k\cos kx + b_k\sin kx) \qquad (x\in\mathbb{R}), \qquad M_p = \max_{x\in\mathbb{R}} f_p(x). \tag{4.7}$$

The series converges absolutely and uniformly, so $f_p$ is continuous and $2\pi$-periodic, and $M_p$ exists. Grouping the prime powers $n = p^k$ in (4.4) according to $p$, with the common phase $x = T\log p$, the series in the first line is $\sum_p f_p(T\log p) \leq \sum_p M_p$, and the series in the second line is $\sum_p(-f_p(-T\log p)) \geq -\sum_p M_p$, because $\sum_k \lambda_k(-a_k\cos kx + b_k\sin kx) = -f_p(-x)$. This is the observation of Amberger [1, Lemma 9] that the lower bound is obtained from the upper one by changing the sign of the odd part. With

$$F_u(T) = \mathcal{E}_u(T) - A_1\log T, \qquad F_\ell(T) = -\mathcal{E}_\ell(T) - A_1\log T, \tag{4.8}$$

Proposition 4.1 gives, for every $T > 0$ not an ordinate and every $A_1$ and $C_\zeta \geq \sum_p M_p$,

$$\big|N(T) - \mathcal{M}(T)\big| \leq A_1\log T + C_\zeta + \max\big(F_u(T), F_\ell(T)\big). \tag{4.9}$$

Sections 4.2–4.4 bound the three ingredients: $C_\zeta$, $\sup_{T\geq 1}\max(F_u, F_\ell)$, and the verification of (K).

### 4.2. **The Euler product.**

**Lemma 4.2** (Bounds for $M_p$)**.** *Let $h(L) = \sqrt{a(L)^2 + b(L)^2}$ and $H_{p,k} = \lambda_k h(k\log p)$. Then*

*(a) $M_p \leq \sum_{k\geq 1} H_{p,k}$; more precisely, for every $K \geq 1$, $M_p \leq \max_x \sum_{k\leq K}\lambda_k(a_k\cos kx + b_k\sin kx) + \sum_{k>K} H_{p,k}$;*

*(b) for every integer $\mathsf{M} \geq 1$ and $x_i = 2\pi i/\mathsf{M}$ $(0 \leq i < \mathsf{M})$, and every $K \geq 1$,*

$$\max_x \sum_{k\leq K}\lambda_k(a_k\cos kx + b_k\sin kx) \leq \max_{0\leq i<\mathsf{M}}\sum_{k\leq K}\lambda_k(a_k\cos kx_i + b_k\sin kx_i) + \frac{\pi}{\mathsf{M}}\sum_{k\leq K} kH_{p,k}.$$

*Proof.* (a) $|a\cos\theta + b\sin\theta| \leq \sqrt{a^2+b^2}$ for real $a, b, \theta$, and the maximum of a sum is at most the sum of the maxima. (b) The finite sum $\phi(x) = \sum_{k\leq K}\lambda_k(a_k\cos kx + b_k\sin kx)$ is differentiable with $|\phi'(x)| \leq \sum_{k\leq K} k\lambda_k\sqrt{a_k^2+b_k^2}$, every real $x$ is within $\pi/\mathsf{M}$ of some $x_i$ modulo $2\pi$, and $\phi$ is $2\pi$-periodic; the mean value theorem gives the claim. □

**Lemma 4.3** (The tail)**.** *Let $N_0 \geq 2$, $g(x) = h(\log x)\,x^{-\sigma_1}$ for $x \geq N_0$, and let $G\colon [N_0,\infty) \to [0,\infty)$ be non-increasing, integrable, with $G \geq g$. Then*

$$\sum_{n>N_0}\Lambda(n)g(n) \leq G(N_0)\big(1.03883\,N_0 - \psi(N_0)\big) + 1.03883\int_{N_0}^{\infty} G(x)\,dx, \tag{4.10}$$

*where $\psi(x) = \sum_{n\leq x}\Lambda(n)$.*

*Proof.* For $0 < y < G(N_0)$ put $X_y = \sup\{x \geq N_0 : G(x) > y\}$, which is finite because $G$ is integrable and non-increasing. If $n > N_0$ and $G(n) > y$ then $n \leq X_y$, so $\sum_{n>N_0,\, G(n)>y} \Lambda(n) \leq \psi(X_y) - \psi(N_0) < 1.03883 X_y - \psi(N_0)$ by $\psi(x) < 1.03883x$ for all $x > 0$ [22, Thm. 12]; for $y \geq G(N_0)$ the set is empty. By Tonelli's theorem,

$$\sum_{n>N_0} \Lambda(n) g(n) \leq \sum_{n>N_0} \Lambda(n) G(n) = \int_0^\infty \sum_{\substack{n>N_0 \\ G(n)>y}} \Lambda(n)\, dy \leq \int_0^{G(N_0)} (1.03883 X_y - \psi(N_0))\, dy.$$

Finally $\int_0^{G(N_0)} X_y\, dy = N_0 G(N_0) + \int_0^{G(N_0)} (X_y - N_0)\, dy$, and for $x > N_0$ one has $X_y \geq x$ if $G(x) > y$, while $X_y \geq x$ implies $G(x) \geq y$ (there are $x' \geq x$ with $G(x') > y$, and $G(x) \geq G(x')$); hence $\{y : X_y \geq x\}$ and $\{y : y < G(x)\}$ differ by a set of measure zero, and $\int_0^{G(N_0)} (X_y - N_0)\, dy = \int_{N_0}^\infty \int_0^{G(N_0)} \mathbf{1}_{X_y \geq x}\, dy\, dx = \int_{N_0}^\infty G(x)\, dx$. This is (4.10). □

The majorant $G$ is built as follows. Let $L_0 = \log N_0$ and let $L_{\mathrm{B}} > L_0$ be such that $g$ is non-increasing on $[e^{L_{\mathrm{B}}}, \infty)$; then on $[e^{L_{\mathrm{B}}}, \infty)$ take $G = g$. On $[L_0, L_{\mathrm{B}}]$ choose points $L_0 = u_0 < u_1 < \cdots < u_r = L_{\mathrm{B}}$, let $\bar{g}_i$ be an upper bound of $g$ on $[e^{u_i}, e^{u_{i+1}}]$ computed in ball arithmetic, and put $G = \max(\bar{g}_i, \bar{g}_{i+1}, \ldots, \bar{g}_{r-1}, g(e^{L_{\mathrm{B}}}))$ on $[e^{u_i}, e^{u_{i+1}})$; this $G$ is non-increasing and $G \geq g$, and $\int_{N_0}^{e^{L_{\mathrm{B}}}} G\, dx$ is a finite sum. The condition on $L_{\mathrm{B}}$ is verified by the next lemma. Write $a(L) = \frac{1}{\pi} P_a(L)$ and $b(L) = \frac{1}{\pi}\big(P_b(L) + \frac{e^{-dL} - 2}{L}\big)$ with polynomials $P_a = \sum_i \mathsf{a}_i L^i$, $P_b = \sum_i \mathsf{b}_i L^i$ read off from (4.1), let $D$ be the larger of their degrees, $P$ the one of $P_a$, $P_b$ of degree $D$ (if both have degree $D$, take $P = P_a$), and $q_i = |\mathsf{a}_i| + |\mathsf{b}_i|$.

**Lemma 4.4** (Monotonicity of $g$)**.** *Let $u = 1/L$ and*

$$\mathrm{num}(u) = u \sum_{i \geq 1} i q_i u^{D-i} + d\, u^{D+1} + 2u^{D+2}, \qquad \mathrm{low}(u) = |\mathsf{p}_D| - \sum_{i<D} |\mathsf{p}_i| u^{D-i} - \varepsilon_P u^{D+1},$$

*where $\mathsf{p}_i$ are the coefficients of $P$ and $\varepsilon_P = 2$ if $P = P_b$, $\varepsilon_P = 0$ if $P = P_a$. If $\mathrm{low}(u) > 0$ and $\mathrm{num}(u)/\mathrm{low}(u) < \sigma_1$ for all $u \in [0, 1/L_{\mathrm{B}}]$, then $g$ is strictly decreasing on $[e^{L_{\mathrm{B}}}, \infty)$.*

*Proof.* On $[e^{L_{\mathrm{B}}}, \infty)$ one has $h > 0$ (because low $> 0$ gives $P \neq 0$ there), so $\log g$ is differentiable in $L = \log x$ with $\frac{d}{dL} \log g = \frac{aa' + bb'}{a^2 + b^2} - \sigma_1 \leq \frac{|a'| + |b'|}{h} - \sigma_1$, using $|a|, |b| \leq h$. Further $h \geq \frac{1}{\pi}|P(L)| - \frac{\varepsilon_P}{\pi L} \geq \frac{L^D}{\pi} \mathrm{low}(1/L)$, since $|\frac{e^{-dL} - 2}{L}| \leq \frac{2}{L}$, and $|a'| + |b'| \leq \frac{1}{\pi}\big(\sum_{i \geq 1} i q_i L^{i-1} + \frac{d}{L} + \frac{2}{L^2}\big) = \frac{L^D}{\pi} \mathrm{num}(1/L)$, because $|\frac{d}{dL} \frac{e^{-dL} - 2}{L}| \leq \frac{d e^{-dL}}{L} + \frac{|e^{-dL} - 2|}{L^2} \leq \frac{d}{L} + \frac{2}{L^2}$. Hence $\frac{d}{dL} \log g \leq \mathrm{num}/\mathrm{low} - \sigma_1 < 0$. □

**Proposition 4.5** (Certified value of $C_\zeta$)**.** *Let $N_0 = 10^8$, $P_0 = 1000$, $\mathsf{M} = 4096$. Then $\sum_p M_p \leq C_\zeta$ with*

$$C_\zeta = \sum_{p \leq P_0} \min\Big( \max_{i < \mathsf{M}} \sum_{k:\, p^k \leq N_0} \lambda_k (a_k \cos k x_i + b_k \sin k x_i) + \frac{\pi}{\mathsf{M}} \sum_{k:\, p^k \leq N_0} k H_{p,k},\ \sum_{k:\, p^k \leq N_0} H_{p,k} \Big) + \sum_{\substack{P_0 < p \leq N_0 \\ p^k \leq N_0}} H_{p,k} + \mathrm{Tail}, \tag{4.11}$$

*where* Tail *is the right side of* (4.10) *with $L_{\mathrm{B}}$ verified by Lemma 4.4. The program* `cz_gen_pp.py` *(Appendix B) evaluates every term of* (4.11) *in ball arithmetic and reports an upper bound of the resulting ball. For the six kernels of Theorem 2.1 the values are in Table 2.*

*Proof.* For $p \leq P_0$, Lemma 4.2(a) with $K$ the largest $k$ with $p^k \leq N_0$, then (b) for the finite sum, and the powers $p^k > N_0$ go to the tail; for $P_0 < p \leq N_0$, Lemma 4.2(a) with the same split; for $p > N_0$, Lemma 4.2(a) with $K = 0$. All prime powers $n > N_0$ are covered by $\sum_{n>N_0} \Lambda(n) g(n)$, bounded by Lemma 4.3 with the majorant described before Lemma 4.4. The last integral $\int_{e^{L_{\mathrm{B}}}}^\infty g\, dx$

is bounded as follows. Let $L_1 = 800$ and let $\tilde g \ge g$ be the majorant of $g$ on $[e^{L_1},\infty)$ given below. In the six runs $L_{\mathrm B} = 2000$ or $5000$, so $L_{\mathrm B} > L_1$ and

$$\int_{e^{L_{\mathrm B}}}^{\infty} g\,dx = \int_{e^{L_1}}^{\infty} g\,dx - \int_{e^{L_1}}^{e^{L_{\mathrm B}}} g\,dx \le \int_{e^{L_1}}^{\infty} \tilde g\,dx - \int_{e^{L_1}}^{e^{L_{\mathrm B}}} g\,dx$$

(the program writes the last integral as $\int_{e^{L_{\mathrm B}}}^{e^{L_1}} g\,dx$, which is negative; if $L_{\mathrm B} < L_1$ the same expression is still an upper bound, since then it equals $\int_{e^{L_{\mathrm B}}}^{e^{L_1}} g\,dx + \int_{e^{L_1}}^{\infty} \tilde g\,dx \ge \int_{e^{L_{\mathrm B}}}^{\infty} g\,dx$). The integral $\int_{e^{L_1}}^{e^{L_{\mathrm B}}} g\,dx$ is computed by the rigorous quadrature of Arb [15] in the variable $L$, which evaluates the integrand $\sqrt{a(L)^2+b(L)^2}\,e^{(1-\sigma_1)L}$ on complex balls and uses Gauss–Legendre rules only on sub-intervals where the implementation certifies that the integrand is holomorphic (the principal square root returns an indeterminate ball when its argument ball meets $(-\infty,0]$, and elsewhere it is holomorphic and coincides with $h$ on the real segment, where $a^2+b^2 \ge 0$), falling back to the enclosure of the integrand on the sub-interval otherwise, and the majorant is $\tilde g(x) = \frac{1}{\pi}\bigl(\sum_i q_i L^i + 3\bigr)e^{-\sigma_1 L}$ with $L = \log x$, which satisfies $\tilde g \ge g$ on $[e^{L_1},\infty)$ because $h \le |a| + |b| \le \frac{1}{\pi}\bigl(\sum_i q_i L^i + \frac{2}{L}\bigr)$ and $\frac{2}{L} \le 3$; its integral $\int_{e^{L_1}}^{\infty} \tilde g\,dx = \frac{1}{\pi}\int_{L_1}^{\infty}\bigl(\sum_i q_i L^i + 3\bigr)e^{(1-\sigma_1)L}\,dL$ is a finite combination of incomplete gamma functions evaluated in closed form. The partial sums over $n \le N_0$ are exact ball sums over the $5\,761\,455$ primes below $10^8$ and their powers. □

### 4.3. **The elementary part.**

**Lemma 4.6** (Stirling's formula with remainder)**.** *For $|\operatorname{ph} z| < \pi$, $\operatorname{Log}\Gamma(z) = (z-\frac12)\operatorname{Log} z - z + \frac12\log(2\pi) + \mathcal R(z)$ with $|\mathcal R(z)| \le \sec^2(\frac12 \operatorname{ph} z)/(12|z|)$ [9, §5.11(ii)], [17, (8.4.06)]. Consequently, for $\operatorname{Re} z > 0$ and every integer $k \ge 0$,*

$$\bigl|\mathcal R^{(k+1)}(z)\bigr| \le (k+1)!\,\frac{2}{3|z|}\Bigl(\frac{2}{|z|}\Bigr)^{k+1}. \tag{4.12}$$

*Proof.* On the circle $|w-z| = \frac12|z|$ one has $|w| \ge \frac12|z|$ and $|\operatorname{ph} w - \operatorname{ph} z| \le \arcsin\frac12 = \frac{\pi}{6}$, so $|\operatorname{ph} w| < \frac{\pi}{2} + \frac{\pi}{6} = \frac{2\pi}{3} < \pi$: the closed disc $|w-z| \le \frac12|z|$ lies in the cut plane $|\operatorname{ph} w| < \pi$ where $\mathcal R$ is holomorphic (it may enter the left half-plane, but stays away from the negative real axis), and $\sec^2(\frac12 \operatorname{ph} w) < \sec^2\frac{\pi}{3} = 4$; hence $|\mathcal R(w)| \le \frac{4}{12\cdot|z|/2} = \frac{2}{3|z|}$ there, and Cauchy's estimate for the $(k+1)$-st derivative on this circle gives (4.12). □

Since $\psi_\Gamma = (\operatorname{Log}\Gamma)'$, Lemma 4.6 gives $\psi_\Gamma(z) = \operatorname{Log} z - \frac{1}{2z} + \mathcal R'(z)$ and, for $k \ge 1$,

$$\psi_\Gamma^{(k)}(z) = \frac{(-1)^{k-1}(k-1)!}{z^k} + \frac{(-1)^k k!}{2z^{k+1}} + \mathcal R^{(k+1)}(z). \tag{4.13}$$

**Lemma 4.7** (The elementary part for large $T$)**.** *Let $T \ge T_0 > 0$, $z_1 = s_1/2$, $C_1 = c_{\mathrm R,1}/(2\pi)$ and suppose $A_1 \ge C_1$. Then*

$$F_u(T) \le \frac78 - C_1\log(2\pi) + \frac{\mathsf K}{T}, \qquad F_\ell(T) \le -\frac78 - C_1\log(2\pi) + \frac{\mathsf K}{T}, \tag{4.14}$$

*where*

$$\begin{aligned}\mathsf K = {}& \frac{1}{\pi}\Bigl(4d + \frac{\sigma_1^2}{2} + \frac{\sigma_2(\sigma_2-1)}{2} + 1 + \frac{\sigma_2^2}{2} + \sigma_1(\sigma_1-1)\Bigr) + \frac{|c_{\mathrm R,1}|}{\pi T_0}\Bigl((2\sigma_1-1) + \frac{\sigma_1^2}{4} + \frac83\Bigr)\\ &+ \frac{T_0}{\pi}\sum_{\substack{\iota=(\mathrm X,j)\in\mathcal B\\ j\ge2}} |c_\iota|\Bigl(\frac{2}{T_0^{\,j}} + \frac{1}{(j-1)!\,2^j}\Bigl[(j-2)!\Bigl(\frac{2}{T_0}\Bigr)^{j-1} + \frac{(j-1)!}{2}\Bigl(\frac{2}{T_0}\Bigr)^{j} + j!\,\frac{4}{3T_0}\Bigl(\frac{4}{T_0}\Bigr)^{j}\Bigr]\Bigr).\end{aligned} \tag{4.15}$$

*The three summands of $\mathsf K$ are, in this order, the contribution of the term $\mathcal E(T)$ (arguments and Stirling remainders), that of the kernel term $j=1$, and that of the kernel terms $j \ge 2$; the proof*

*treats them in the same order, and the program* `elem_gen.py` *(Appendix B) accumulates them in the same order (with the additional term of Lemma 4.8 in mode S).*

*Proof. The term $\mathcal{E}(T)$.* For $s=\sigma+iT$ with $\sigma>1$, $\arg s=\frac{\pi}{2}-\arctan\frac{\sigma}{T}$ and $\arg(s-1)=\frac{\pi}{2}-\arctan\frac{\sigma-1}{T}$, so $\arg s+\arg(s-1)=\pi-\theta_\sigma$ with $0\le\theta_\sigma\le\frac{2\sigma-1}{T}$. With $z=s/2$, Lemma 4.6 gives

$$\operatorname{Im}\operatorname{Log}\Gamma(z)=\Big(\frac{\sigma}{2}-\frac{1}{2}\Big)\arg z+\frac{T}{2}\log|z|-\frac{T}{2}+\operatorname{Im}\mathcal{R}(z),$$

with $\arg z=\frac{\pi}{2}-\arctan\frac{\sigma}{T}$ and $\log|z|=\log\frac{T}{2}+\frac12\log(1+\frac{\sigma^2}{T^2})$, and $|\mathcal{R}(z)|\le\frac{1}{3T}$ since $|\operatorname{ph} z|<\frac{\pi}{2}$ and $|z|\ge\frac{T}{2}$. Hence

$$\mathcal{A}(s)=\pi+\frac{T}{2}\log\frac{T}{2\pi}-\frac{T}{2}+\frac{(\sigma-1)\pi}{4}+\delta_\sigma,\quad \delta_\sigma=-\theta_\sigma-\frac{\sigma-1}{2}\arctan\frac{\sigma}{T}+\frac{T}{4}\log\Big(1+\frac{\sigma^2}{T^2}\Big)+\operatorname{Im}\mathcal{R}(z),$$

with $0\le\frac{\sigma-1}{2}\arctan\frac{\sigma}{T}\le\frac{\sigma(\sigma-1)}{2T}$ and $0\le\frac{T}{4}\log(1+\frac{\sigma^2}{T^2})\le\frac{\sigma^2}{4T}$. Since $2(\sigma_1-1)-(\sigma_2-1)=-\frac12$ and $2(2\sigma_1-1)=2\sigma_2-1=4d$,

$$\frac{1}{\pi}\big(2\mathcal{A}(s_1)-\mathcal{A}(s_2)\big)=1+\mathcal{M}(T)-\frac18+\frac{2\delta_{\sigma_1}-\delta_{\sigma_2}}{\pi},$$

and, collecting the bounds (the arguments contribute at most $\frac{4d}{T}$ in absolute value, the three remainders at most $\frac1T$),

$$|\mathcal{E}(T)-\tfrac78|\le\frac{1}{\pi T}\Big(4d+\sigma_1(\sigma_1-1)+\frac{\sigma_1^2}{2}+\frac{\sigma_2(\sigma_2-1)}{2}+\frac{\sigma_2^2}{4}+1\Big)\le\frac{\mathsf{K}_\Gamma}{T},$$

where $\mathsf{K}_\Gamma$ is the first bracket of (4.15) (which is larger, having $\frac{\sigma_2^2}{2}$ in place of $\frac{\sigma_2^2}{4}$).

*The term $j=1$.* By $\psi_\Gamma(z)=\operatorname{Log}z-\frac{1}{2z}+\mathcal{R}'(z)$ (the formula preceding (4.13)), $\operatorname{Re}D_1(s_1)=\operatorname{Re}\frac{1}{s_1}+\operatorname{Re}\frac{1}{s_1-1}-\frac12\log\pi+\frac12\log|z_1|-\frac12\operatorname{Re}\frac{1}{s_1}+\frac12\operatorname{Re}\mathcal{R}'(z_1)$, and $\log|z_1|=\log\frac{T}{2}+\frac12\log(1+\frac{\sigma_1^2}{T^2})$, so

$$\operatorname{Re}D_1(s_1)=\frac12\log\frac{T}{2\pi}+\delta_1,\qquad \delta_1=\frac12\operatorname{Re}\frac{1}{s_1}+\operatorname{Re}\frac{1}{s_1-1}+\frac14\log\Big(1+\frac{\sigma_1^2}{T^2}\Big)+\frac12\operatorname{Re}\mathcal{R}'(z_1),$$

and, using $0\le\operatorname{Re}\frac1s\le\frac{\sigma}{T^2}$, $0\le\operatorname{Re}\frac{1}{s-1}\le\frac{\sigma-1}{T^2}$, $0\le\log(1+y)\le y$ and $|\mathcal{R}'(z_1)|\le\frac{4}{3|z_1|^2}\le\frac{16}{3T^2}$ from (4.12),

$$|\delta_1|\le\frac{1}{T^2}\Big(\frac{3\sigma_1}{2}-1+\frac{\sigma_1^2}{4}+\frac83\Big)\le\frac{1}{T^2}\Big((2\sigma_1-1)+\frac{\sigma_1^2}{4}+\frac83\Big)\le\frac{1}{T_0T}\Big((2\sigma_1-1)+\frac{\sigma_1^2}{4}+\frac83\Big)$$

for $T\ge T_0$, so $|c_{\mathrm{R},1}|\,|\delta_1|/\pi$ is at most the second term of (4.15) divided by $T$.

*The terms $j\ge2$.* $|D_j(s_1)|\le\frac{2}{T^j}+\frac{|\psi_\Gamma^{(j-1)}(z_1)|}{(j-1)!\,2^j}$, and by (4.13) with $k=j-1\ge1$, (4.12) and $|z_1|\ge\frac{T}{2}$,

$$|\psi_\Gamma^{(j-1)}(z_1)|\le(j-2)!\Big(\frac{2}{T}\Big)^{j-1}+\frac{(j-1)!}{2}\Big(\frac{2}{T}\Big)^{j}+j!\,\frac{4}{3T}\Big(\frac{4}{T}\Big)^{j}.$$

The resulting bound for $|D_j(s_1)|$ is of the form $\sum_{\nu\ge1}\alpha_\nu T^{-\nu}$ with $\alpha_\nu\ge0$ (for $j=2$ the first term is $\frac{1}{2T}$), so $T$ times it is non-increasing in $T$ and $|D_j(s_1)|\le T_0\,\mathrm{bound}(T_0)/T$ for $T\ge T_0$; this gives the third term of (4.15).

*Conclusion.* $\mathcal{E}_u(T)=\mathcal{E}(T)+\frac{c_{\mathrm{R},1}}{\pi}\operatorname{Re}D_1(s_1)+\frac1\pi\sum_{j\ge2}c_\iota\Phi_{\mathrm{X}}(D_j(s_1))=\frac78+C_1\log\frac{T}{2\pi}+\Delta(T)$ with $|\Delta(T)|\le\mathsf{K}/T$, so $F_u(T)=\frac78-C_1\log(2\pi)+(C_1-A_1)\log T+\Delta(T)\le\frac78-C_1\log(2\pi)+\mathsf{K}/T$, because $A_1\ge C_1$ and $\log T\ge0$. Likewise $\mathcal{E}_\ell(T)=\frac78-C_1\log\frac{T}{2\pi}+\Delta_\ell(T)$ with $|\Delta_\ell|\le\mathsf{K}/T$, and $F_\ell(T)=-\frac78-C_1\log(2\pi)+(C_1-A_1)\log T-\Delta_\ell(T)\le-\frac78-C_1\log(2\pi)+\mathsf{K}/T$. □

**Lemma 4.8** (The theta function)**.** *For $T>0$, $\frac1\pi\vartheta(T)+1=\mathcal{M}(T)+\frac78+\frac{\varpi(T)}{\pi}$ with $|\varpi(T)|\le(\frac{1}{16}+\frac18+\frac13)\frac1T$.*

*Proof.* With $z=\frac14+\frac{iT}{2}$, Lemma 4.6 gives $\vartheta(T)=-\frac14\arg z+\frac{T}{2}\log|z|-\frac{T}{2}-\frac{T}{2}\log\pi+\operatorname{Im}\mathcal{R}(z)$, and $\arg z=\frac{\pi}{2}-\arctan\frac{1}{2T}$, $\log|z|=\log\frac{T}{2}+\frac12\log(1+\frac{1}{4T^2})$. Hence $\vartheta(T)=\frac{T}{2}\log\frac{T}{2\pi}-\frac{T}{2}-\frac{\pi}{8}+\varpi(T)$ with $\varpi=\frac14\arctan\frac{1}{2T}+\frac{T}{4}\log(1+\frac{1}{4T^2})+\operatorname{Im}\mathcal{R}(z)$, and the three terms are at most $\frac{1}{8T}$, $\frac{1}{16T}$ and $\frac{1}{3T}$ ($|z|\ge\frac{T}{2}$, $|\operatorname{ph} z|<\frac{\pi}{2}$). Dividing by $\pi$ and using $\pi\mathcal{M}(T)=\frac{T}{2}\log\frac{T}{2\pi}-\frac{T}{2}$ gives the claim. □

For the argument, write $\mathcal{E}_u^S=\mathcal{E}_u-\frac78-\frac{\varpi}{\pi}$, $\mathcal{E}_\ell^S=\mathcal{E}_\ell-\frac78-\frac{\varpi}{\pi}$ and $F_u^S=\mathcal{E}_u^S-A_1\log T$, $F_\ell^S=-\mathcal{E}_\ell^S-A_1\log T$. Since $S(T)=N(T)-\mathcal{M}(T)-\frac78-\frac{\varpi(T)}{\pi}$ by (1.5) and Lemma 4.8, Proposition 4.1 gives

$$|S(T)|\le A_1\log T+C_\zeta+\max\bigl(F_u^S(T),F_\ell^S(T)\bigr),\tag{4.16}$$

and Lemma 4.7 gives, for $T\ge T_0$, $\max(F_u^S,F_\ell^S)\le -C_1\log(2\pi)+(\mathsf{K}+\frac{1}{\pi}(\frac{1}{16}+\frac18+\frac13))/T$.

**Proposition 4.9** (Certified suprema of the elementary part)**.** *Let $T_0=10^4$. The program* `elem_gen.py` *(Appendix B) covers $[1,T_0]$ by 9215 intervals $[\alpha,\beta]$ with $\beta/\alpha\le1.001$ and, on each, encloses $F_u$ and $F_\ell$ (or $F_u^S$, $F_\ell^S$) by the centred form $F(\frac{\alpha+\beta}{2})+F'([\alpha,\beta])\cdot[-\frac{\beta-\alpha}{2},\frac{\beta-\alpha}{2}]$, where $F'([\alpha,\beta])$ is a ball enclosing the derivative on the interval; for $T\ge T_0$ it uses Lemma 4.7. The resulting upper bounds for $\sup_{T\ge1}\max(F_u,F_\ell)$ and $\sup_{T\ge1}\max(F_u^S,F_\ell^S)$ for the six kernels are in Table 2.*

*Proof.* The functions $F_u,F_\ell$ are real-analytic in $T>0$; the centred form is the mean value theorem $F(T)=F(T_{\mathrm m})+F'(T_*)(T-T_{\mathrm m})$ for some $T_*$ between $T$ and $T_{\mathrm m}$. The derivative is computed from $\frac{d}{dT}\arg s=\operatorname{Re}\frac1s$, $\frac{d}{dT}\operatorname{Im}\operatorname{Log}\Gamma(\frac{s}{2})=\frac12\operatorname{Re}\psi_\Gamma(\frac{s}{2})$, $\frac{d}{dT}\mathcal{M}(T)=\frac{1}{2\pi}\log\frac{T}{2\pi}$, $\frac{d}{dT}D_j(s)=iD_j'(s)$ with $D_1'(s)=-s^{-2}-(s-1)^{-2}+\frac14\psi_\Gamma'(\frac{s}{2})$ and $D_j'(s)=-js^{-j-1}-j(s-1)^{-j-1}+\frac{(-1)^{j-1}}{(j-1)!\,2^{j+1}}\psi_\Gamma^{(j)}(\frac{s}{2})$ for $j\ge2$, and, in the $S$-mode, $\frac{d}{dT}\vartheta(T)=\frac12\operatorname{Re}\psi_\Gamma(\frac14+\frac{iT}{2})-\frac12\log\pi$. The polygamma values on an interval are enclosed by $\psi_\Gamma^{(k)}(z_{\mathrm m})+\psi_\Gamma^{(k+1)}(Z)\cdot i[-\frac{r}{2},\frac{r}{2}]$, where $Z$ is the ball of the segment and $r$ its length in $T$, which is valid because $\psi_\Gamma^{(k)}(z+\eta)-\psi_\Gamma^{(k)}(z)=\int_0^1\psi_\Gamma^{(k+1)}(z+u\eta)\eta\,du$ lies in $\eta$ times the convex hull of the enclosure. All other operations are ball operations, so each interval bound is rigorous, and the supremum over $[1,T_0]$ is the maximum of the 9215 upper bounds. For $T\ge T_0$ the bound (4.14) is monotone in $T$, so it is evaluated at $T_0$. □

4.4. **The kernel inequality.** Condition (K) is an inequality between explicit elementary functions of two real variables on a half-strip. It is certified by bisection, as follows. For $t\ne0$, by $\arctan y+\arctan\frac1y=\operatorname{sgn}(y)\frac{\pi}{2}$ and since $d\pm b>0$ and $2d\pm b>0$ for $0\le b\le\frac12$,

$$f(b,t)=\operatorname{sgn}(t)\,\pi-2\arctan\frac{t}{d+b}-2\arctan\frac{t}{d-b}+\arctan\frac{t}{2d+b}+\arctan\frac{t}{2d-b},\tag{4.17}$$

so on each closed half-box $[0,\frac12]\times[0,T_1]$ (with $\operatorname{sgn}=+1$) and $[0,\frac12]\times[-T_1,0]$ (with $\operatorname{sgn}=-1$) the right side of (4.17) is real-analytic, including at $t=0$, and so is $U$, because $d\pm b\ge d-\frac12>0$.

**Lemma 4.10** (Tail of the kernel inequality)**.** *Let $T_1\ge2d+\frac12$, $0\le b\le\frac12$, $|t|\ge T_1$ and $u=1/t$. Then*

$$t^2f(b,t)\le4d^3u+\mathsf{K}_5|u|^3,\qquad \mathsf{K}_5=\frac15\Bigl(2\bigl(d+\tfrac12\bigr)^5+2d^5+\bigl(2d+\tfrac12\bigr)^5+(2d)^5\Bigr),\tag{4.18}$$

*and*

$$t^2U(b,t)=c_{\mathrm R,1}\Bigl(\frac{d+b}{1+(d+b)^2u^2}+\frac{d-b}{1+(d-b)^2u^2}\Bigr)+\sum_{\substack{\iota\in\mathcal{B}\\ j\ge2}}c_\iota\sum_{x=d\pm b}\Phi_{\mathrm X}\Bigl(u^{j-2}(xu+i)^{-j}\Bigr),\tag{4.19}$$

*a real-analytic function of $(b,u)$ on $[0,\frac12]\times[-1/T_1,1/T_1]$ with value $2dc_{\mathrm R,1}-2c_{\mathrm R,2}$ at $u=0$.*

*Proof.* Write $f=\sum_{i=1}^4\alpha_i\arctan(y_iu)$ with $(\alpha_i)=(2,2,-1,-1)$ and $(y_i)=(d+b,d-b,2d+b,2d-b)$; then $|y_iu|\le(2d+\frac12)/T_1\le1$. For $|y|\le1$, $\arctan y=y-\frac{y^3}{3}+r(y)$ with $|r(y)|\le\frac{|y|^5}{5}$ (alternating series). Since $\sum\alpha_iy_i=0$ and $\sum\alpha_iy_i^3=2(d+b)^3+2(d-b)^3-(2d+b)^3-(2d-b)^3=-12d^3$,

$$f=4d^3u^3+\sum_i\alpha_ir(y_iu),\qquad\Big|\sum_i\alpha_ir(y_iu)\Big|\le\frac{|u|^5}{5}\sum_i|\alpha_i||y_i|^5\le\mathsf{K}_5|u|^5,$$

using $|y_1|\le d+\frac12$, $|y_2|\le d$, $|y_3|\le2d+\frac12$, $|y_4|\le2d$. Multiplying by $t^2=u^{-2}$ gives (4.18). For (4.19): $t^2\operatorname{Re}(x+it)^{-1}=t^2\frac{x}{x^2+t^2}=\frac{x}{1+x^2u^2}$, and for $j\ge2$, $x+it=(xu+i)/u$, so $t^2(x+it)^{-j}=u^{j-2}(xu+i)^{-j}$. At $u=0$ the terms with $j=1$ give $2dc_{\mathrm{R},1}$, the terms with $j=2$ give $c_{\mathrm{R},2}\cdot2\operatorname{Re}(i^{-2})+c_{\mathrm{I},2}\cdot2(-\operatorname{Im}i^{-2})=-2c_{\mathrm{R},2}$, and the terms with $j\ge3$ vanish. □

**Proposition 4.11** (Certification of (K))**.** *The program* `certify_hybrid.py` *(Appendix B), run with an empty list of discs, certifies* (K) *for a kernel as follows. (a) On* $[0,\frac12]\times[0,T_1]$ *and on* $[0,\frac12]\times[-T_1,0]$ *it evaluates* $H=U-f$*, with* $f$ *in the form* (4.17)*, on boxes by the centred form* $H(\text{centre})+\nabla H(\text{box})\cdot(\text{half-widths})$*, and bisects every box on which the ball of* $H$ *does not lie in* $(0,\infty)$*; it stops when all boxes are certified. (b) For* $|t|\ge T_1$ *it certifies* $t^2U-4d^3u-\mathsf{K}_5|u|^3>0$ *on* $[0,\frac12]\times(0,1/T_1]$ *and on* $[0,\frac12]\times[-1/T_1,0)$ *in the same way, with the expressions of Lemma 4.10; by* (4.18) *this implies* $U>f$ *there. The driver tries* $T_1=100$ *first and* $T_1=2000$ *if the first run fails; both satisfy* $T_1\ge2d+\frac12$*. For each of the six kernels of Theorem 2.1 the run terminated with every box certified (Appendix C).*

*Proof.* The centred form is the two-variable mean value theorem for the real-analytic function $H$ on a convex box, with the gradient enclosed by balls; a positive lower bound of the enclosure on a box proves $U>f$ on the box. The union of the certified boxes is the whole half-strip $0\le b\le\frac12$, $0<|t|\le T_1$ (the boxes are closed and include $t=0$, where (4.17) is the continuous extension), together with $|t|\ge T_1$ by (b). □

| line | $d$ | $m$ | $A_1$ | $C_\zeta<$ | $\sup\max(F_u,F_\ell)\le$ | $A_3^N$ | $\sup\max(F_u^S,F_\ell^S)\le$ | $A_3^S$ |
|---|---|---|---|---|---|---|---|---|
| $\ell_5$ | 1 | 5 | 0.20176 | 0.70360002 | 0.66536440 | 1.369 | 0.13949090 | 0.844 |
| $\ell_{20}$ | 1.2 | 5 | 0.10823 | 1.46807534 | 0.70207071 | 2.171 | −0.18278649 | 1.286 |
| $\ell_{60}$ | 1 | 5 | 0.09473 | 2.10426949 | 0.72894156 | 2.834 | −0.15847575 | 1.946 |
| $\ell_{150}$ | 0.85 | 5 | 0.09349 | 2.25578346 | 0.73031685 | 2.987 | −0.15757634 | 2.099 |
| $\ell_{400}$ | 0.85 | 10 | 0.08399 | 4.46374163 | 0.73756551 | 5.202 | −0.14582673 | 4.318 |
| $\ell_{6000}$ | 0.85 | 10 | 0.08382 | 4.58951035 | 0.73573753 | 5.326 | −0.14641036 | 4.444 |

TABLE 2. The six kernels of Theorem 2.1: parameters, certified upper bounds for $C_\zeta$ and for the suprema of the elementary part, and the resulting constants $A_3^N=\lceil C_\zeta+\sup\max(F_u,F_\ell)\rceil_3$, $A_3^S=\lceil C_\zeta+\sup\max(F_u^S,F_\ell^S)\rceil_3$ and $A_1=\lceil c_{\mathrm{R},1}/(2\pi)\rceil_5$, where $\lceil x\rceil_r=10^{-r}\lceil10^rx\rceil$ denotes rounding up to $r$ decimals. The coefficients $c_\iota$ are in Appendix A.

## 4.5. **Proof of Theorem 2.1 and of Corollary 2.2.**

*Proof of Theorem 2.1.* Each line $A_1L+A_3$ of (2.1) corresponds to a kernel listed in Table 2 and Appendix A, with $A_1=\lceil C_1\rceil_5\ge C_1=c_{\mathrm{R},1}/(2\pi)$. For it, (K) holds by Proposition 4.11; $\sum_pM_p\le C_\zeta$ by Proposition 4.5; and $\sup_{T\ge1}\max(F_u,F_\ell)$ is bounded by Proposition 4.9, with $A_1\ge C_1$ as required in Lemma 4.7. By (4.9), for every $T\ge1$ that is not an ordinate, $|N(T)-\mathcal{M}(T)|\le A_1\log T+C_\zeta+\sup\max(F_u,F_\ell)\le A_1\log T+A_3^N$. At an ordinate $T$, $N(T)=\lim_{t\downarrow T}N(t)$ (convention (1.2)), the limit being taken through non-ordinates, and $\mathcal{M}$ is continuous, so the inequality persists. The same argument with (4.16) gives (2.2), with $S(T)=S(T^+)$ at the ordinates. □

*Proof of Corollary 2.2.* The difference between the minimum of finitely many lines and the minimum of finitely many other lines is a piecewise linear function of $L \geq 0$, whose minimum over $[0, \infty)$ is attained at $L = 0$, at a breakpoint of one of the two envelopes, or, if the slopes allow, approached as $L \to \infty$. Here the smallest slope of the bounds of [5] and of [1] is 0.0946 and 0.097 respectively, larger than 0.08382, so the differences tend to $+\infty$. Evaluating at $L = 0$ and at all breakpoints of both envelopes, in exact rational arithmetic, gives the minimal differences 2.231 (attained at a breakpoint $L \approx 49.11$) and 0.291 (at a breakpoint $L \approx 8.57$), both positive; the gaps are truncated downwards and the positions rounded to two decimals. The statements about single lines are the comparisons of two lines each. When both the slope and the constant of the new line are smaller, the inequality holds for all $L \geq 0$: this is the case for $0.08399L + 5.202$ and for $0.08382L + 5.326$ against $0.0946L + 5.746$. When the new line has the smaller slope but the larger constant, the inequality holds exactly for $L$ larger than the quotient of the difference of the constants by the difference of the slopes: $(5.202 - 4.954)/(0.097 - 0.08399)$ and $(5.326 - 4.954)/(0.097 - 0.08382)$ against $0.097L + 4.954$. □

## 5. Jensen discs and admissible profiles

**Definition 5.1** (Admissible profile)**.** An *admissible profile* is a continuous function $\mathfrak{e}\colon [\frac{1}{2}, 1] \to [0, \infty)$ with $\mathfrak{e}(1) = 0$ such that, for every $\varepsilon > 0$,

$$|\zeta(\sigma + it)| \leq K_\varepsilon |t|^{\mathfrak{e}(\sigma)+\varepsilon} \qquad \text{for } \tfrac{1}{2} \leq \sigma \leq 1,\ |t| \geq 2, \tag{5.1}$$

with a constant $K_\varepsilon$ independent of $\sigma$ and $t$. It is extended by $\mathfrak{e}(\sigma) = 0$ for $\sigma > 1$.

For $\sigma \in \mathbb{R}$ let $\mu(\sigma)$ be the infimum of the $\xi \geq 0$ such that $\zeta(\sigma + it) \ll |t|^\xi$ as $|t| \to \infty$ [25, §5.1]; $\mu$ is convex on $\mathbb{R}$, hence continuous (in particular on $[\frac{1}{2}, 1]$), and $\mu(\sigma) = 0$ for $\sigma \geq 1$. By definition, for every $\varepsilon > 0$ and each fixed $\sigma$, $\zeta(\sigma + it) \ll_{\sigma,\varepsilon} |t|^{\mu(\sigma)+\varepsilon}$ for $|t| \geq 2$.

**Lemma 5.2** (Interpolation of pointwise bounds)**.** *Let $\frac{1}{2} = \varsigma_0 < \varsigma_1 < \cdots < \varsigma_n = 1$ and $e_0 \geq e_1 \geq \cdots \geq e_n = 0$ be such that, for each $i$ and each $\varepsilon > 0$, $|\zeta(\varsigma_i + it)| \leq K_{i,\varepsilon}|t|^{e_i+\varepsilon}$ for $|t| \geq 2$. Then the piecewise linear function $\mathfrak{e}$ with $\mathfrak{e}(\varsigma_i) = e_i$ is an admissible profile.*

*Proof.* Fix $\varepsilon > 0$ and $i < n$, and put $\mathcal{Z}(s) = (s-1)\zeta(s)$, $Q = 10$. On the strip $\varsigma_i \leq \operatorname{Re} s \leq \varsigma_{i+1}$, $\mathcal{Z}$ is holomorphic and of polynomial growth: by [25, (2.1.4)], $\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \{x\} x^{-s-1}\,dx$ for $\operatorname{Re} s > 0$, so $|\mathcal{Z}(s)| \leq |s| + |s||s-1|/\operatorname{Re} s \leq 3|s|(1 + |s|)$ there. On the line $\operatorname{Re} s = \varsigma_i$: for $|t| \geq 2$, $|s - 1| \leq |Q + s|$ (as $Q + \varsigma_i \geq |\varsigma_i - 1|$) and $|t| \leq |Q + s|$, so $|\mathcal{Z}(s)| \leq K_{i,\varepsilon}|Q + s|^{1+e_i+\varepsilon}$; for $|t| \leq 2$, $\mathcal{Z}$ is bounded on the compact segment and $|Q + s| \geq 10$, so the same bound holds with a larger constant $A_i$. Likewise $|\mathcal{Z}(s)| \leq B_i|Q+s|^{1+e_{i+1}+\varepsilon}$ on $\operatorname{Re} s = \varsigma_{i+1}$. Since $1+e_i+\varepsilon \geq 1+e_{i+1}+\varepsilon \geq 0$, the theorem of Phragmén–Lindelöf type of Rademacher [19] (also [10, Cor. 3] with $\alpha_2 = 0$) gives, on the strip,

$$|\mathcal{Z}(s)| \leq \big(A_i|Q+s|^{1+e_i+\varepsilon}\big)^{\frac{\varsigma_{i+1}-\sigma}{\varsigma_{i+1}-\varsigma_i}} \big(B_i|Q+s|^{1+e_{i+1}+\varepsilon}\big)^{\frac{\sigma-\varsigma_i}{\varsigma_{i+1}-\varsigma_i}} \leq \max(A_i, B_i)\,|Q+s|^{1+\mathfrak{e}(\sigma)+\varepsilon},$$

because the exponent interpolates linearly. For $|t| \geq 2$, $|Q + s|^2 = (Q + \sigma)^2 + t^2 \leq 121 + t^2 \leq 36t^2$ and $|s - 1| \geq |t|$, so $|\zeta(s)| \leq \max(A_i, B_i)6^{1+e_0+\varepsilon}|t|^{\mathfrak{e}(\sigma)+\varepsilon}$ on the strip. Taking the maximum over the $n$ strips gives (5.1); $\mathfrak{e}$ is continuous, non-negative, and $\mathfrak{e}(1) = 0$. □

An *exponent pair* $(k, \ell)$ in the sense of [25, §5.20] satisfies $0 \leq k \leq \frac{1}{2} \leq \ell \leq 1$, and

$$\mu(\ell - k) \leq k \tag{5.2}$$

[23, Lemma 23], [25, (5.20.1)].

**Lemma 5.3** (The profile $\mathfrak{e}_{\mathrm{P}}$)**.** *Let $\mathcal{N}$ be the set of the following points $(\varsigma, e)$ with $\frac{1}{2} \leq \varsigma \leq 1$, for each of which $\mu(\varsigma) \leq e$ is a theorem:*

*(i)* $(\frac{1}{2}, \frac{13}{84})$ [6, Thm. 5]*;*

*(ii) the breakpoints of the piecewise linear bound for $\mu$ of Trudgian and Yang* [28, Thm. 2.4]*, that is, $(\varsigma, \mu^{\mathrm{TY}}(\varsigma))$ for $\varsigma \in \{\frac{88225}{153852}, \frac{521}{796}, \frac{53141}{76066}, \frac{3620}{5119}, \frac{52209}{69128}, \frac{1389}{1736}, \frac{134765}{163248}, \frac{18193}{21906}, \frac{249}{280}, \frac{277}{300}\}$, where $\mu^{\mathrm{TY}}$ is the bound of that theorem (its ten pieces agree at the common endpoints, and the last piece is stated there up to $\frac{277}{300}$);*

*(iii) $(\ell - k, k)$ for the four exponent pairs $(k, \ell)$ of Tao, Trudgian and Yang* [23, Thm. 20]*: $(\frac{89}{1282}, \frac{997}{1282})$, $(\frac{652397}{9713986}, \frac{7599781}{9713986})$, $(\frac{10769}{351096}, \frac{609317}{702192})$, $(\frac{89}{3478}, \frac{15327}{17390})$, by* (5.2) *(both references write exponent pairs in the order $(k, \ell)$ with $k \le \frac{1}{2} \le \ell$, as here);*

*(iv) $(\varsigma_n, \mu_n)$ with $\varsigma_n = 1 - \frac{3n^2-3n+2}{n(n-1)^2(n+2)}$ and $\mu_n = \frac{2}{(n-1)^2(n+2)}$ for $6 \le n \le 399$: by Heath-Brown* [14, Thm. 2]*, $(\frac{2}{(n-1)^2(n+2)}, 1 - \frac{3n-2}{n(n-1)(n+2)} + \varepsilon')$ is an exponent pair for every $n \ge 3$ and $\varepsilon' > 0$, so* (5.2) *gives $\mu(\varsigma_n + \varepsilon') \le \mu_n$ for every $\varepsilon' > 0$, and $\mu(\varsigma_n) \le \mu_n$ by the continuity of $\mu$ (the points with $n = 3, 4$ lie above the hull of the others and the point with $n = 5$ on a chord, so they would add no vertex; the points with $n > 399$ would lower the hull by less than $5 \cdot 10^{-9}$, on an interval of length below $2 \cdot 10^{-5}$ next to $\varsigma = 1$, and are not used);*

*(v) $(1, 0)$, since $\zeta(1 + it) \ll \log|t|$* [25, Thm. 3.5].

*Let $\mathfrak{e}_{\mathrm{P}}$ be the piecewise linear function through the vertices of the lower convex hull of $\mathcal{N}$ (the hull only selects the points; each chord is then validated by Lemma 5.2, not by the convexity of $\mu$) (Table 5 in Appendix A lists them, except the vertices of (iv) with $10 \le n \le 398$, which are given by the formula in (iv) and generated by `profile_published.py`; the four points of (ii) with $\varsigma \in \{\frac{3620}{5119}, \frac{134765}{163248}, \frac{18193}{21906}, \frac{249}{280}\}$ lie on or above chords of other points and are not vertices). Then $\mathfrak{e}_{\mathrm{P}}$ is an admissible profile.*

*Proof.* At each vertex $(\varsigma_i, e_i)$ of the hull, $\mu(\varsigma_i) \le e_i$, so $|\zeta(\varsigma_i + it)| \le K_{i,\varepsilon}|t|^{e_i+\varepsilon}$ for $|t| \ge 2$ and every $\varepsilon > 0$, by the definition of $\mu$. The vertices have $\varsigma_0 = \frac{1}{2}$, $\varsigma_{\mathrm{last}} = 1$, and non-increasing ordinates (the values in Table 5 decrease, and the last is 0). Lemma 5.2 applies. □

The points of $\mathcal{N}$ that are not vertices are not needed: the chord between two vertices is a valid bound by Lemma 5.2, and it is smaller. No use is made of the convexity of $\mu$ beyond its continuity in (iv).

A *disc datum* is a triple $(c, R, v)$ of real numbers with $c > 1$, $R > 0$ and $c - R \ge \frac{1}{2}$. Its weight is

$$W(b, t) = \log^+ \frac{R}{\sqrt{(c - \frac{1}{2} - |b|)^2 + (t + v)^2}} \qquad (|b| < \tfrac{1}{2},\ t \in \mathbb{R}), \tag{5.3}$$

and, for an admissible profile $\mathfrak{e}$,

$$J(\mathfrak{e}; c, R) = \frac{1}{2\pi} \int_0^{2\pi} \mathfrak{e}(c + R\cos\theta)\, d\theta = \frac{1}{\pi} \int_0^{\pi} \mathfrak{e}(c + R\cos\theta)\, d\theta. \tag{5.4}$$

Since $\mathfrak{e} = 0$ on $[1, \infty)$ and $c - R \ge \frac{1}{2}$, only the values of $\mathfrak{e}$ on $[c - R, 1] \subset [\frac{1}{2}, 1]$ enter.

**Lemma 5.4** (Jensen bound)**.** *Let $\mathfrak{e}$ be an admissible profile, $(c, R, v)$ a disc datum and $\varepsilon > 0$. There is $T_\varepsilon$ such that for all $T \ge T_\varepsilon$,*

$$\sum_{\rho} W(b_\rho, t_\rho) \le 2\big(J(\mathfrak{e}; c, R) + \varepsilon\big) \log T, \tag{5.5}$$

*and the same bound holds for $\sum_\rho W(b_\rho, -t_\rho)$.*

*Proof.* Let $s_0 = c + i(T + v)$ and, for a zero $\rho$, let $\rho^*$ be the one of $\rho$, $1 - \bar\rho$ with real part $\ge \frac{1}{2}$ (their common value when $\beta = \frac{1}{2}$), so $\rho^* = \frac{1}{2} + |b_\rho| + i\gamma$. Then $|\rho^* - s_0|^2 = (c - \frac{1}{2} - |b_\rho|)^2 + (t_\rho + v)^2$ and $W(b_\rho, t_\rho) = \log^+(R/|\rho^* - s_0|)$. The point $\rho^*$ is a zero of $\zeta$, and each zero $w$ with $\operatorname{Re} w > \frac{1}{2}$ is $\rho^*$ for exactly two zeros ($w$ and $1 - \bar w$), which have the same multiplicity, while a zero $w$ with $\operatorname{Re} w = \frac{1}{2}$

is $\rho^*$ only for $\rho = w$. Since the open disc $\mathbb{D} = \{|s - s_0| < R\}$ lies in $\operatorname{Re} s > c - R \geq \frac{1}{2}$,

$$\sum_{\rho} W(b_\rho, t_\rho) \leq 2 \sum_{w \in \mathbb{D}} \log \frac{R}{|w - s_0|}, \tag{5.6}$$

the sum over the zeros of $\zeta$ in $\mathbb{D}$ with multiplicity. For $T > R + |v| + 1$ the closed disc $\bar{\mathbb{D}}$ does not contain $s = 1$, so $\zeta$ is holomorphic on it, and $\zeta(s_0) \neq 0$ because $\operatorname{Re} s_0 = c > 1$. Jensen's formula [24, §3.61], [20, Thm. 15.18] (which allows zeros on the circle) gives

$$\sum_{w \in \mathbb{D}} \log \frac{R}{|w - s_0|} = \frac{1}{2\pi} \int_0^{2\pi} \log |\zeta(s_0 + Re^{i\theta})| \, d\theta - \log |\zeta(s_0)|. \tag{5.7}$$

By the Euler product, $|\zeta(s_0)| \geq \prod_p (1 + p^{-c})^{-1} = \zeta(2c)/\zeta(c) > 0$, a bound independent of $T$ and $v$. On the circle, $s = \sigma + it$ with $\sigma = c + R\cos\theta \geq \frac{1}{2}$ and $t = T + v + R\sin\theta \in [T - R - |v|, T + R + |v|]$, so $\log t = \log T + O(1/T)$ and $t \geq 2$ for large $T$. Where $\sigma \leq 1$, (5.1) gives $\log |\zeta(s)| \leq (\mathfrak{e}(\sigma) + \varepsilon/2) \log t + \log K_{\varepsilon/2}$. Where $1 \leq \sigma \leq 2$, $|\zeta(s)| \leq K \log t$ [25, Thm. 3.5], and where $\sigma \geq 2$, $|\zeta(s)| \leq \zeta(2)$; in both cases $\log |\zeta(s)| \leq \log\log t + O(1) = o(\log T)$. Inserting in (5.7) and (5.6),

$$\sum_{\rho} W(b_\rho, t_\rho) \leq 2\big(J(\mathfrak{e}; c, R) + \tfrac{\varepsilon}{2}\big) \log T + O_\varepsilon(\log\log T) + 2 \log \frac{\zeta(c)}{\zeta(2c)},$$

and the last two terms are at most $\varepsilon \log T$ for $T$ large. Replacing $t_\rho$ by $-t_\rho$ in $W$ amounts to replacing $v$ by $-v$, which changes neither $J$ nor the argument. □

## 6. The asymptotic estimate

**Lemma 6.1** (Main term, asymptotic form)**.** *For fixed $d > \frac{1}{2}$, as $T \to \infty$, $\frac{1}{\pi}\big(2 \arg \xi(s_1) - \arg \xi(s_2)\big) = \frac{1}{\pi}\vartheta(T) + 1 + O(1)$.*

*Proof.* By (4.5), $\arg \xi(s) = \mathcal{A}(s) + O(1)$ for $\sigma \in \{\sigma_1, \sigma_2\}$, since $|\sum_{n \geq 2} \frac{\Lambda(n)}{\log n} n^{-\sigma} \sin(T \log n)| \leq \log \zeta(\sigma)$. The proof of Lemma 4.7 gives $\frac{1}{\pi}(2\mathcal{A}(s_1) - \mathcal{A}(s_2)) = \mathcal{M}(T) + \frac{7}{8} + O(1/T)$, and Lemma 4.8 gives $\mathcal{M}(T) + \frac{7}{8} = \frac{1}{\pi}\vartheta(T) + 1 + O(1/T)$. □

**Lemma 6.2** (Sums of the kernel functions, asymptotic form)**.** *For fixed $d > \frac{1}{2}$ and a kernel $U$, as $T \to \infty$,*

$$\sum_{\rho} U(b_\rho, t_\rho) = c_{\mathrm{R},1} \log T + O(1), \qquad \sum_{\rho} U(b_\rho, -t_\rho) = c_{\mathrm{R},1} \log T + O(1).$$

*Proof.* By Lemma 3.2, $\sum_\rho G_\iota(b_\rho, t_\rho) = 2\Phi_{\mathrm{X}}(D_j(s_1)) + O(1)$, the arithmetic sum being bounded by $\frac{2}{(j-1)!} \sum_{n \geq 2} \Lambda(n) (\log n)^{j-1} n^{-\sigma_1} < \infty$. By the proof of Lemma 4.7, $\operatorname{Re} D_1(s_1) = \frac{1}{2} \log \frac{T}{2\pi} + O(T^{-2})$, $\operatorname{Im} D_1(s_1) = O(1)$ and $D_j(s_1) = O(T^{-(j-1)}) = o(1)$ for $j \geq 2$, because $|\psi_\Gamma^{(j-1)}(z_1)| = O(T^{-(j-1)})$ by (4.13) and (4.12); for $j = 2$ this is only $O(T^{-1})$, which suffices. Hence $\sum_\rho U(b_\rho, t_\rho) = c_{\mathrm{R},1} \log T + O(1)$. The second statement follows from $U(b, -t) = \sum_\iota \varepsilon_{\mathrm{X}} c_\iota G_\iota(b, t)$ and the same bounds. □

Fix $d = \frac{9}{10}$, $m = 10$, the 19 coefficients $c_\iota$ ($\iota \in \mathcal{B}_{10}$) of Table 3, and the $K = 90$ disc data $(c_k, R_k, v_k)$ with weights $\kappa_k \geq 0$ of Table 4 (Appendix A). The $c_\iota$ and $\kappa_k$ are the binary floating-point numbers written as decimal strings in the file `hyb_m10_P.json` (Appendix D; the strings round-trip to the binary coefficients, which the certifier converts to exact rationals); the disc data are the rational numbers of the table, obtained from the file by rounding to denominators $\leq 10^4$ and replacing $R_k$ by $\min(R_k, c_k - \frac{1}{2})$, so that $c_k - R_k \geq \frac{1}{2}$ exactly. Let $W_k$ be the weight (5.3) of $(c_k, R_k, v_k)$ and put

$$U = \sum_{\iota \in \mathcal{B}_{10}} c_\iota G_\iota, \qquad C_{\mathrm{H}} = \frac{c_{\mathrm{R},1}}{2\pi} + \frac{1}{\pi} \sum_{k=1}^{K} \kappa_k J(\mathfrak{e}_{\mathrm{P}}; c_k, R_k). \tag{6.1}$$

**Proposition 6.3** (Certified hybrid inequality)**.** *For $0 \le b \le \frac{1}{2}$ and $t \ne 0$,*

$$f(b,t) \le U(b,t) + \sum_{k=1}^{K} \kappa_k W_k(b,t). \tag{6.2}$$

*Moreover*

$$C_{\mathrm{H}} \in [0.074859614271090718755,\ 0.074859614271090718756]$$

*(the endpoints of the computed ball, rounded outwards to twenty digits, as printed by the certificate), so $C_{\mathrm{H}} < 0.0748596142711$; its two parts are $c_{\mathrm{R},1}/(2\pi) = 0.0655821827845\ldots$ and $\frac{1}{\pi}\sum_k \kappa_k J_k = 0.00927743148654\ldots$.*

*Proof.* The program `certify_hybrid.py` with the file `hyb_m10_P.json` and $T_1 = 40$ (Appendix C) proceeds as in Proposition 4.11, with these changes. (a) On the two half-boxes $[0,\frac{1}{2}]\times[0,T_1]$ and $[0,\frac{1}{2}]\times[-T_1,0]$ it bounds $H = U - f + \sum_k \kappa_k W_k^{\mathrm{low}}$ from below, where $U - f$ is enclosed by the centred form and $W_k^{\mathrm{low}}$ is the lower endpoint of the ball of $\log \frac{R_k}{\sqrt{(c_k-\frac{1}{2}-b)^2+(t+v_k)^2}}$ over the box when that endpoint is positive, and 0 otherwise; since $W_k = \max(0,x_k)$ with $x_k$ that logarithm, $\max(0,x_k) \ge \max(0,\inf_{\mathrm{box}} x_k) \ge W_k^{\mathrm{low}}$ on the box, so with $\kappa_k \ge 0$ a positive lower bound for $H$ proves (6.2) on the box. It terminated with all boxes certified (2 028 149 boxes for $t>0$, 2 294 869 for $t<0$). (b) For $|t| \ge T_1 = 40$ every $W_k$ vanishes, because $W_k(b,t) > 0$ requires $|t+v_k| < R_k$, hence $|t| < R_k + |v_k| \le \max_k(R_k + |v_k|) = \frac{11}{5} = 2.2 < T_1$ (the maximum is computed in exact rational arithmetic from the disc data of Table 4, and the program asserts $R_k + |v_k| \le T_1$ for every $k$ before certifying), and $T_1 \ge 2d + \frac{1}{2} = 2.3$; the tail is certified as in Proposition 4.11(b) (277 098 boxes). (c) The profile $\mathfrak{e}_{\mathrm{P}}$ is piecewise linear with rational vertices and $c_k - R_k \ge \frac{1}{2}$, so by (5.4) each $J(\mathfrak{e}_{\mathrm{P}}; c_k, R_k)$ is a finite sum, over the pieces $[\varsigma_i, \varsigma_{i+1}] \cap [c_k - R_k, 1] = [p,q]$ with $\mathfrak{e}_{\mathrm{P}} = \alpha + \beta\sigma$ there, of

$$\frac{1}{\pi}\Big[(\alpha+\beta c_k)(\theta_p - \theta_q) + \beta R_k(\sin\theta_p - \sin\theta_q)\Big], \qquad \theta_\sigma = \arccos\frac{\sigma - c_k}{R_k} \quad (\theta_p \ge \theta_q),$$

since $\sigma = c_k + R_k\cos\theta$ and $d\sigma = -R_k\sin\theta\,d\theta$; the arguments of arccos lie in $[-1,1]$ because $p,q \in [c_k - R_k, c_k + R_k]$. The program evaluates these sums in ball arithmetic and reports the ball of $C_{\mathrm{H}}$ stated. □

*Proof of Theorem 2.3.* Let $\varepsilon > 0$ and let $T \to \infty$ through values that are not ordinates. All functions in (6.2) are even in $b$, and $|b_\rho| < \frac{1}{2}$, so (6.2) holds at $(b_\rho, t_\rho)$ and at $(b_\rho, -t_\rho)$ for every zero; with $f(b,-t) = -f(b,t)$,

$$-U(b_\rho,-t_\rho) - \sum_k \kappa_k W_k(b_\rho,-t_\rho) \le f(b_\rho,t_\rho) \le U(b_\rho,t_\rho) + \sum_k \kappa_k W_k(b_\rho,t_\rho).$$

Summing over $\rho$ (all series converge absolutely: those of $U$ by Lemma 3.2, those of $W_k$ because only finitely many zeros lie in each disc) and using Lemmas 6.2 and 5.4 with the admissible profile $\mathfrak{e}_{\mathrm{P}}$ of Lemma 5.3,

$$\Big|\frac{1}{2\pi}\sum_\rho f(b_\rho,t_\rho)\Big| \le \Big(\frac{c_{\mathrm{R},1}}{2\pi} + \frac{1}{\pi}\sum_k \kappa_k\big(J(\mathfrak{e}_{\mathrm{P}}; c_k, R_k) + \varepsilon\big)\Big)\log T + O(1)$$

for $T \ge \max_k T_\varepsilon(k)$. By Lemma 3.1 and Lemma 6.1, $S(T) = \frac{1}{2\pi}\sum_\rho f(b_\rho,t_\rho) + O(1)$, so $|S(T)| \le (C_{\mathrm{H}} + \varepsilon\sum_k \kappa_k/\pi)\log T + O(1)$. Dividing by $\log T$ and letting $T \to \infty$ through non-ordinates gives $\limsup |S(T)|/\log T \le C_{\mathrm{H}} + \varepsilon\sum_k \kappa_k/\pi$ over non-ordinates; this limit superior equals $\mathfrak{a}$, because at an ordinate $\gamma$ the value $S(\gamma) = S(\gamma^+)$ (convention (1.2)) is the limit of $S(T_n)$ along non-ordinates $T_n \downarrow \gamma$, and $\log T_n \to \log\gamma$. Letting $\varepsilon \to 0$ gives $\mathfrak{a} \le C_{\mathrm{H}}$, and Proposition 6.3 gives (2.5). Finally $C_{\mathrm{H}} < 0.0748596142711 < 0.07486 < 0.07862 < \Lambda(0) = 0.0786252\ldots$ by (1.7). □

## 7. Methods and remarks

7.1. **Where the kernel inequality binds.** For fixed coefficients, $U - f$ is harmonic in $(b,t)$ on each of the half-strips $|b| < d$, $t > 0$ and $|b| < d$, $t < 0$: each $\arctan\frac{y}{t}$ is an argument of $t + iy$, hence harmonic in $(y,t)$ and in $(b,t)$, and each $F_\iota(d \pm b, t)$ is the real or imaginary part of a holomorphic function of $(d \pm b) + it$. By the minimum principle on the rectangles $[-\frac12,\frac12] \times [\varepsilon, R]$ and $[-\frac12,\frac12] \times [-R, -\varepsilon]$, followed by $\varepsilon \to 0$ and $R \to \infty$, $f \le U$ on $|b| \le \frac12$, $t \neq 0$ holds as soon as $U - f \ge 0$ on the lines $b = \pm\frac12$ and $\liminf(U - f) \ge 0$ as $t \to 0^{\pm}$ and as $|t| \to \infty$, uniformly in $b$. The last condition is automatic, because $U - f \to 0$ uniformly in $b$ as $|t| \to \infty$; more precisely $t^2(U - f) \to 2dc_{\mathrm{R},1} - 2c_{\mathrm{R},2}$ (Lemma 4.10), so the condition on the lines $b = \pm\frac12$ forces this constant to be non-negative, and for every kernel certified here it is positive, since the boxes of part (b) of the certification contain $u = 0$. (This paragraph is a structural remark on where the constraints bind; the certification of Proposition 4.11 bisects the whole box and does not rely on it, and the passage from the truncated rectangles to the half-strips, which requires the uniform limits just mentioned, is not needed for any result of this article.) The validity of the kernel inequality is thus controlled by its boundary $b = \pm\frac12$, that is, by hypothetical zeros with $\beta$ arbitrarily close to 0 or 1; in the optimised kernels considered here, the active and nearly active constraints of the cutting-plane iteration indeed lie near $b = \pm\frac12$. One interpretation is that the values of $\zeta$ on $\operatorname{Re} s > 1$, which are all the method uses, cannot exclude clusters of order $\log T$ of such hypothetical zeros near height $T$; the numerical optima of Remark 2.5 are consistent with this. The weights $W_k$ relax precisely these boundary constraints: in Table 4 every disc has $c_k - R_k \ge \frac12$, and only the values of $\mathfrak{e}_{\mathrm{P}}$ on $[\frac12, 1]$ enter $C_{\mathrm{H}}$.

7.2. **Optimisation.** For fixed $d$ the set of admissible $(c_\iota)$ in (K), and of admissible $(c_\iota, \kappa_k)$ in (6.2), is convex, being defined by linear inequalities indexed by $(b,t)$. For Theorem 2.1, the quantity minimised at a target height $L^*$ is $A_1 L^* + C_\zeta + \sup F$, where $C_\zeta$ is approximated by $\sum_{n\ge2} \Lambda(n) n^{-\sigma_1} h(\log n)$ over the prime powers $n \le 2 \cdot 10^4$ plus a quadrature of the tail (for $\ell_{6000}$, with the maximum over a grid of 96 phases in place of $h$ for the primes $p \le 300$), a convex function of the coefficients (a sum of Euclidean norms and of pointwise maxima of linear functions), and $\sup F$ by its maximum on a grid of 260 heights in $[1, 10^6]$ and its limit at $T = \infty$. The problem is a second-order cone programme, solved with `cvxpy` and Clarabel; the constraint (K) is imposed on a grid of $26 \times 2200$ points $(b,t)$ with a margin $\eta/(1+t^2)$, $\eta = 2 \cdot 10^{-4}$, and the points where the current solution violates it on a finer grid ($201 \times 24000$) are added as cutting planes, until no violation remains. The optimisation was run for $d \in \{0.65, 0.75, 0.8, 0.85, 0.9, 0.95, 1, 1.2\}$, $m \in \{2,3,4,5,7,10\}$ and $L^* \in \{5, 20, 60, 150, 400, 1000, 2500, 6000, 20000\}$; the six kernels of Theorem 2.1 are the best found for $L^* = 5, 20, 60, 150, 400, 6000$. Longer kernels did not lower the leading constant below 0.0838: for $L^* = 20000$ the optimum is again $A_1 = 0.08381$.

For Theorem 2.3 the objective is (6.1), linear in $(c_\iota, \kappa_k)$, so the problem is a linear programme (HiGHS through `scipy`; program `explore_discs.py` of Appendix B), with the profile $\mathfrak{e}_{\mathrm{P}}$ in the coefficients $J_k$, a margin $\eta = 10^{-3}$, the constraint (6.2) imposed on a grid of $26 \times 1600$ points and refined by cutting planes. The functions $W_k$ have a logarithmic singularity at the centre of the disc and a kink on its boundary, and $U + \sum_k \kappa_k W_k - f$ can have narrow dips that a grid misses: the violations are therefore searched on a finer grid (101 values of $b$; in $t$, 12 000 geometrically spaced points in $10^{-7} \le |t| \le 3000$ together with 60 000 equally spaced points in $|t| \le 3$) and the positive local maxima along $t$, together with the three largest values for each $b$, are refined by a local maximisation in $(b,t)$ (Nelder–Mead, at most 40 starting points per iteration), the maxima being added as cutting planes. The candidate discs were all $(c, R, v)$ with

$$c \in \{1.03, 1.06, 1.1, \log 3, 1.15, 1.2, 1.25, 1.3, 1.4, 1.5, \log 5, 2, e, 3\},$$
$$R = c - 1 + x, \quad x \in \{0.05, 0.1, \dots, 0.5\}, \quad R \le c - \tfrac12,$$

and $v \in \{0, \pm 0.2, \pm 0.35, \pm 0.5, \pm 0.65, \pm 0.8, \pm 1, \pm 1.3, \pm 1.6, \pm 2\}$; the optimum uses 90 of the 2 660 candidates, its value on the grid is 0.07486, and the certified value of (6.1) for the same data is $C_{\mathrm{H}} = 0.0748596142710907\ldots$ (the difference comes from the rational rounding of the disc data and from the exact $J_k$). The certification of this solution succeeded at the first attempt. (An earlier run of the same programme, with the coarser violation search of a grid alone and with the profile of the table of [2] in place of $\mathfrak{e}_{\mathrm{P}}$, had produced a solution with a numerically detected violation of (6.2) (floating point, not enclosed in intervals) of size $-8.9 \cdot 10^{-5}$ at $(b,t) \approx (0.467, -0.450)$, between two cutting planes, which had to be repaired by adding $2 \cdot 10^{-4}$ to $c_{\mathrm{R},1}$; the refinement of the violation search described above was introduced for this reason, and the result is $2.9 \cdot 10^{-5}$ smaller.) With the Bourgain profile (the piecewise linear function through the points $(\lambda_j - \kappa_j, \kappa_j)$, $(\kappa_j, \lambda_j) = A^j(\frac{13}{84}, \frac{55}{84})$, $0 \le j \le 7$, and $(1,0)$, where $A$ is the $A$-process) in place of $\mathfrak{e}_{\mathrm{P}}$ the same programme gives, on the grid and without margin, 0.07531; with $\mathfrak{e}_{\mathrm{P}}$ and without margin it gives 0.07478; neither is certified.

7.3. **Certification.** All certificates use `python-flint` 0.9.0 (Arb [16], FLINT [11]) at 256 bits of precision; a ball computed by Arb contains the exact value of the expression evaluated, and the programs only ever use the endpoints of balls in the direction that makes the final statement weaker. The endpoints are binary numbers; when a bound is printed or written to a file it is converted to decimal with directed rounding (towards $+\infty$ for an upper bound), so that every decimal asserted as a final upper or lower bound in the outputs of Appendix C is a valid bound; ball displays, radii, intermediate values and running times printed there are diagnostic. The constants of Theorem 2.1 are sums of such final decimals, rounded upwards. The kernel inequality for the hybrid ($T_1 = 40$) needed 2 028 149, 2 294 869 and 277 098 boxes for the three parts of Proposition 4.11 and 838 seconds; for the kernel of $\ell_{400}$ the three parts needed 11 835 193, 9 801 985 and 4 194 302 boxes and 2319 seconds, and for that of $\ell_{6000}$, 14 760 921, 15 866 017 and 4 194 302 boxes and 3123 seconds. The constant $C_\zeta$ for each kernel takes between 52 and 81 seconds (the sum over the 5 761 455 primes below $10^8$ and their powers, and the tail), and the elementary part between 61 and 81 seconds per mode. The coefficients are passed to the certifiers as decimal strings that round-trip to the binary floating-point numbers written by the optimiser (the JSON files of Appendix D), so the coefficients are recovered exactly from the JSON decimals; the only conversions between search and certificate are the rational rounding of the disc data described before (6.1) and that of the parameter $d$ to a denominator $\le 10^6$ (in the seven runs $d \in \{0.85, 0.9, 1, 1.2\}$, already exact), both of which are part of the certified data. All 25 certificates were rerun on a second machine (MacBook Air M2, macOS 26.6, Python 3.14.5, `python-flint` 0.9.0) with the same published programs and data, with identical bounds, enclosures and box counts; the report and the evidence are in the folder `docs/reproduction/` of the repository.

7.4. **Comparison of the two methods.** Backlund's method bounds the variation of $\arg\zeta$ on a horizontal segment ending at $\frac12 + iT$ by the number of zeros of $\operatorname{Re}\zeta$ on it, which Jensen's formula controls through the growth of $\zeta$ on a disc centred at $c + iT$, $c > 1$, that reaches $\frac12 - \delta$; the growth to the left of the critical line is supplied by the functional equation, and the factor $\frac{1}{\log 2}$ in (1.7) comes from the geometry of a disc that reaches $\frac12$ from a centre at distance about 1. The method of Turing type bounds a sum over all the zeros, each weighted by $f(b_\rho, t_\rho)$, by its exact evaluation at points of $\operatorname{Re} s > 1$, and needs no information inside the strip; its limit, in the computations of Remark 2.5, is near 0.079. The hybrid keeps the exact evaluation for most of the sum and uses Jensen's formula only where the kernel fails, on small discs that lie entirely in $\operatorname{Re} s \ge \frac12$, so that only the growth of $\zeta$ to the right of the critical line, where it is smallest, is paid for. Any improvement of the bounds for $\mu(\sigma)$ on $(\frac12, 1)$ lowers $C_{\mathrm{H}}$ through $J(\mathfrak{e}_{\mathrm{P}}; c_k, R_k)$ without any other change in the proof. Making Theorem 2.3 explicit is the question of Section 7.5.

7.5. **An open question.** Theorem 2.1 uses no disc and Theorem 2.3 gives no height. An explicit hybrid, with explicit subconvexity bounds in place of the profile $\mathfrak{c}_{\mathrm{P}}$ and the explicit interpolation of Lemma 5.2, is possible in principle, since the certificate of (6.2) does not depend on the profile; it would carry a term in $\log\log T$ from the logarithmic factors of the explicit bounds. In a preliminary computation at $\log T = 1000$ the discs did not lower the bound of the kernel alone. The question is left open.

**Data availability.** The programs, the coefficient files and the outputs of all certificates are reproduced in Appendices B–D and are available at `https://github.com/vcasana1/PaperMath/tree/explicit-S-turing/numerics/explicit_S_turing`; the article is deposited at Zenodo, doi:10.5281/zenodo.23858827 (first deposit 1 November 2026).

**AI declaration.** The ideas of this article, the constructions and the strategy of the proofs are the author's. Generative AI tools were used during the work and the preparation of the manuscript: Claude (Anthropic), models Claude Fable 5.1, Claude Opus 5.5 and Claude Sonnet 5.5; ChatGPT (OpenAI), model Astra 6; and Microsoft 365 Copilot, based on the GPT-5 reasoning model. Under the author's direction, Claude and ChatGPT were used to draft and revise the LaTeX text, to check proofs, constants and citations against the sources, and to search the literature. The Python programs of Appendix B were written with Claude. Microsoft 365 Copilot was used only to review drafts of the manuscript. The author has checked the whole article and takes full responsibility for its content. No AI tool is an author.

**Mathematical software.** The computations of Appendices B–C use Python 3.13 with `numpy`, `scipy`, `cvxpy` (solver Clarabel), `mpmath` and `python-flint` 0.9.0 (Arb ball arithmetic [16], FLINT 3.6.0 [11]).

## References


[1] V. S. G. Amberger, Estimating the number of zeros of Dedekind zeta-functions, arXiv:2510.27444v5, 12 October 2026 (v1: 31 October 2025), cited with the numbering of v5, doi:10.48550/arXiv.2510.27444. (Cited on pp. 3, 4, 5, 6, 10, and 16.)

[2] T. Tao, T. Trudgian, A. Yang et al., Analytic Number Theory Exponent Database, `https://teorth.github.io/expdb/` (table of bounds for $\mu(\sigma)$ in the chapter on the growth of $\zeta$; version of October 2026). (Cited on p. 21.)

[3] R. J. Backlund, Über die Nullstellen der Riemannschen Zetafunktion, *Acta Math.* **41** (1916) 345–375, doi:10.1007/BF02422950. (Cited on pp. 2 and 5.)

[4] C. Bellotti and P.-J. Wong, Improved estimates for the argument and zero-counting function of the Riemann zeta-function, *Math. Comp.* (2025, published online 14 November 2025), statements numbered as in arXiv:2412.15470v2 (doi:10.48550/arXiv.2412.15470), doi:10.1090/mcom/4133. (Cited on pp. 2, 4, and 5.)

[5] C. Bellotti and P.-J. Wong, Counting zeros of Artin $L$-functions, arXiv:2606.31441v1, 30 June 2026, doi:10.48550/arXiv.2606.31441. (Cited on pp. 3, 4, 5, 6, and 16.)

[6] J. Bourgain, Decoupling, exponential sums and the Riemann zeta function, *J. Amer. Math. Soc.* **30** (2017) 205–224, theorems numbered as in arXiv:1408.5794v2 (doi:10.48550/arXiv.1408.5794), doi:10.1090/jams/860. (Cited on pp. 16 and 26.)

[7] V. Casaña-Giner, Bounds for the argument of the Riemann zeta function and a prime-corrected Riemann–von Mangoldt approximation of its zeros, manuscript, version 57 (31 October 2026), submitted; statements numbered as in that version. (Cited on pp. 3, 4, and 5.)

[8] H. Davenport, *Multiplicative Number Theory*, 3rd edn., revised by H. L. Montgomery, Graduate Texts in Mathematics, Vol. 74 (Springer, New York, 2000), doi:10.1007/978-1-4757-5927-3. (Cited on pp. 2, 7, and 8.)

[9] NIST Digital Library of Mathematical Functions, §5.11(ii), Release 1.2.4 of 2025-03-15, F. W. J. Olver et al., eds., `https://dlmf.nist.gov/5.11.ii`. (Cited on p. 12.)

[10] A. Fiori, A note on the Phragmén–Lindelöf theorem, arXiv:2502.13282v3, 2026, doi:10.48550/arXiv.2502.13282. (Cited on p. 16.)

[11] The FLINT team, FLINT: Fast Library for Number Theory, version 3.6.0, `https://flintlib.org`. (Cited on pp. 21, 22, and 29.)

[12] J. Grossmann, Über die Nullstellen der Riemannschen Zeta-Funktion und der Dirichletschen $L$-Funktionen, Ph.D. thesis, Georg-August-Universität Göttingen (1913). (Cited on pp. 2 and 5.)

[13] E. Hasanalizade, Q. Shen and P.-J. Wong, Counting zeros of the Riemann zeta function, *J. Number Theory* **235** (2022) 219–241, doi:10.1016/j.jnt.2021.06.032. (Cited on pp. 2 and 5.)

[14] D. R. Heath-Brown, A new $k$th derivative estimate for exponential sums via Vinogradov's mean value, *Proc. Steklov Inst. Math.* **296** (2017) 88–103, theorems numbered as in arXiv:1601.04493v3 (doi:10.48550/arXiv.1601.04493), doi:10.1134/S0081543817010084. (Cited on pp. 17 and 26.)

[15] F. Johansson, Numerical integration in arbitrary-precision ball arithmetic, in: *Mathematical Software – ICMS 2018*, Lecture Notes in Computer Science, Vol. 10931 (Springer, Cham, 2018) 255–263, doi:10.1007/978-3-319-96418-8_30. (Cited on p. 12.)

[16] F. Johansson, Arb: efficient arbitrary-precision midpoint-radius interval arithmetic, *IEEE Trans. Comput.* **66** (2017) 1281–1292, doi:10.1109/TC.2017.2690633. (Cited on pp. 4, 21, 22, and 29.)

[17] F. W. J. Olver, *Asymptotics and Special Functions* (Academic Press, New York, 1974); reprinted (A K Peters, Wellesley, MA, 1997), doi:10.1201/9781439864548. (Cited on p. 12.)

[18] D. J. Platt and T. S. Trudgian, An improved explicit bound on $|\zeta(\frac{1}{2}+it)|$, *J. Number Theory* **147** (2015) 842–851, doi:10.1016/j.jnt.2014.08.019. (Cited on pp. 2 and 5.)

[19] H. Rademacher, On the Phragmén–Lindelöf theorem and some applications, *Math. Z.* **72** (1959) 192–204, doi:10.1007/BF01162949. (Cited on p. 16.)

[20] W. Rudin, *Real and Complex Analysis*, 3rd edn. (McGraw-Hill, New York, 1987). (Cited on p. 18.)

[21] J. B. Rosser, Explicit bounds for some functions of prime numbers, *Amer. J. Math.* **63** (1941) 211–232, doi:10.2307/2371291. (Cited on pp. 2 and 5.)

[22] J. B. Rosser and L. Schoenfeld, Approximate formulas for some functions of prime numbers, *Illinois J. Math.* **6** (1962) 64–94, doi:10.1215/ijm/1255631807. (Cited on pp. 3 and 11.)

[23] T. Tao, T. S. Trudgian and A. Yang, New exponent pairs, zero density estimates, and zero additive energy estimates: a systematic approach, arXiv:2501.16779v1, 28 January 2025, doi:10.48550/arXiv.2501.16779. (Cited on pp. 4, 16, 17, and 26.)

[24] E. C. Titchmarsh, *The Theory of Functions*, 2nd edn. (Oxford University Press, Oxford, 1939). (Cited on p. 18.)

[25] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd edn., revised by D. R. Heath-Brown (Clarendon Press, Oxford, 1986). (Cited on pp. 2, 7, 8, 16, 17, 18, and 26.)

[26] T. S. Trudgian, An improved upper bound for the argument of the Riemann zeta-function on the critical line, *Math. Comp.* **81** (2012) 1053–1061, doi:10.1090/S0025-5718-2011-02537-8. (Cited on pp. 2 and 5.)

[27] T. S. Trudgian, An improved upper bound for the argument of the Riemann zeta-function on the critical line II, *J. Number Theory* **134** (2014) 280–292, doi:10.1016/j.jnt.2013.07.017. (Cited on pp. 2 and 5.)

[28] T. S. Trudgian and A. Yang, Toward optimal exponent pairs, arXiv:2306.05599v3, 15 July 2024, doi:10.48550/arXiv.2306.05599. (Cited on pp. 4, 17, and 26.)

[29] A. M. Turing, Some calculations of the Riemann zeta-function, *Proc. London Math. Soc. (3)* **3** (1953) 99–117, doi:10.1112/plms/s3-3.1.99. (Cited on pp. 3 and 6.)

[30] H. von Mangoldt, Zur Verteilung der Nullstellen der Riemannschen Funktion $\xi(t)$, *Math. Ann.* **60** (1905) 1–19, doi:10.1007/BF01447494. (Cited on pp. 2 and 5.)

## Appendix A. Data tables

| $\iota$ | $c_\iota$ | $\iota$ | $c_\iota$ |
|---|---|---|---|
| $(\mathrm{R},1)$ | 0.412065007285 | $(\mathrm{I},2)$ | 0.733460748935 |
| $(\mathrm{R},2)$ | 0.370154287951 | $(\mathrm{I},3)$ | −0.136130059477 |
| $(\mathrm{R},3)$ | 0.28440158115 | $(\mathrm{I},4)$ | 0.151360502416 |
| $(\mathrm{R},4)$ | 0.158618575787 | $(\mathrm{I},5)$ | −0.116929743682 |
| $(\mathrm{R},5)$ | 0.0695422300653 | $(\mathrm{I},6)$ | −0.0365689491866 |
| $(\mathrm{R},6)$ | −0.0615776499388 | $(\mathrm{I},7)$ | 0.106014270436 |
| $(\mathrm{R},7)$ | −0.040570135781 | $(\mathrm{I},8)$ | −0.0622716309967 |
| $(\mathrm{R},8)$ | 0.075267182926 | $(\mathrm{I},9)$ | 0.0166909270918 |
| $(\mathrm{R},9)$ | −0.0323507305873 | $(\mathrm{I},10)$ | −0.00179414655284 |
| $(\mathrm{R},10)$ | 0.00563260523722 | | |

Table 3. Coefficients $c_\iota$ of the hybrid kernel ($d = \frac{9}{10}$, $m = 10$), to twelve significant digits; the exact binary values are in Appendix D.

Table 4: The 90 disc data $(c_k, R_k, v_k)$ (rational, as used in the certificate), the weights $\kappa_k$ (ten significant digits; the full-precision decimal strings of the binary floating-point weights are in Appendix D). The rationals $c_k, R_k, v_k$ are the data used in the certificate: a centre such as $\log 3$ in the search grid of Section 7.2 appears here as its rounding to denominator $\le 10^4$, not as a symbolic value, $J_k = J(\mathfrak{e}_{\mathrm{P}}; c_k, R_k)$ and the contributions $\kappa_k J_k/\pi$ to $C_{\mathrm{H}}$.

| $k$ | $c_k$ | $R_k$ | $v_k$ | $\kappa_k$ | $J_k$ | $\kappa_k J_k/\pi$ |
|---|---|---|---|---|---|---|
| 1 | 103/100 | 9/50 | −1/2 | 0.04645216861 | 0.0069138770 | 0.000102230 |
| 2 | 103/100 | 9/50 | 1/2 | 0.04300701195 | 0.0069138770 | $9.46479 \cdot 10^{-5}$ |
| 3 | 103/100 | 23/100 | −13/20 | 0.04706304324 | 0.010786963 | 0.000161596 |
| 4 | 103/100 | 23/100 | 13/20 | 0.1466908127 | 0.010786963 | 0.000503677 |
| 5 | 103/100 | 7/25 | −13/20 | 0.02597995484 | 0.015048973 | 0.000124450 |
| 6 | 103/100 | 7/25 | 13/20 | 0.1037916655 | 0.015048973 | 0.000497187 |
| 7 | 103/100 | 33/100 | −13/20 | 0.02381800473 | 0.019571517 | 0.000148382 |
| 8 | 103/100 | 33/100 | 13/20 | 0.1041736978 | 0.019571517 | 0.000648982 |
| 9 | 103/100 | 19/50 | −13/20 | 0.0145395318 | 0.024737703 | 0.000114488 |
| 10 | 103/100 | 19/50 | 13/20 | 0.06487216198 | 0.024737703 | 0.000510820 |
| 11 | 103/100 | 43/100 | 1 | 0.01585298753 | 0.030311094 | 0.000152955 |
| 12 | 103/100 | 12/25 | −1 | 0.003361796325 | 0.036143828 | $3.86773 \cdot 10^{-5}$ |
| 13 | 103/100 | 12/25 | 1 | 0.006488928258 | 0.036143828 | $7.46547 \cdot 10^{-5}$ |
| 14 | 103/100 | 53/100 | −8/5 | $5.414487565 \cdot 10^{-5}$ | 0.042238079 | $7.27967 \cdot 10^{-7}$ |
| 15 | 103/100 | 53/100 | −13/10 | 0.000301871607 | 0.042238079 | $4.05860 \cdot 10^{-6}$ |
| 16 | 103/100 | 53/100 | −1 | 0.0009151843659 | 0.042238079 | $1.23045 \cdot 10^{-5}$ |
| 17 | 103/100 | 53/100 | 1 | 0.004457480243 | 0.042238079 | $5.99299 \cdot 10^{-5}$ |
| 18 | 103/100 | 53/100 | 13/10 | 0.001466000015 | 0.042238079 | $1.97101 \cdot 10^{-5}$ |
| 19 | 53/50 | 4/25 | −1/2 | 0.04752584139 | 0.0032593311 | $4.93070 \cdot 10^{-5}$ |
| 20 | 53/50 | 4/25 | 1/2 | 0.159072944 | 0.0032593311 | 0.000165035 |
| 21 | 53/50 | 21/100 | −1/2 | 0.02538968149 | 0.0063616476 | $5.14135 \cdot 10^{-5}$ |
| 22 | 53/50 | 21/100 | 1/2 | 0.2223325253 | 0.0063616476 | 0.000450218 |
| 23 | 53/50 | 13/50 | −1/2 | 0.08631620266 | 0.010091730 | 0.000277273 |
| 24 | 53/50 | 13/50 | 1/2 | 0.1538221037 | 0.010091730 | 0.000494122 |
| 25 | 53/50 | 13/50 | 13/20 | 0.01149832481 | 0.010091730 | $3.69360 \cdot 10^{-5}$ |

| $k$ | $c_k$ | $R_k$ | $v_k$ | $\kappa_k$ | $J_k$ | $\kappa_k J_k/\pi$ |
|---|---|---|---|---|---|---|
| 26 | 53/50 | 41/100 | −4/5 | 0.01072845742 | 0.023727069 | $8.10273 \cdot 10^{-5}$ |
| 27 | 53/50 | 41/100 | −13/20 | 0.02409446047 | 0.023727069 | 0.000181975 |
| 28 | 53/50 | 41/100 | 13/20 | 0.08996010049 | 0.023727069 | 0.000679429 |
| 29 | 53/50 | 41/100 | 4/5 | 0.01877223007 | 0.023727069 | 0.000141778 |
| 30 | 53/50 | 23/50 | 4/5 | 0.01359832436 | 0.029208025 | 0.000126426 |
| 31 | 53/50 | 14/25 | −8/5 | $9.669233393 \cdot 10^{-5}$ | 0.040975795 | $1.26116 \cdot 10^{-6}$ |
| 32 | 53/50 | 14/25 | −13/10 | $4.802780697 \cdot 10^{-5}$ | 0.040975795 | $6.26427 \cdot 10^{-7}$ |
| 33 | 53/50 | 14/25 | 13/10 | 0.0001824982988 | 0.040975795 | $2.38033 \cdot 10^{-6}$ |
| 34 | 11/10 | 3/20 | −7/20 | 0.05671657488 | 0.00078557774 | $1.41824 \cdot 10^{-5}$ |
| 35 | 11/10 | 1/4 | −1/2 | 0.05147231926 | 0.0057973918 | $9.49853 \cdot 10^{-5}$ |
| 36 | 11/10 | 1/4 | 1/2 | 0.1883380387 | 0.0057973918 | 0.000347553 |
| 37 | 11/10 | 2/5 | 4/5 | 0.07545736056 | 0.017623649 | 0.000423299 |
| 38 | 11/10 | 9/20 | −4/5 | 0.001627486755 | 0.022555747 | $1.16849 \cdot 10^{-5}$ |
| 39 | 11/10 | 9/20 | 4/5 | 0.009838802011 | 0.022555747 | $7.06398 \cdot 10^{-5}$ |
| 40 | 11/10 | 1/2 | −4/5 | 0.001256180045 | 0.027910450 | $1.11601 \cdot 10^{-5}$ |
| 41 | 11/10 | 1/2 | 4/5 | 0.005913769697 | 0.027910450 | $5.25389 \cdot 10^{-5}$ |
| 42 | 11/10 | 3/5 | −1 | 0.0006183338066 | 0.039458156 | $7.76622 \cdot 10^{-6}$ |
| 43 | 11/10 | 3/5 | 1 | 0.002725978105 | 0.039458156 | $3.42381 \cdot 10^{-5}$ |
| 44 | 11/10 | 3/5 | 13/10 | 0.0002112464591 | 0.039458156 | $2.65324 \cdot 10^{-6}$ |
| 45 | 5225/4756 | 229/1153 | −1/2 | 0.05954976707 | 0.0029087233 | $5.51357 \cdot 10^{-5}$ |
| 46 | 5225/4756 | 229/1153 | 1/2 | 0.2546846193 | 0.0029087233 | 0.000235806 |
| 47 | 5225/4756 | 2329/9368 | −1/2 | 0.01670054759 | 0.0058145056 | $3.09096 \cdot 10^{-5}$ |
| 48 | 5225/4756 | 2329/9368 | 1/2 | 0.07440766183 | 0.0058145056 | 0.000137715 |
| 49 | 5225/4756 | 1743/5837 | −1/2 | 0.01266957078 | 0.0093687838 | $3.77829 \cdot 10^{-5}$ |
| 50 | 5225/4756 | 1743/5837 | 1/2 | 0.1891814931 | 0.0093687838 | 0.000564173 |
| 51 | 5225/4756 | 2667/5945 | −4/5 | 0.001477775558 | 0.022593493 | $1.06278 \cdot 10^{-5}$ |
| 52 | 5225/4756 | 2667/5945 | 4/5 | 0.006728141614 | 0.022593493 | $4.83870 \cdot 10^{-5}$ |
| 53 | 5225/4756 | 4132/8287 | −4/5 | 0.001479398897 | 0.027952580 | $1.31631 \cdot 10^{-5}$ |
| 54 | 5225/4756 | 4132/8287 | 4/5 | 0.005460180548 | 0.027952580 | $4.85824 \cdot 10^{-5}$ |
| 55 | 5225/4756 | 2847/4756 | −13/10 | $5.562368483 \cdot 10^{-5}$ | 0.039507983 | $6.99511 \cdot 10^{-7}$ |
| 56 | 23/20 | 1/5 | −1/2 | 0.0542433989 | 0.00067784859 | $1.17039 \cdot 10^{-5}$ |
| 57 | 23/20 | 1/5 | −7/20 | 0.03475948982 | 0.00067784859 | $7.49991 \cdot 10^{-6}$ |
| 58 | 23/20 | 1/5 | 7/20 | 0.2143837181 | 0.00067784859 | $4.62567 \cdot 10^{-5}$ |
| 59 | 23/20 | 9/20 | −4/5 | 0.006316370699 | 0.016543682 | $3.32621 \cdot 10^{-5}$ |
| 60 | 23/20 | 1/2 | −4/5 | 0.003332642063 | 0.021311247 | $2.26072 \cdot 10^{-5}$ |
| 61 | 23/20 | 11/20 | −4/5 | 0.006954420904 | 0.026509943 | $5.86840 \cdot 10^{-5}$ |
| 62 | 23/20 | 11/20 | 4/5 | 0.02859280204 | 0.026509943 | 0.000241277 |
| 63 | 23/20 | 13/20 | −13/10 | $5.258824036 \cdot 10^{-5}$ | 0.037780806 | $6.32426 \cdot 10^{-7}$ |
| 64 | 23/20 | 13/20 | −1 | 0.0004219252908 | 0.037780806 | $5.07408 \cdot 10^{-6}$ |
| 65 | 23/20 | 13/20 | 1 | 0.001832673786 | 0.037780806 | $2.20397 \cdot 10^{-5}$ |
| 66 | 23/20 | 13/20 | 13/10 | 0.0002062650784 | 0.037780806 | $2.48054 \cdot 10^{-6}$ |
| 67 | 6/5 | 1/4 | −1/2 | 0.03706538771 | 0.00060498120 | $7.13774 \cdot 10^{-6}$ |
| 68 | 6/5 | 1/4 | −7/20 | 0.0305999968 | 0.00060498120 | $5.89269 \cdot 10^{-6}$ |
| 69 | 6/5 | 1/4 | 7/20 | 0.06841653548 | 0.00060498120 | $1.31751 \cdot 10^{-5}$ |
| 70 | 6/5 | 1/4 | 1/2 | 0.3323254238 | 0.00060498120 | $6.39964 \cdot 10^{-5}$ |
| 71 | 6/5 | 7/10 | −13/10 | $4.579439123 \cdot 10^{-5}$ | 0.036302130 | $5.29169 \cdot 10^{-7}$ |
| 72 | 6/5 | 7/10 | −1 | 0.0003653681966 | 0.036302130 | $4.22195 \cdot 10^{-6}$ |
| 73 | 6/5 | 7/10 | 1 | 0.00157216072 | 0.036302130 | $1.81668 \cdot 10^{-5}$ |
| 74 | 6/5 | 7/10 | 13/10 | 0.0001785005556 | 0.036302130 | $2.06263 \cdot 10^{-6}$ |
| 75 | 5/4 | 3/10 | −7/20 | 0.05227534138 | 0.00055148537 | $9.17658 \cdot 10^{-6}$ |
| 76 | 5/4 | 3/10 | 7/20 | 0.2040492823 | 0.00055148537 | $3.58195 \cdot 10^{-5}$ |
| 77 | 5/4 | 3/4 | −13/10 | $6.871033612 \cdot 10^{-5}$ | 0.034985511 | $7.65174 \cdot 10^{-7}$ |

| $k$ | $c_k$ | $R_k$ | $v_k$ | $\kappa_k$ | $J_k$ | $\kappa_k J_k/\pi$ |
|---|---|---|---|---|---|---|
| 78 | 5/4 | 3/4 | $-1$ | 0.000318403076 | 0.034985511 | $3.54581 \cdot 10^{-6}$ |
| 79 | 5/4 | 3/4 | 1 | 0.001362286751 | 0.034985511 | $1.51707 \cdot 10^{-5}$ |
| 80 | 5/4 | 3/4 | 13/10 | 0.0002746990133 | 0.034985511 | $3.05911 \cdot 10^{-6}$ |
| 81 | 13/10 | 4/5 | $-1$ | 0.0008212433178 | 0.033803191 | $8.83649 \cdot 10^{-6}$ |
| 82 | 13/10 | 4/5 | 1 | 0.003961667359 | 0.033803191 | $4.26271 \cdot 10^{-5}$ |
| 83 | 7/5 | 9/20 | $-7/20$ | 0.3861919019 | 0.00044923233 | $5.52235 \cdot 10^{-5}$ |
| 84 | 7/5 | 9/20 | 7/20 | 0.8551654873 | 0.00044923233 | 0.000122284 |
| 85 | 7/5 | 9/10 | $-13/10$ | 0.0001493724531 | 0.031760037 | $1.51009 \cdot 10^{-6}$ |
| 86 | 7/5 | 9/10 | 13/10 | 0.00143209911 | 0.031760037 | $1.44779 \cdot 10^{-5}$ |
| 87 | 3/2 | 11/20 | $-7/20$ | 0.0979914707 | 0.00040600316 | $1.26639 \cdot 10^{-5}$ |
| 88 | 3/2 | 11/20 | 7/20 | 0.3388231655 | 0.00040600316 | $4.37878 \cdot 10^{-5}$ |
| 89 | 9993/6209 | 2980/4519 | $-7/20$ | 0.1671407202 | 0.00037055405 | $1.97144 \cdot 10^{-5}$ |
| 90 | 9993/6209 | 2980/4519 | 7/20 | 0.5061513201 | 0.00037055405 | $5.97011 \cdot 10^{-5}$ |
| | | | | | $\frac{1}{\pi}\sum_k \kappa_k J_k =$ | 0.009277431487 |

Table 5: The vertices $(\varsigma_i, e_i)$ of the profile $\mathfrak{e}_{\mathrm{P}}$ of Lemma 5.3, and the source of the bound $\mu(\varsigma_i) \le e_i$. The Heath-Brown vertices $n = 10, \dots, 398$ are omitted from the table; they are generated by `profile_published.py` (Appendix B).

| $i$ | $\varsigma_i$ | $e_i$ | source |
|---|---|---|---|
| 0 | 1/2 | 13/84 | Bourgain [6, Thm. 5] |
| 1 | 88225/153852 | 4742/38463 | Trudgian–Yang [28, Thm. 2.4] |
| 2 | 521/796 | 18/199 | Trudgian–Yang [28, Thm. 2.4] |
| 3 | 53141/76066 | 2779/38033 | Trudgian–Yang [28, Thm. 2.4] |
| 4 | 454/641 | 89/1282 | Tao–Trudgian–Yang [23, Thm. 20, Lemma 23] |
| 5 | 3473692/4856993 | 652397/9713986 | Tao–Trudgian–Yang [23, Thm. 20, Lemma 23] |
| 6 | 52209/69128 | 2371/43205 | Trudgian–Yang [28, Thm. 2.4] |
| 7 | 1389/1736 | 9/217 | Trudgian–Yang [28, Thm. 2.4] |
| 8 | 587779/702192 | 10769/351096 | Tao–Trudgian–Yang [23, Thm. 20, Lemma 23] |
| 9 | 7441/8695 | 89/3478 | Tao–Trudgian–Yang [23, Thm. 20, Lemma 23] |
| 10 | 277/300 | 1/100 | Heath-Brown [14, Thm. 2], $n = 6$ |
| 11 | 535/567 | 1/162 | Heath-Brown [14, Thm. 2], $n = 7$ |
| 12 | 375/392 | 1/245 | Heath-Brown [14, Thm. 2], $n = 8$ |
| 13 | 3059/3168 | 1/352 | Heath-Brown [14, Thm. 2], $n = 9$ |
| | ⋮ | (Heath-Brown nodes $n = 10, \dots, 398$) | ⋮ |
| 403 | 6336001297/6336120399 | 1/31760002 | Heath-Brown [14, Thm. 2], $n = 399$ |
| 404 | 1 | 0 | $\zeta(1 + it) \ll \log t$ [25, Thm. 3.5] |

| $\iota$ | $c_\iota$ | $\iota$ | $c_\iota$ |
|---|---|---|---|
| $(\mathrm{R},1)$ | 1.26767379327 | $(\mathrm{I},2)$ | −0.37840733351 |
| $(\mathrm{R},2)$ | 0.700929451456 | $(\mathrm{I},3)$ | 0.134730628931 |
| $(\mathrm{R},3)$ | −0.494633048662 | $(\mathrm{I},4)$ | −0.017083827788 |
| $(\mathrm{R},4)$ | 0.097515550525 | $(\mathrm{I},5)$ | 0.000100343368477 |
| $(\mathrm{R},5)$ | −0.000589396293163 | | |

TABLE 6. Coefficients of the kernel of the line $\ell_5$ ($d = 1$, $m = 5$), to twelve significant digits.

| $\iota$ | $c_\iota$ | $\iota$ | $c_\iota$ |
|---|---|---|---|
| $(\mathrm{R},1)$ | 0.679978474949 | $(\mathrm{I},2)$ | 1.18435020971 |
| $(\mathrm{R},2)$ | 0.804110639165 | $(\mathrm{I},3)$ | −0.576537489831 |
| $(\mathrm{R},3)$ | 0.777427090742 | $(\mathrm{I},4)$ | 0.00610293149501 |
| $(\mathrm{R},4)$ | −0.0084486134969 | $(\mathrm{I},5)$ | $-2.10769863064 \cdot 10^{-5}$ |
| $(\mathrm{R},5)$ | $2.95597215845 \cdot 10^{-5}$ | | |

TABLE 7. Coefficients of the kernel of the line $\ell_{20}$ ($d = 1.2$, $m = 5$), to twelve significant digits.

| $\iota$ | $c_\iota$ | $\iota$ | $c_\iota$ |
|---|---|---|---|
| $(\mathrm{R},1)$ | 0.595153203515 | $(\mathrm{I},2)$ | 0.80794661471 |
| $(\mathrm{R},2)$ | 0.591131249545 | $(\mathrm{I},3)$ | −0.34004800791 |
| $(\mathrm{R},3)$ | 0.388921642826 | $(\mathrm{I},4)$ | 0.00364916717989 |
| $(\mathrm{R},4)$ | −0.00432516763692 | $(\mathrm{I},5)$ | $-1.27504674131 \cdot 10^{-5}$ |
| $(\mathrm{R},5)$ | $1.53789024752 \cdot 10^{-5}$ | | |

TABLE 8. Coefficients of the kernel of the line $\ell_{60}$ ($d = 1$, $m = 5$), to twelve significant digits.

| $\iota$ | $c_\iota$ | $\iota$ | $c_\iota$ |
|---|---|---|---|
| $(\mathrm{R},1)$ | 0.587356111095 | $(\mathrm{I},2)$ | 0.529298607056 |
| $(\mathrm{R},2)$ | 0.495527358923 | $(\mathrm{I},3)$ | −0.269742540483 |
| $(\mathrm{R},3)$ | 0.171910239516 | $(\mathrm{I},4)$ | 0.0395067096637 |
| $(\mathrm{R},4)$ | −0.0451542915632 | $(\mathrm{I},5)$ | −0.000234590737695 |
| $(\mathrm{R},5)$ | 0.000271746755402 | | |

TABLE 9. Coefficients of the kernel of the line $\ell_{150}$ ($d = 0.85$, $m = 5$), to twelve significant digits.

| $\iota$ | $c_\iota$ | $\iota$ | $c_\iota$ |
|---|---|---|---|
| $(\mathrm{R},1)$ | $0.527699716861$ | $(\mathrm{I},2)$ | $0.579501653809$ |
| $(\mathrm{R},2)$ | $0.448444622334$ | $(\mathrm{I},3)$ | $-0.243041549546$ |
| $(\mathrm{R},3)$ | $0.2073491544$ | $(\mathrm{I},4)$ | $0.0959846058781$ |
| $(\mathrm{R},4)$ | $-0.0517088136271$ | $(\mathrm{I},5)$ | $-0.0223162904392$ |
| $(\mathrm{R},5)$ | $0.0427743920458$ | $(\mathrm{I},6)$ | $0.00101667452746$ |
| $(\mathrm{R},6)$ | $-0.00216867871949$ | $(\mathrm{I},7)$ | $-2.32679097042 \cdot 10^{-5}$ |
| $(\mathrm{R},7)$ | $5.22536634852 \cdot 10^{-5}$ | $(\mathrm{I},8)$ | $3.09645235738 \cdot 10^{-7}$ |
| $(\mathrm{R},8)$ | $-7.19471178778 \cdot 10^{-7}$ | $(\mathrm{I},9)$ | $-2.32748949201 \cdot 10^{-9}$ |
| $(\mathrm{R},9)$ | $5.54823288096 \cdot 10^{-9}$ | $(\mathrm{I},10)$ | $7.769753796 \cdot 10^{-12}$ |
| $(\mathrm{R},10)$ | $-1.89056202487 \cdot 10^{-11}$ | | |

TABLE 10. Coefficients of the kernel of the line $\ell_{400}$ ($d = 0.85$, $m = 10$), to twelve significant digits.

| $\iota$ | $c_\iota$ | $\iota$ | $c_\iota$ |
|---|---|---|---|
| $(\mathrm{R},1)$ | $0.526619965261$ | $(\mathrm{I},2)$ | $0.583424601581$ |
| $(\mathrm{R},2)$ | $0.44752559746$ | $(\mathrm{I},3)$ | $-0.23667212703$ |
| $(\mathrm{R},3)$ | $0.217633359331$ | $(\mathrm{I},4)$ | $0.0956784075046$ |
| $(\mathrm{R},4)$ | $-0.0663261021081$ | $(\mathrm{I},5)$ | $-0.0185896622954$ |
| $(\mathrm{R},5)$ | $0.0492247519465$ | $(\mathrm{I},6)$ | $0.000831384847612$ |
| $(\mathrm{R},6)$ | $-0.0024545979738$ | $(\mathrm{I},7)$ | $-1.89384002059 \cdot 10^{-5}$ |
| $(\mathrm{R},7)$ | $5.84151825308 \cdot 10^{-5}$ | $(\mathrm{I},8)$ | $2.51756719451 \cdot 10^{-7}$ |
| $(\mathrm{R},8)$ | $-7.9615067149 \cdot 10^{-7}$ | $(\mathrm{I},9)$ | $-1.8927048877 \cdot 10^{-9}$ |
| $(\mathrm{R},9)$ | $6.08645994128 \cdot 10^{-9}$ | $(\mathrm{I},10)$ | $6.32254880932 \cdot 10^{-12}$ |
| $(\mathrm{R},10)$ | $-2.0583832928 \cdot 10^{-11}$ | | |

TABLE 11. Coefficients of the kernel of the line $\ell_{6000}$ ($d = 0.85$, $m = 10$), to twelve significant digits.

## Appendix B. Programs

The programs are written in Python 3 and use `python-flint` (Arb ball arithmetic [16], FLINT [11]) at 256 bits, `numpy`, `scipy`, `cvxpy` with the solver Clarabel, and `mpmath` (in `opt_line.py` only). The certification programs are those of Sections B–B; every bound they print or write to a file is converted to decimal with directed rounding (`up_str` in `certutil.py`), so that the printed decimal is itself a valid bound. Each certificate begins with the SHA-256 digests of its input file and of the programs that produced it, and the versions of Python and `python-flint`; the build script of this note checks that these digests are those of the files reproduced in Appendices B and D. The search programs that produced the coefficients are reproduced in Section B for completeness, but no statement of this article depends on them. All files are in the repository cited in Section 7, in the folders `hybrid_asymptotic/` and `family/`; the deposited archive is flat, and the programs locate their helper modules in either layout.

**`certutil.py`: shared helpers, directed decimal rounding of bounds**

```
# Shared helpers of the certification programs (python-flint / Arb ball arithmetic at 256 bits).
#   A(q)            exact ball of a Fraction or of an integer (floats are refused)
#   box(lo, hi)     ball containing the interval [lo, hi]
#   exact(x)        the rational value of an exact (zero-radius) arb
#   up_str(x, n)    decimal string with n significant digits that is >= every point of the ball x (directed
     rounding)
#   down_str(x, n)  decimal string with n significant digits that is <= every point of the ball x
#   provenance(...) lines with the SHA-256 of the input and program files and the library versions, for the
     record
# The .str() method of arb rounds to the nearest decimal, so it must not be used to print an upper or lower
     bound;
# every bound printed or written to a JSON file by the certification programs goes through up_str / down_str.
import hashlib, os, sys, platform
from fractions import Fraction as Fr
from decimal import Decimal, ROUND_CEILING, ROUND_FLOOR, localcontext
import flint
from flint import arb, ctx
ctx.prec = 256

def A(q):
    # exact ball of a Fraction or an integer; floats are refused, so that no binary approximation of a
     decimal enters silently
    if isinstance(q, Fr): return arb(q.numerator)/arb(q.denominator)
    assert isinstance(q, int), 'A() takes a Fraction or an integer, got %r' % (q,)
    return arb(q)

def box(lo, hi):
    return A(lo).union(A(hi))

def exact(x):
    assert x.is_exact() and x.is_finite(), x
    m, e = x.man_exp()
    return Fr(int(m))*Fr(2)**int(e)

def _dir_str(q, n, rounding):
    with localcontext() as c:
        c.prec = n; c.rounding = rounding
        dec = Decimal(q.numerator)/Decimal(q.denominator)      # correctly rounded to n significant digits
    return format(dec, 'f')

def up_str(x, n):
    u = exact(x.upper()); s = _dir_str(u, n, ROUND_CEILING)
    assert Fr(s) >= u, (s, u)
    return s

def down_str(x, n):
```

```
    l = exact(x.lower()); s = _dir_str(l, n, ROUND_FLOOR)
    assert Fr(s) <= l, (s, l)
    return s

def sha256(path):
    h = hashlib.sha256()
    with open(path, 'rb') as fh: h.update(fh.read())
    return h.hexdigest()

def provenance(*paths):
    # one line per file (name and SHA-256), then the versions of Python and python-flint and the working
     precision
    lines = ['%s sha256 %s' % (os.path.basename(p), sha256(p)) for p in paths]
    lines.append('python %s, python-flint %s, precision %d bits' % (platform.python_version(), getattr(flint,
      '__version__', '?'), ctx.prec))
    return '\n'.join(lines)

if __name__ == '__main__':
    x = arb(1)/3
    print(x.str(12, radius=False), up_str(x, 12), down_str(x, 12))
    y = -arb(1)/7
    print(y.str(12, radius=False), up_str(y, 12), down_str(y, 12))
```

## `certify_hybrid.py`: certification of the kernel inequality (with or without discs) and of $C_{\mathrm{H}}$

```
# Certification (Arb ball arithmetic, python-flint, 256 bits) of the kernel inequality
#   f(b,t) <= U(b,t) + sum_k kappa_k W_k(b,t)     for 0 <= b <= 1/2, t != 0,
# with f(b,t) = 2 atan((d+b)/t) + 2 atan((d-b)/t) - atan((2d+b)/t) - atan((2d-b)/t),
#      U(b,t) = sum_iota c_iota [F_iota(d+b,t) + F_iota(d-b,t)],  F_Rj = Re (x+it)^(-j),  F_Ij = -Im (x+it)
     ^(-j),
#      W_k(b,t) = log^+( R_k / sqrt((c_k-1/2-b)^2 + (t+v_k)^2) ),
# and of the constant C_H = c_R1/(2 pi) + (1/pi) sum_k kappa_k J_k,  J_k = (1/pi) int_0^pi e_P(c_k + R_k cos
     th) dth,
# where e_P is the piecewise linear profile of profile_published.py.  With an empty list of discs the program
      certifies
# the kernel inequality f <= U alone and prints C_1 = c_R1/(2 pi).
#
# Usage:  python3 certify_hybrid.py <solution.json> [T1]
#   solution.json: d, B (basis), c (coefficients, exact binary values), discs [(c_k,R_k,v_k)], kappa; T1
     defaults to 40.
# Method (Proposition "Certification of the kernel inequality" and Section "The hybrid" of the article):
#   (a) on [0,1/2] x (0,T1] and [0,1/2] x [-T1,0): centred form H(centre) + grad H(box) . half-widths,
     bisection;
#   (b) for |t| >= T1, u = 1/t: t^2 U - 4 d^3 u - K5 |u|^3 > 0 on [0,1/2] x [0,1/T1] and [0,1/2] x [-1/T1,0],
      bisection;
#       this needs T1 >= 2d + 1/2 (convergence of the arctan series) and W_k = 0 for |t| >= T1.
import sys, os, json, time
from fractions import Fraction as Fr
from flint import arb, acb
from certutil import A, box, up_str, down_str, exact, provenance

sol = json.load(open(sys.argv[1])); T1 = Fr(sys.argv[2]) if len(sys.argv) > 2 else Fr(40)
d = Fr(sol['d']).limit_denominator(10**6); B = [tuple(x) for x in sol['B']]
c = [Fr(x) for x in sol['c']]                                          # exact binary values written by the
     optimiser
discs = [(Fr(cc).limit_denominator(10**4), Fr(R).limit_denominator(10**4), Fr(v).limit_denominator(10**4))
     for cc, R, v in sol['discs']]
discs = [(cc, min(R, cc-Fr(1, 2)), v) for cc, R, v in discs]           # rational rounding must keep c_k -
     R_k >= 1/2 (a smaller R_k only lowers W_k)
kap = [Fr(x) for x in sol['kappa']]
# integrity of the data
assert len(c) == len(B) and len(discs) == len(kap)
```

```
assert B[0] == ('R', 1) and ('R', 2) in B and all(k in ('R', 'I') and j >= 1 for k, j in B) and ('I', 1) not
    in B
assert d > Fr(1, 2)
assert all(k >= 0 for k in kap) and all(R > 0 and cc-R >= Fr(1, 2) and cc > 1 for cc, R, v in discs)
assert T1 >= 2*d+Fr(1, 2)
assert all(abs(v)+R <= T1 for cc, R, v in discs)                          # then W_k = 0 for |t| >= T1
JMAX = max(j for _, j in B)
dA = A(d); half = A(Fr(1, 2)); PI = arb.pi()
cA = [A(x) for x in c]; kA = [A(x) for x in kap]; dsA = [(A(cc), A(R).log(), A(v)) for cc, R, v in discs]

def powers(x, t, n):
    # [(x+it)^0, (x+it)^-1, ..., (x+it)^-n]
    w = acb(1)/acb(x, t); pw = [acb(1)]
    for _ in range(n): pw.append(pw[-1]*w)
    return pw
def kernel(b, t):
    U = arb(0)
    for x in (dA+b, dA-b):
        pw = powers(x, t, JMAX)
        for i, (k, j) in enumerate(B):
            U += cA[i]*(pw[j].real if k == 'R' else -pw[j].imag)
    return U
def fbox(b, t, sgn):
    # f on the closure of t > 0 (sgn = +1) or of t < 0 (sgn = -1), in the form regular at t = 0
    return sgn*PI-2*(t/(dA+b)).atan()-2*(t/(dA-b)).atan()+(t/(2*dA+b)).atan()+(t/(2*dA-b)).atan()
def Wlow(b, t):
    # lower bound of sum_k kappa_k W_k on the box (W_k >= 0, so dropping a term is safe)
    s = arb(0)
    for (cc, lR, v), k in zip(dsA, kA):
        val = lR-((cc-half-b)**2+(t+v)**2).log()/2
        lo = val.lower()
        if lo > 0: s += k*lo
    return s
def grad(b, t):
    # partial derivatives of U - f with respect to b and t, on a box
    gb = arb(0); gt = arb(0)
    for x, sg in ((dA+b, 1), (dA-b, -1)):
        pw = powers(x, t, JMAX+1)
        for i, (k, j) in enumerate(B):
            dx = -j*pw[j+1]; dt = acb(0, -j)*pw[j+1]                       # d/dx (x+it)^-j,  d/dt (x+it)^-j
            if k == 'R': gb += cA[i]*sg*dx.real; gt += cA[i]*dt.real
            else: gb -= cA[i]*sg*dx.imag; gt -= cA[i]*dt.imag
    y1 = dA+b; y2 = dA-b; y3 = 2*dA+b; y4 = 2*dA-b; t2 = t*t
    ft = -2*y1/(y1*y1+t2)-2*y2/(y2*y2+t2)+y3/(y3*y3+t2)+y4/(y4*y4+t2)
    fb = 2*t/(y1*y1+t2)-2*t/(y2*y2+t2)-t/(y3*y3+t2)+t/(y4*y4+t2)
    return gb-fb, gt-ft
def Hcentered(b0, b1, t0, t1, sgn):
    bm = (b0+b1)/2; tm = (t0+t1)/2; b = box(b0, b1); t = box(t0, t1)
    H0 = kernel(A(bm), A(tm))-fbox(A(bm), A(tm), sgn)
    gb, gt = grad(b, t)
    H = H0+gb*box(-(b1-b0)/2, (b1-b0)/2)+gt*box(-(t1-t0)/2, (t1-t0)/2)
    return H+Wlow(b, t)
def cert_main(blo, bhi, tlo, thi, sgn, maxdepth=60):
    stack = [(blo, bhi, tlo, thi, 0)]; n = 0
    while stack:
        b0, b1, t0, t1, dep = stack.pop(); n += 1
        H = Hcentered(b0, b1, t0, t1, sgn)
        if H > 0: continue
        if dep >= maxdepth: raise RuntimeError('fail at b=[%s,%s] t=[%s,%s] H=%s' % (float(b0), float(b1),
     float(t0), float(t1), H))
        wb = (b1-b0)/Fr(1, 20); tm = max(abs(t0), abs(t1)); wt = (t1-t0)/max(Fr(1, 20), tm*Fr(3, 10))
        if wb >= wt: m_ = (b0+b1)/2; stack += [(b0, m_, t0, t1, dep+1), (m_, b1, t0, t1, dep+1)]
        else: m_ = (t0+t1)/2; stack += [(b0, b1, t0, m_, dep+1), (b0, b1, m_, t1, dep+1)]
    return n
```

```
# tail |t| >= T1, u = 1/t:  t^2 (U - f) >= t^2 U - 4 d^3 u - K5 |u|^3,  with t^2 U = sum_iota c_iota [
    Phi_iota(d+b,u) + Phi_iota(d-b,u)],
# Phi_R1 = x/(1+x^2 u^2),  Phi_iota = Re / -Im of u^(j-2) (xu+i)^(-j) for j >= 2;  K5 = (1/5) sum_i |alpha_i|
     y_i^5 (Lemma "The tail of the kernel").
def tailval(b, u):
    V = arb(0)
    for x in (dA+b, dA-b):
        for i, (k, j) in enumerate(B):
            if j == 1: V += cA[i]*x/(1+x*x*u*u)
            else:
                z = acb(u)**(j-2)*(acb(1)/acb(x*u, 1))**j
                V += cA[i]*(z.real if k == 'R' else -z.imag)
    return V
K5 = (2*(dA+half)**5+2*dA**5+(2*dA+half)**5+(2*dA)**5)/5
def cert_tail(u0, u1, maxdepth=40):
    assert max(abs(u0), abs(u1))*(2*d+Fr(1, 2)) <= 1                  # |y u| <= 1 for every y in {d+b, d-b,
     2d+b, 2d-b}
    stack = [(Fr(0), Fr(1, 2), u0, u1, 0)]; n = 0
    while stack:
        b0, b1, a0, a1, dep = stack.pop(); n += 1
        b = box(b0, b1); u = box(a0, a1)
        H = tailval(b, u)-4*dA**3*u-K5*abs(u)**3
        if H > 0: continue
        if dep >= maxdepth: raise RuntimeError('tail fail b=[%s,%s] u=[%s,%s] H=%s' % (float(b0), float(b1),
    float(a0), float(a1), H))
        if (b1-b0)*8 >= (a1-a0)*T1: m_ = (b0+b1)/2; stack += [(b0, m_, a0, a1, dep+1), (m_, b1, a0, a1, dep
    +1)]
        else: m_ = (a0+a1)/2; stack += [(b0, b1, a0, m_, dep+1), (b0, b1, m_, a1, dep+1)]
    return n

def Jexact(nodes, cc, R):
    # (1/pi) int_0^pi e(cc + R cos th) dth for e piecewise linear through nodes, e = 0 for sigma >= 1; needs
     cc - R >= 1/2
    assert nodes[0][0] == Fr(1, 2) and nodes[-1] == (Fr(1), Fr(0)) and all(p[0] < q[0] for p, q in zip(nodes,
     nodes[1:]))
    tot = arb(0); ccA = A(cc); RA = A(R)
    lo, hi = cc-R, min(cc+R, Fr(1))
    for (p, ep), (q, eq) in zip(nodes, nodes[1:]):
        a, bq = max(p, lo), min(q, hi)
        if a >= bq: continue
        Bs = (eq-ep)/(q-p); As = ep-Bs*p                                  # e = As + Bs sigma on [p, q]
        tha = A((a-cc)/R).acos(); thb = A((bq-cc)/R).acos()               # tha >= thb
        tot += (A(As)+A(Bs)*ccA)*(tha-thb)+A(Bs)*RA*(tha.sin()-thb.sin())
    return tot/PI

if __name__ == '__main__':
    t_ = time.time()
    print(provenance(sys.argv[1], __file__, os.path.join(os.path.dirname(os.path.abspath(__file__)), '
    certutil.py'), *([os.path.join(os.path.dirname(os.path.abspath(__file__)), 'profile_published.py')] if
    sol['discs'] else [])))
    if discs:
        assert sol['prof'] == 'P', 'only the published-source profile e_P is supported'
        from profile_published import nodes_published
        nodes = nodes_published()
        J = [Jexact(nodes, cc, R) for cc, R, v in discs]
    else: J = []
    C = cA[0]/(2*PI)+sum((k*j for k, j in zip(kA, J)), arb(0))/PI
    print('d =', d, ' m =', JMAX, ' discs used:', len(discs), (' profile e_P (profile_published.py)' if discs
     else ''))
    print('C = c_R1/(2pi) + sum kappa_k J_k/pi =', C.str(25), ' (ball, midpoint and radius)')
    print('   radius of the ball:', C.rad().str(3), '  enclosure of C with directed rounding: [%s, %s]' % (
    down_str(C, 20), up_str(C, 20)))
    print('kernel part c_R1/(2pi) =', (cA[0]/(2*PI)).str(20))
    lim = 2*dA*cA[0]-2*cA[B.index(('R', 2))]; print('limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 =', lim.
    str(10), ' positive:', lim > 0)
```

```
    n1 = cert_main(Fr(0), Fr(1, 2), Fr(0), T1, +1); print('t in (0,T1]: certified, boxes', n1, flush=True)
    n2 = cert_main(Fr(0), Fr(1, 2), -T1, Fr(0), -1); print('t in [-T1,0): certified, boxes', n2, flush=True)
    n3 = cert_tail(Fr(0), 1/T1); n4 = cert_tail(-1/T1, Fr(0)); print('|t|>=T1: certified, boxes', n3+n4)
    Cup = up_str(C, 12); assert exact(C.upper()) <= Fr(Cup)
    print('all certified;  C <', Cup, '  time %.0fs' % (time.time()-t_))
```

### `profile_published.py`: the nodes of the profile $\mathfrak{e}_{\mathrm{P}}$

```
# Upper bound for mu(sigma) on [1/2,1] from published sources only, as a piecewise linear function through
    nodes
# (sigma_i, e_i) with mu(sigma_i) <= e_i a theorem in each case:
#  - 1/2: 13/84 (Bourgain 2017, Thm 5);
#  - breakpoints of Trudgian-Yang, Toward optimal exponent pairs, arXiv:2306.05599v3, Thm 2.4 (pieces are
    continuous);
#  - l-k for the exponent pairs (k,l) of Tao-Trudgian-Yang, arXiv:2501.16779v1, Thm 20, via Lemma 23 (mu(l-k)
     <= k);
#  - Heath-Brown 2017 (Proc. Steklov Inst. Math. 296), Thm 2: pairs (2/((n-1)^2(n+2)), 1-(3n-2)/(n(n-1)(n+2))
    +eps), n >= 3,
#    giving mu(sigma_n) <= mu_n, sigma_n = 1-(3n^2-3n+2)/(n(n-1)^2(n+2)), mu_n = 2/((n-1)^2(n+2)) (continuity
     of mu), 6 <= n <= 399;
#  - (1, 0) (zeta(1+it) << log t).
# The profile is the lower convex hull of these points (nodes above a chord of two others are dropped: the
    chord is a
# valid bound by the Phragmen-Lindelof interpolation lemma, and smaller).
from fractions import Fraction as Fr
TY = [(Fr(1,2),Fr(88225,153852),Fr(31,84),Fr(3,7)),(Fr(88225,153852),Fr(521,796),Fr(220633,620612),Fr
    (62831,155153)),
 (Fr(521,796),Fr(53141,76066),Fr(1333,3825),Fr(1508,3825)),(Fr(53141,76066),Fr(3620,5119),Fr(405,1202),Fr
    (227,601)),
 (Fr(3620,5119),Fr(52209,69128),Fr(49318855,170145110),Fr(52938216,170145110)),(Fr(52209,69128),Fr(1389,1736)
    ,Fr(471957,1682490),Fr(502648,1682490)),
 (Fr(1389,1736),Fr(134765,163248),Fr(2841,10316),Fr(3016,10316)),(Fr(134765,163248),Fr(18193,21906),Fr
    (859,3214),Fr(908,3214)),
 (Fr(18193,21906),Fr(249,280),Fr(5*8707,180277),Fr(5*9067,180277)),(Fr(249,280),Fr(277,300),Fr(29,130),Fr
    (30,130))]
PAIRS = [(Fr(89,1282),Fr(997,1282)),(Fr(652397,9713986),Fr(7599781,9713986)),(Fr(10769,351096),Fr
    (609317,702192)),(Fr(89,3478),Fr(15327,17390))]
def hb(n): return 1-Fr(3*n*n-3*n+2, n*(n-1)**2*(n+2)), Fr(2, (n-1)**2*(n+2))
def raw_nodes(nmax=400):
    for (l1,h1,a1,b1),(l2,h2,a2,b2) in zip(TY,TY[1:]): assert a1-b1*h1 == a2-b2*h1
    nodes = {}
    for lo,hi,a,b in TY: nodes[lo] = a-b*lo; nodes[hi] = a-b*hi
    for k,l in PAIRS: nodes[l-k] = min(nodes.get(l-k, Fr(1)), k)
    for n in range(6, nmax): s,m = hb(n); nodes[s] = min(nodes.get(s, Fr(1)), m)
    nodes[Fr(1)] = Fr(0)
    return sorted(nodes.items())
def nodes_published(nmax=400):
    P = raw_nodes(nmax); hull = []
    def cross(o,a,b): return (a[0]-o[0])*(b[1]-o[1])-(a[1]-o[1])*(b[0]-o[0])
    for p in P:
        while len(hull) >= 2 and cross(hull[-2], hull[-1], p) <= 0: hull.pop()
        hull.append(p)
    return hull
if __name__ == '__main__':
    H = nodes_published()
    print(len(H), 'nodes; first ones:')
    for s,e in H[:14]: print('  sigma = %-20s  e = %-18s  (%.6f, %.6f)' % (s, e, s, e))
```

### `cz_gen_pp.py`: certification of $C_\zeta$ (per-prime phase maximum)

```
# Rigorous upper bound (Arb ball arithmetic, 256 bits) for the Euler-product constant of a kernel,
#   C_zeta = sum_{p <= P0} M_p + sum_{n = p^k, p > P0} Lambda(n) n^(-sigma1) h(log n) + tail,
# with h(L) = sqrt(a(L)^2 + b(L)^2),  a(L) = -(1/pi) sum_{(R,j)} c_{R,j} L^(j-1)/(j-1)!,
```

```
# b(L) = (1/pi) [ (e^(-dL) - 2)/L - sum_{(I,j)} c_{I,j} L^(j-1)/(j-1)! ],  and M_p the maximum over the phase x of
#   f_p(x) = sum_{k: p^k <= N0} (log p) p^(-k sigma1) [ a(k log p) cos(kx) + b(k log p) sin(kx) ]
# (per-prime phase maximum, Lemma "The contribution of one prime"), bounded by the maximum on a grid of M points plus a
# Lipschitz correction.  The sum over the prime powers n <= N0 is exact; the tail n > N0 uses Lemma "The tail" with the
# non-increasing step majorant G of g(x) = h(log x) x^(-sigma1) on [N0, e^LB], the certified monotonicity of g beyond e^LB
# (Lemma "Monotonicity of g") and psi(x) < 1.03883 x (Rosser-Schoenfeld 1962, Thm 12).
#
# Usage:  python3 cz_gen_pp.py <solution.json> <N0> [P0]      (the article uses N0 = 1e8, P0 = 1000)
# Output: <solution>_czpp.json with the upper bound 'Czeta_upper' (decimal rounded upwards).
import sys, os, json, time, numpy as np
from fractions import Fraction as Fr
from math import factorial
from flint import arb, acb
sys.path.append(os.path.join(os.path.dirname(os.path.abspath(__file__)), '..', 'hybrid_asymptotic'))   # certutil.py lives there (or next to this file in the flat archive)
from certutil import A, up_str, provenance

sol = json.load(open(sys.argv[1])); N0 = int(float(sys.argv[2])); P0 = int(sys.argv[3]) if len(sys.argv) > 3 else 1000
print(provenance(sys.argv[1], __file__, os.path.join(os.path.dirname(os.path.abspath(__file__)), '..', 'hybrid_asymptotic', 'certutil.py') if not os.path.exists(os.path.join(os.path.dirname(os.path.abspath(__file__)), 'certutil.py')) else os.path.join(os.path.dirname(os.path.abspath(__file__)), 'certutil.py')), flush=True)
d = Fr(sol['d']).limit_denominator(10**6); B = [tuple(b) for b in sol['B']]; c = [A(Fr(x)) for x in sol['c']]
assert len(c) == len(B) and B[0] == ('R', 1) and ('I', 1) not in B and d > Fr(1, 2) and P0 <= N0
PI = arb.pi(); dA = A(d); s1 = A(Fr(1, 2)+d); alpha = s1-1                 # alpha = sigma1 - 1 > 0
# polynomial coefficients (in L) of pi*a and of the polynomial part of pi*b
DEG = max(j for _, j in B)-1
pa = [arb(0)]*(DEG+1); pb = [arb(0)]*(DEG+1)
for (k, j), cc in zip(B, c):
    if k == 'R': pa[j-1] -= cc/factorial(j-1)
    else: pb[j-1] -= cc/factorial(j-1)
def poly(p, L): return sum((p[i]*L**i for i in range(len(p))), arb(0))
def h(L):
    a0 = poly(pa, L)/PI; b0 = (((-dA*L).exp()-2)/L+poly(pb, L))/PI
    return (a0*a0+b0*b0).sqrt()

t0 = time.time()
# primes up to N0 (sieve of Eratosthenes)
s = np.ones(N0+1, np.bool_); s[:2] = False
for i in range(2, int(N0**0.5)+1):
    if s[i]: s[i*i::i] = False
P = np.nonzero(s)[0]; del s
# exact sum over the prime powers n <= N0, with the per-prime phase maximum for p <= P0
M = 4096
xs = [2*PI*i/M for i in range(M)]; hx = PI/M                               # grid points and half-spacing
tot = arb(0); psi = arb(0); gain = arb(0)
for p in P.tolist():
    lp = arb(p).log(); n = p; k = 1; coef = []; hyp = arb(0)
    while n <= N0:
        L = k*lp; w = lp*(-s1*L).exp(); a0 = poly(pa, L)/PI; b0 = (((-dA*L).exp()-2)/L+poly(pb, L))/PI
        coef.append((k, w*a0, w*b0)); hyp += w*(a0*a0+b0*b0).sqrt(); psi += lp; n *= p; k += 1
    if p <= P0:
        # max_x f_p(x) <= max_i f_p(x_i) + hx * sum_k k |(a_k, b_k)|   (Lipschitz constant of f_p)
        Lip = sum((kk*(a*a+b*b).sqrt() for kk, a, b in coef), arb(0))
        best = None
        for x in xs:
            v = sum((a*(kk*x).cos()+b*(kk*x).sin() for kk, a, b in coef), arb(0))
            best = v.upper() if best is None or v.upper() > best else best
        mb = best+hx*Lip
```

```
        mb = mb if mb < hyp else hyp                                    # the trivial bound sum_k |(a_k,
    b_k)| is always valid
        gain += hyp-mb; tot += mb
    else:
        tot += hyp
print('per-prime phase maximum for p <=', P0, ': gain', gain.str(8), flush=True)
print('primes', len(P), ' partial sum (n<=%d) =' % N0, tot.str(15), ' psi(N0) =', psi.str(15), ' %.0fs' % (
    time.time()-t0), flush=True)

# monotonicity of g on [e^LB, oo) (Lemma "Monotonicity of g"): in u = 1/L in [0, 1/LB], num(u)/low(u) <
    sigma1 with
#   num(u) = u sum_{i>=1} i q_i u^(D-i) + 2 u^(D+2) + d u^(D+1),   low(u) = |p_D| - sum_{i<D} |p_i| u^(D-i) -
     eps_P u^(D+1),
# where P is the one of pi*a, pi*b of larger degree D (pi*a if equal), q_i = |a_i| + |b_i|, eps_P = 2 if P =
    pi*b else 0.
L0 = arb(N0).log()
DA = max([i for i in range(DEG+1) if pa[i] != 0], default=-1); DB = max([i for i in range(DEG+1) if pb[i] !=
    0], default=-1)
q = [abs(pa[i])+abs(pb[i]) for i in range(DEG+1)]
if DA >= DB: D, pp, extra = DA, pa, 0
else: D, pp, extra = DB, pb, 2
assert D == DEG, 'the top coefficient of the kernel vanishes'
LB = None
for LBc in [2000.0, 5000.0, 2e4, 1e5, 1e6, 1e7]:
    u = arb(0).union(1/arb(LBc))
    num = u*sum((i*q[i]*u**(D-i) for i in range(1, DEG+1)), arb(0))+2*u**(D+2)+dA*u**(D+1)
    low = abs(pp[D])-sum((abs(pp[i])*u**(D-i) for i in range(D)), arb(0))-extra*u**(D+1)
    if low > 0 and num/low < s1: LB = LBc; ratio = num/low; break
assert LB is not None, 'monotonicity of g could not be certified'
print('g monotone on [%g, oo): crude bound %s < sigma1' % (LB, ratio.str(6)))

# step majorant G >= g on [N0, e^LB]: on [e^u_i, e^u_{i+1}] the running maximum, from the right, of upper
    bounds of g on
# the small boxes, started at g(e^LB) (g is non-increasing beyond e^LB), so that G is non-increasing on [N0,
    oo)
def cpoly(p, Lb):   # centred form of a polynomial on a ball: p(m) + p'(Lb)(Lb - m)
    m = arb(Lb.mid()); dp = [p[i]*i for i in range(1, len(p))]
    return poly(p, m)+poly(dp, Lb)*(Lb-m)
def h_up(Lb):
    a0 = cpoly(pa, Lb)/PI; b0 = (((-dA*Lb).exp()-2)/Lb+cpoly(pb, Lb))/PI
    au = abs(a0).upper(); bu = abs(b0).upper()
    return (au*au+bu*bu).sqrt().upper()
edges = [float(L0.lower())-1e-9]
while edges[-1] < LB: edges.append(min(edges[-1]+max(0.003, 0.0002*edges[-1]), LB))
gup = []
for i in range(len(edges)-1):
    Lb = arb(edges[i]).union(arb(edges[i+1])); gup.append(h_up(Lb)*(-s1*arb(edges[i])).exp())
run = h(arb(LB))*(-s1*arb(LB)).exp(); G = [None]*len(gup)
for i in range(len(gup)-1, -1, -1):
    run = run if run > gup[i] else (gup[i] if gup[i] > run else run.union(gup[i]).upper()); G[i] = run.upper
    ()
Istep = sum((G[i]*(arb(edges[i+1]).exp()-arb(edges[i]).exp()) for i in range(len(G))), arb(0))   # int_{N0}^{
    e^LB} G dx

# int_{e^LB}^oo g dx  <=  int_{e^LB}^{e^L1} g dx + int_{e^L1}^oo gtilde dx   with L1 = 800 and the majorant
#   gtilde = (1/pi) (sum_i q_i L^i + 3) e^(-sigma1 L) >= g on [e^L1, oo)   (from h <= |a| + |b| and 2/L <= 3)
    ;
# when LB > L1 the first integral is negative, and the inequality holds because gtilde >= g on [e^L1, e^LB]
    as well.
L1 = arb(800)
assert LB >= 800
def fz(z, analytic):
    a0 = sum((acb(pa[i])*z**i for i in range(len(pa))), acb(0))/PI
    b0 = (((-acb(dA)*z).exp()-2)/z+sum((acb(pb[i])*z**i for i in range(len(pb))), acb(0)))/PI
    return (a0*a0+b0*b0).sqrt(analytic=analytic)*((1-acb(s1))*z).exp()
```

```
Icancel = acb.integral(fz, acb(L1), acb(LB))                                    # int_{L1}^{LB} h(L) e^{(1-sigma1)
     L} dL = int_{e^L1}^{e^LB} g dx
assert Icancel.is_finite() and Icancel.imag.contains(0)
def inc(k, X): return (-alpha*X).exp()*sum((arb(factorial(k))/factorial(i)*X**i/alpha**(k-i+1) for i in range
    (k+1)), arb(0))   # int_X^oo L^k e^{-alpha L} dL
Ifar = (sum((q[i]*inc(i, L1) for i in range(DEG+1)), arb(0))+3*inc(0, L1))/PI   # int_{e^L1}^oo gtilde dx
Itail = Istep-Icancel.real+Ifar                                                 # >= int_{N0}^oo G dx
# Lemma "The tail": sum_{n > N0} Lambda(n) g(n) <= G(N0) (1.03883 N0 - psi(N0)) + 1.03883 int_{N0}^oo G dx
tail = G[0]*(arb('1.03883')*N0-psi)+arb('1.03883')*Itail
Cz = tot+tail
print('tail <=', tail.str(10), ' (int_{N0}^{e^LB} G =', Istep.str(10), ', int_{e^L1}^{e^LB} g =', Icancel.
    real.str(10), ', far tail <=', Ifar.str(5), ')')
print('C_zeta <', up_str(Cz, 12), '   time %.0fs' % (time.time()-t0))
json.dump({'Czeta_upper': up_str(Cz, 15), 'N0': N0, 'P0': P0}, open(sys.argv[1].replace('.json', '_czpp.json
    '), 'w'))
```

## `elem_gen.py`: certification of $\sup F_u$, $\sup F_\ell$ (modes N and S)

```
# Rigorous upper bound (Arb ball arithmetic, 256 bits) for sup_{T >= 1} F_u(T) and sup_{T >= 1} F_l(T), where
#   F_u(T) = E_u(T) - A1 log T,   F_l(T) = -E_l(T) - A1 log T,
#   E_u(T) = (1/pi)(2 Ael(s1) - Ael(s2)) - main(T) + (1/pi) sum_iota c_iota Phi_iota(D_j(s1)),
#   E_l(T) = (1/pi)(2 Ael(s1) - Ael(s2)) - main(T) - (1/pi) [ sum_{(R,j)} c Re D_j(s1) - sum_{(I,j)} c (-Im
    D_j(s1)) ],
#   Ael(s) = arg s + arg(s-1) - (T/2) log pi + Im log Gamma(s/2),   main(T) = (T/2pi) log(T/(2 pi e)),
#   D_1(s) = 1/s + 1/(s-1) - (1/2) log pi + (1/2) psi(s/2),   D_j(s) = s^-j + (s-1)^-j + ((-1)^(j-1)/(j-1)!)
    2^-j psi^(j-1)(s/2) (j >= 2),
#   s1 = 1/2 + d + iT,  s2 = 1/2 + 2d + iT,  Phi_R = Re,  Phi_I = -Im.
# Mode S replaces main(T) by theta(T)/pi + 1 (then the bounds are for |S(T)|).
# [1, T0]: centred form F(mid) + F'(box) * [-half-width, half-width] on geometric intervals of ratio
    1001/1000;
# T >= T0: Lemma "The elementary part for large T": F_u <= 7/8 - C1 log(2 pi) + K/T, F_l <= -7/8 - C1 log(2
    pi) + K/T
# (mode S: -C1 log(2 pi) + (K + (1/16 + 1/8 + 1/3)/pi)/T), with the constant K of that lemma; needs A1 >= C1
    = c_R1/(2 pi).
#
# Usage:  python3 elem_gen.py <solution.json> <A1> [T0] [N|S]      (the article uses T0 = 10000)
# Output: <solution>_elem.json or <solution>_elemS.json with 'Fsup_upper' (decimal rounded upwards).
import sys, os, json, time
from fractions import Fraction as Fr
from math import factorial
from flint import arb, acb
sys.path.append(os.path.join(os.path.dirname(os.path.abspath(__file__)), '..', 'hybrid_asymptotic'))   #
    certutil.py lives there (or next to this file in the flat archive)
from certutil import A, up_str, provenance

sol = json.load(open(sys.argv[1])); A1 = A(Fr(sys.argv[2])); T0 = int(sys.argv[3]) if len(sys.argv) > 3 else
    10**4
MODE = sys.argv[4] if len(sys.argv) > 4 else 'N'
print(provenance(sys.argv[1], __file__, os.path.join(os.path.dirname(os.path.abspath(__file__)), '..', '
    hybrid_asymptotic', 'certutil.py') if not os.path.exists(os.path.join(os.path.dirname(os.path.abspath(
    __file__)), 'certutil.py')) else os.path.join(os.path.dirname(os.path.abspath(__file__)), 'certutil.py')
    ), flush=True)
assert MODE in ('N', 'S') and T0 >= 1
d = Fr(sol['d']).limit_denominator(10**6); B = [tuple(b) for b in sol['B']]; c = [A(Fr(x)) for x in sol['c']]
assert len(c) == len(B) and B[0] == ('R', 1) and ('I', 1) not in B and d > Fr(1, 2)
J = max(j for _, j in B)
PI = arb.pi(); s1r = A(Fr(1, 2)+d); s2r = A(Fr(1, 2)+2*d); LOGPI = PI.log()
cR1 = c[0]; C1 = cR1/(2*PI)
assert A1 >= C1, 'A1 must be >= c_R1/(2 pi)'

def psi(k, z): return z.polygamma(k)
def Dj(j, s, ps):   # ps[k] = psi^(k)(s/2)
    if j == 1: return 1/s+1/(s-1)-LOGPI/2+ps[0]/2
    return s**(-j)+(s-1)**(-j)+arb((-1)**(j-1))/factorial(j-1)/2**j*ps[j-1]
```

```
def dDj(j, s, ps):  # d/ds D_j(s)
    if j == 1: return -s**-2-(s-1)**-2+ps[1]/4
    return -j*s**(-j-1)-j*(s-1)**(-j-1)+arb((-1)**(j-1))/factorial(j-1)/2**(j+1)*ps[j]
def combine(vals):  # sum_{(R,j)} c Re(.) and sum_{(I,j)} c (-Im(.))
    R = sum((cc*vals[i].real for i, ((k, j), cc) in enumerate(zip(B, c)) if k == 'R'), arb(0))
    I = sum((cc*(-vals[i].imag) for i, ((k, j), cc) in enumerate(zip(B, c)) if k == 'I'), arb(0))
    return R, I
def F_at(T):
    # (F_u(T), F_l(T)) at an exact T
    s1 = acb(s1r, T); s2 = acb(s2r, T)
    Ael = lambda s: (s.log().imag+(s-1).log().imag)-T/2*LOGPI+(s/2).lgamma().imag
    base = (2*Ael(s1)-Ael(s2))/PI-T/(2*PI)*(T/(2*PI*arb(1).exp())).log()
    if MODE == 'S':   # replace main(T) by theta(T)/pi + 1
        theta = acb(arb(1)/4, T/2).lgamma().imag-T/2*LOGPI
        base = base+T/(2*PI)*(T/(2*PI*arb(1).exp())).log()-theta/PI-1
    ps = [psi(k, s1/2) for k in range(J)]
    R, I = combine([Dj(j, s1, ps) for (_, j) in B])
    return base+(R+I)/PI-A1*T.log(), -(base-(R-I)/PI)-A1*T.log()
def dF_ball(Tb):
    # (F_u'(T), F_l'(T)) on a ball Tb of heights; the polygammas on the ball by the mean value form psi^(k)(
     z_m) + psi^(k+1)(z_ball) (z - z_m)
    Tm = arb(Tb.mid()); rad = Tb-Tm
    def P(k, sr):
        zm = acb(sr, Tm)/2; zb = acb(sr, Tb)/2
        return psi(k, zm)+psi(k+1, zb)*acb(0, rad/2)
    s1 = acb(s1r, Tb); s2 = acb(s2r, Tb)
    dAel = lambda s, sr: ((1/s).real+(1/(s-1)).real)-LOGPI/2+P(0, sr).real/2      # d/dT Ael(sr + iT)
    dbase = (2*dAel(s1, s1r)-dAel(s2, s2r))/PI-(Tb/(2*PI)).log()/(2*PI)
    if MODE == 'S':   # d/dT [main - theta/pi] = (1/2pi) log(T/2pi) - (1/2pi) Re psi(1/4+iT/2) + (1/2pi) log
     pi
        zm = acb(arb(1)/4, Tm/2); zb = acb(arb(1)/4, Tb/2)
        P0 = psi(0, zm)+psi(1, zb)*acb(0, rad/2)
        dbase = dbase+(Tb/(2*PI)).log()/(2*PI)-(P0.real-LOGPI)/(2*PI)
    ps = [P(k, s1r) for k in range(J+1)]
    R, I = combine([acb(0, 1)*dDj(j, s1, ps) for (_, j) in B])              # d/dT = i d/ds
    return dbase+(R+I)/PI-A1/Tb, -(dbase-(R-I)/PI)-A1/Tb

# [1, T0]
t0 = time.time(); ratio = Fr(1001, 1000); a = Fr(1); n = 0; supu = None; supl = None
while a < T0:
    b = min(a*ratio, Fr(T0)); m = (a+b)/2; dl = (b-a)/2
    fu, fl = F_at(A(m)); gu, gl = dF_ball(A(a).union(A(b)))
    eu = (fu+gu*A(dl)*arb(0, 1)).upper(); el = (fl+gl*A(dl)*arb(0, 1)).upper()
    supu = eu if supu is None or eu > supu else supu; supl = el if supl is None or el > supl else supl
    a = b; n += 1
print('intervals', n, ' sup F_u on [1,T0] <=', up_str(supu, 10), '  sup F_l <=', up_str(supl, 10))
# T >= T0 (Lemma "The elementary part for large T")
T = A(Fr(T0)); s1 = s1r; s2 = s2r
K = (2*s2-1)/PI+(s1*s1/2+(s2-1)*s2/2+1)/PI+(2*s2*s2/4+(s1-1)*s1)/PI        # arguments and Gamma factors
K += abs(cR1)/PI*((2*s1-1)+s1*s1/4+arb(8)/3)/T                             # j = 1: |delta_1| <= B/T^2 <= B/(
     T0 T)
for (k, j), cc in zip(B, c):
    if j == 1: continue
    # |D_j(s1)| <= 2/T^j + |psi^(j-1)(z1)| / ((j-1)! 2^j),  |psi^(k)(z1)| <= (k-1)! (2/T)^k + k! (2/T)^(k+1)
     /2 + (k+1)! (4/(3T)) (4/T)^(k+1);
    # T times the bound is non-increasing in T, so the bound at T0 times T0/T is valid for T >= T0
    kk = j-1
    ps = arb(factorial(kk-1))*(2/T)**kk+arb(factorial(kk))*(2/T)**(kk+1)/2+arb(factorial(kk+1))*(4/(3*T))*(4/
     T)**(kk+1)
    Dbound = 2/T**j+ps/(factorial(j-1)*2**j)
    K += abs(cc)/PI*Dbound*T
Finf_u = arb(7)/8-C1*(2*PI).log(); Finf_l = -arb(7)/8-C1*(2*PI).log()
if MODE == 'S':   # theta/pi + 1 - main = 7/8 + varpi/pi, |varpi(T)| <= (1/16 + 1/8 + 1/3)/T (Lemma "The
     theta function")
    Finf_u = -C1*(2*PI).log(); Finf_l = -C1*(2*PI).log(); K = K+(arb(1)/16+arb(1)/8+arb(1)/3)/PI
```

```
tu = (Finf_u+K/T).upper(); tl = (Finf_l+K/T).upper()
S = max(supu, supl, tu, tl)
print('T>=T0: F_u <=', up_str(tu, 10), ' F_l <=', up_str(tl, 10), ' (K =', K.str(6), ')')
print('SUP over T>=1 of max(F_u,F_l) <=', up_str(S, 12), '   time %.0fs' % (time.time()-t0))
json.dump({'A1': sys.argv[2], 'T0': T0, 'mode': MODE, 'Fsup_upper': up_str(S, 15)}, open(sys.argv[1].replace
    ('.json', '_elem%s.json' % ('' if MODE == 'N' else 'S')), 'w'))
```

### certify_line_pp.py: driver for one explicit line

```
# Driver: certify one explicit line |N(T) - main(T)| <= A1 log T + A3_N and |S(T)| <= A1 log T + A3_S for all
     T >= 1.
#   1. certify_hybrid.py (no discs): the kernel inequality f <= U, with T1 = 100 (T1 = 2000 if that run fails
    );
#   2. cz_gen_pp.py: C_zeta < Czeta_upper  (N0 = 1e8, per-prime phase maximum for p <= 1000);
#   3. elem_gen.py, modes N and S: sup_{T>=1} max(F_u, F_l) < Fsup_upper  (T0 = 1e4);
#   4. A1 = C1 rounded upwards to 5 decimals, A3 = Czeta_upper + Fsup_upper rounded upwards to 3 decimals (
    exact decimal arithmetic).
# Usage:  python3 certify_line_pp.py <tag> <solution.json>                  (writes line_<tag>/sol.json and
    the certificates there)
#         python3 certify_line_pp.py <tag> <L*> [sweep*.jsonl pattern]      (picks the best sweep candidate
    for the target height L*)
import json, sys, os, subprocess, glob
from decimal import Decimal, ROUND_CEILING, localcontext
HERE = os.path.dirname(os.path.abspath(__file__))
def locate(name):   # repository layout (../hybrid_asymptotic/) or flat archive (same folder)
    for p in (os.path.join(HERE, name), os.path.join(HERE, '..', 'hybrid_asymptotic', name)):
        if os.path.exists(p): return p
    raise FileNotFoundError(name)

tag = sys.argv[1]; src = sys.argv[2]; pat = sys.argv[3] if len(sys.argv) > 3 else 'sweep*.jsonl'
if src.endswith('.json'): r = json.load(open(src))
else:
    L = float(src); rs = [json.loads(l) for f in glob.glob(pat) for l in open(f)]
    r = min((x for x in rs if x['Lstar'] == L), key=lambda x: x['val'])
D = 'line_'+tag; os.makedirs(D, exist_ok=True); sol = D+'/sol.json'; json.dump(r, open(sol, 'w'))
A1s = str(Decimal(repr(r['C1'])).quantize(Decimal('0.00001'), rounding=ROUND_CEILING))   # elem_gen.py checks
     A1 >= C1 exactly
print('candidate', tag, 'd =', r['d'], 'C1 =', r['C1'], 'A3 (estimate) =', r['A3_est'], '-> A1 =', A1s, flush
    =True)

def run(cmd, log):
    with open(D+'/'+log, 'w') as fh: p = subprocess.run(cmd, stdout=fh, stderr=subprocess.STDOUT)
    out = open(D+'/'+log).read(); lines = out.strip().splitlines(); print(lines[-1] if lines else '(no output
    )', flush=True)
    return p.returncode, out
rc1, o1 = run(['python3', locate('certify_hybrid.py'), sol, '100'], 'cert_kernel.txt')
if not (rc1 == 0 and 'all certified' in o1): rc1, o1 = run(['python3', locate('certify_hybrid.py'), sol,
    '2000'], 'cert_kernel.txt')
rc2, o2 = run(['python3', os.path.join(HERE, 'cz_gen_pp.py'), sol, '1e8', '1000'], 'cert_czeta_pp.txt')
rc3, o3 = run(['python3', os.path.join(HERE, 'elem_gen.py'), sol, A1s, '10000', 'N'], 'cert_elem.txt')
rc4, o4 = run(['python3', os.path.join(HERE, 'elem_gen.py'), sol, A1s, '10000', 'S'], 'cert_elem_S.txt')

def finish():
    # the certified upper bounds are decimal strings rounded upwards; add them in exact decimal arithmetic
    and round upwards
    with localcontext() as dc:
        dc.prec = 60
        cz = Decimal(json.load(open(sol.replace('.json', '_czpp.json')))['Czeta_upper'])
        fN = Decimal(json.load(open(sol.replace('.json', '_elem.json')))['Fsup_upper'])
        fS = Decimal(json.load(open(sol.replace('.json', '_elemS.json')))['Fsup_upper'])
        AN = (cz+fN).quantize(Decimal('0.001'), rounding=ROUND_CEILING); AS = (cz+fS).quantize(Decimal
    ('0.001'), rounding=ROUND_CEILING)
    json.dump({'A1': A1s, 'A3_N': str(AN), 'A3_S': str(AS), 'Czeta_pp_upper': str(cz), 'Fsup_N': str(fN), '
    Fsup_S': str(fS),
```

```
            'd': r['d'], 'B': r['B'], 'Lstar': r['Lstar']}, open(D+'/summary_pp.json', 'w'), indent=1)
    print('LINE %s CERTIFIED: |N(T)-main(T)| <= %s log T + %s ;  |S(T)| <= %s log T + %s   (T >= 1)' % (tag,
     A1s, AN, A1s, AS), flush=True)
if rc1 == 0 and 'all certified' in o1 and rc2 == 0 and rc3 == 0 and rc4 == 0: finish()
else: print('LINE %s FAILED' % tag, flush=True); sys.exit(1)
```

## Search programs

### lpcommon.py: helpers of the search programs

```
# Floating-point helpers shared by the search programs (explore_discs.py, kernel_only.py, opt_line.py).
# Nothing here is used by the certification programs.
import numpy as np
from fractions import Fraction as Fr

def f(b, t, d):
    # f(b,t) = 2 atan((d+b)/t) + 2 atan((d-b)/t) - atan((2d+b)/t) - atan((2d-b)/t)
    return 2*np.arctan((b+d)/t)+2*np.arctan((d-b)/t)-np.arctan((b+2*d)/t)-np.arctan((2*d-b)/t)

def F(x, t, k, j):
    # F_Rj = Re (x+it)^-j,  F_Ij = -Im (x+it)^-j
    z = (x+1j*t)**(-j); return z.real if k == 'R' else -z.imag

def basis(m): return [('R', j) for j in range(1, m+1)]+[('I', j) for j in range(2, m+1)]

def tgrid(tmax, ngeom=700, nlin=400):
    # symmetric grid of heights t: geometric on [1e-4, tmax] plus a linear part on (0, 4], and a point near 0
    p = np.unique(np.concatenate([[1e-9], np.geomspace(1e-4, tmax, ngeom), np.linspace(0, 4, nlin)[1:]]));
     return np.concatenate([-p[::-1], p])

def bourgain_nodes():
    # piecewise linear profile through (lambda_j - kappa_j, kappa_j), (kappa_j, lambda_j) = A^j (13/84,
     55/84), j = 0..7, and (1, 0)
    k, l = Fr(13, 84), Fr(55, 84); nodes = []
    for _ in range(8):
        nodes.append((l-k, k)); k, l = k/(2*k+2), Fr(1, 2)+l/(2*k+2)
    nodes.append((Fr(1), Fr(0))); return nodes

def J_float(nodes, c, R, n=40001):
    # (1/pi) int_0^pi e(c + R cos th) dth, e piecewise linear through nodes on [1/2, 1], e = 0 for sigma >= 1
      (trapezoidal rule)
    xs = np.array([float(a) for a, _ in nodes]); es = np.array([float(e) for _, e in nodes])
    th = np.linspace(0, np.pi, n); s = c+R*np.cos(th); v = np.where(s >= 1, 0, np.interp(s, xs, es))
    return np.trapezoid(v, th)/np.pi
```

### explore_discs.py: the hybrid linear programme

```
# Search program (not a certificate): the hybrid of the kernel majorant with Jensen discs.
# Linear programme in (c_iota, kappa_k):
#   minimise  c_R1/(2 pi) + (1/pi) sum_k kappa_k J_k
#   subject to  f(b,t) + eta/(1+t^2) <= U(b,t) + sum_k kappa_k W_k(b,t)  on a grid of (b,t), refined by
     cutting planes,
#               2 d c_R1 - 2 c_R2 >= eta  (limit of t^2 U at |t| -> oo),  kappa_k >= 0,
# with J_k = (1/pi) int_0^pi e(c_k + R_k cos th) dth for the profile e (PROF=P: e_P of profile_published.py;
     PROF=B: the
# Bourgain profile of lpcommon.bourgain_nodes).  Candidate discs: the family 'vwide' described in the article
     .
# Usage:  PROF=P python3 explore_discs.py <m> <d> [eta]      -> expl_m<m>_d<d>_<PROF>_eta<eta>.json (input of
      certify_hybrid.py)
import numpy as np, sys, os, json, time, math
from scipy.optimize import linprog, minimize
from scipy import sparse
from lpcommon import f, F, basis, tgrid, bourgain_nodes, J_float
```

```
PROF = os.environ.get('PROF', 'P')
if PROF == 'P':
    from profile_published import nodes_published; nodes = nodes_published()
elif PROF == 'B': nodes = bourgain_nodes()
else: raise SystemExit('PROF must be P or B')

def Wmat(discs, bb, tt):
    # sparse matrix of W_k(b,t) = log^+(R_k / sqrt((c_k - 1/2 - b)^2 + (t + v_k)^2))
    rows = []; cols = []; vals = []
    for k, (c, R, v) in enumerate(discs):
        dist = np.sqrt((c-0.5-bb)**2+(tt+v)**2)
        w = np.log(R/np.maximum(dist, 1e-300)); i = np.nonzero(w > 0)[0]
        rows.append(i); cols.append(np.full(len(i), k)); vals.append(w[i])
    r = np.concatenate(rows) if rows else np.zeros(0, int)
    return sparse.csr_matrix((np.concatenate(vals) if vals else [], (r, np.concatenate(cols) if cols else [])
    ), shape=(len(bb), len(discs)))
def Gmat(d, B, bb, tt): return np.array([F(d+bb, tt, k, j)+F(d-bb, tt, k, j) for (k, j) in B]).T

def solve(d, m, discs, eta=0.0, maxit=30, nb=26):
    B = basis(m); nB = len(B)
    sc = np.array([(d-0.5)**j for (_, j) in B])                # scaling of the coefficients, for
    conditioning
    J = np.array([J_float(nodes, c, R) for c, R, v in discs])
    obj = np.concatenate([[sc[0]/(2*np.pi)], np.zeros(nB-1), J/np.pi])
    bb, tt = [a.ravel() for a in np.meshgrid(np.linspace(0, 0.5, nb), tgrid(300., 500, 300), indexing='ij')]
    def block(bb, tt): return sparse.hstack([sparse.csr_matrix(Gmat(d, B, bb, tt)*sc), Wmat(discs, bb, tt)]).
    tocsr(), f(bb, tt, d)+eta/(1+tt**2)
    A, Fv = block(bb, tt)
    arow = np.zeros(nB+len(discs)); arow[0] = 2*d*sc[0]; arow[1] = -2*sc[1]
    bf = np.linspace(0, 0.5, 101); tp = np.unique(np.concatenate([np.geomspace(1e-7, 3000., 6000), np.
    linspace(1e-4, 3., 30000)])); tf = np.concatenate([-tp[::-1], tp])
    def violation(x, used, du, bb, tt):   # f + margin - U - sum kappa W at the points (bb, tt); positive =
    violated
        v = f(bb, tt, d)+eta/(1+tt**2)-Gmat(d, B, bb, tt)@(x[:nB]*sc)
        if len(used): v = v-Wmat(du, bb, tt)@x[nB+used]
        return v
    for it in range(maxit):
        for method in ('highs', 'highs-ds', 'highs-ipm'):   # the dual simplex and the interior point method
    are tried if HiGHS' default fails
            r = linprog(obj, A_ub=-sparse.vstack([A, sparse.csr_matrix(arow)]), b_ub=-np.concatenate([Fv, [
    eta]]),
                        bounds=[(None, None)]*nB+[(0, None)]*len(discs), method=method)
            if r.status == 0: break
        assert r.status == 0, 'LP solver: '+r.message
        x = r.x; used = np.nonzero(x[nB:] > 1e-12)[0]; du = [discs[k] for k in used]
        nbl = []; ntl = []; worst = -1; cand = []
        for b in bf:   # violations on a finer grid -> new cutting planes (local maxima of the violation
    along t)
            v = violation(x, used, du, np.full_like(tf, b), tf)
            worst = max(worst, v.max()); idx = np.nonzero((v > 0) & (v >= np.roll(v, 1)) & (v >= np.roll(v,
    -1)))[0]
            nbl += [b]*len(idx); ntl += list(tf[idx])
            top = np.argsort(v)[-3:]; cand += [(v[i], b, tf[i]) for i in top]
        # the grid can miss narrow dips of U + sum kappa W - f (near the edges of the discs): refine the
    largest grid values
        # of the violation, including negative ones, by a local maximisation in (b, t), and add the maxima
    found as cutting planes
        cand.sort(reverse=True)
        for v0, b0, t0 in cand[:40]:
            if t0 == 0: continue
            g = lambda p: -violation(x, used, du, np.array([min(max(p[0], 0.0), 0.5)]), np.array([p[1] if p
    [1]*t0 > 0 else t0]))[0]
            res = minimize(g, [b0, t0], method='Nelder-Mead', options={'xatol': 1e-7, 'fatol': 1e-12, '
    maxiter': 400})
            bb_, tt_ = min(max(res.x[0], 0.0), 0.5), (res.x[1] if res.x[1]*t0 > 0 else t0)
```

```
            vv = -res.fun; worst = max(worst, vv)
            if vv > 0: nbl.append(bb_); ntl.append(tt_)
        print('  it %d  objective %.6f  worst violation %.2e  new cutting planes %d  rows %d' % (it, r.fun,
     worst, len(nbl), A.shape[0]), flush=True)
        if not nbl: break
        # new cutting planes, without near-duplicates (which make the linear programme ill-conditioned)
        pts = []
        for b_, t_ in zip(nbl, ntl):
            if all(abs(b_-b2) > 1e-6 or abs(t_-t2) > 1e-6*max(1.0, abs(t_)) for b2, t2 in pts): pts.append((
     b_, t_))
        A2, F2 = block(np.array([p[0] for p in pts]), np.array([p[1] for p in pts])); A = sparse.vstack([A,
     A2]).tocsr(); Fv = np.concatenate([Fv, F2])
    xs = x.copy(); xs[:nB] *= sc
    return r.fun, xs, B, J, worst, it

def family_vwide():
    # centres c, radii R = c - 1 + x with R <= c - 1/2, vertical offsets v
    C = [1.03, 1.06, 1.1, math.log(3), 1.15, 1.2, 1.25, 1.3, 1.4, 1.5, math.log(5), 2.0, math.e, 3.0]
    out = []
    for c in C:
        for R in [c-1+x for x in [0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5]]:
            if R <= 0 or R > c-0.5+1e-12: continue
            for v in [-2, -1.6, -1.3, -1.0, -0.8, -0.65, -0.5, -0.35, -0.2, 0, 0.2, 0.35, 0.5, 0.65, 0.8,
     1.0, 1.3, 1.6, 2]:
                if abs(v) <= R+2.5: out.append((c, R, float(v)))
    return out

if __name__ == '__main__':
    m = int(sys.argv[1]); d = float(sys.argv[2]); eta = float(sys.argv[3]) if len(sys.argv) > 3 else 0.0
    discs = family_vwide(); t0 = time.time()
    val, x, B, J, w, it = solve(d, m, discs, eta)
    nB = len(B); keep = [k for k in range(len(discs)) if x[nB+k] > 1e-10]
    used_c = sorted(set(round(discs[k][0], 4) for k in keep))
    print('m=%d d=%.3f prof=%s eta=%g ndiscs=%d C(grid)=%.6f kernel=%.6f worst violation=%.2e it=%d used=%d
     centres=%s  %.0fs'
          % (m, d, PROF, eta, len(discs), val, x[0]/(2*np.pi), w, it, len(keep), used_c, time.time()-t0),
     flush=True)
    json.dump({'m': m, 'd': d, 'prof': PROF, 'eta': eta, 'B': B, 'c': list(x[:nB]), 'discs': [discs[k] for k
     in keep], 'kappa': [x[nB+k] for k in keep],
               'J_grid': [J[k] for k in keep], 'C_grid_objective': val}, open('expl_m%d_d%.2f_%s_eta%g.json'
     % (m, d, PROF, eta), 'w'))
```

## kernel_only.py: kernel-only optima (Remark on the infimum)

```
# Search program (not a certificate): the smallest leading constant c_R1/(2 pi) of a kernel majorant without
     discs,
#   minimise c_R1/(2 pi)  subject to  f(b,t) <= U(b,t) on a grid of (b,t), refined by cutting planes, and 2 d
      c_R1 - 2 c_R2 >= 0,
# for the basis of order m and the given d; with the option 'rh' the constraint is imposed only at b = 0 (
     beta = 1/2), which is
# what would be needed if all zeros were on the critical line.  Used for Remark "On the infimum of the
     leading constant".
# Usage:  python3 kernel_only.py <m> <d1,d2,...> [rh]
import numpy as np, sys, time
from scipy.optimize import linprog
from lpcommon import f, F, basis, tgrid

def Gmat(d, B, bb, tt): return np.array([F(d+bb, tt, k, j)+F(d-bb, tt, k, j) for (k, j) in B]).T

def solve(d, m, rh=False, maxit=60, nb=26, tol=1e-9):
    B = basis(m); nB = len(B)
    sc = np.array([(d-0.5)**j for (_, j) in B])                # scaling of the coefficients, for
     conditioning
    obj = np.zeros(nB); obj[0] = sc[0]/(2*np.pi)
```

```
    bgrid = np.array([0.0]) if rh else np.linspace(0, 0.5, nb)
    bb, tt = [a.ravel() for a in np.meshgrid(bgrid, tgrid(300.), indexing='ij')]
    A = Gmat(d, B, bb, tt)*sc; Fv = f(bb, tt, d)
    arow = np.zeros(nB); arow[0] = 2*d*sc[0]; arow[1] = -2*sc[1]
    bf = np.array([0.0]) if rh else np.linspace(0, 0.5, 201); tp = np.geomspace(1e-8, 3000., 12000); tf = np.
     concatenate([-tp[::-1], tp])
    for it in range(maxit):
        r = linprog(obj, A_ub=-np.vstack([A, arow]), b_ub=-np.concatenate([Fv, [0.0]]), bounds=[(None, None)
     ]*nB, method='highs')
        assert r.status == 0, 'LP solver: '+r.message
        x = r.x; nbl = []; ntl = []; worst = -1
        for b in bf:   # violations on a finer grid -> new cutting planes (local maxima of the violation)
            v = f(b, tf, d)-Gmat(d, B, np.full_like(tf, b), tf)@(x*sc)
            worst = max(worst, v.max()); idx = np.nonzero((v > tol) & (v >= np.roll(v, 1)) & (v >= np.roll(v,
      -1)))[0]
            nbl += [b]*len(idx); ntl += list(tf[idx])
        if not nbl: break
        A = np.vstack([A, Gmat(d, B, np.array(nbl), np.array(ntl))*sc]); Fv = np.concatenate([Fv, f(np.array(
     nbl), np.array(ntl), d)])
    return r.fun, worst, it

if __name__ == '__main__':
    m = int(sys.argv[1]); ds = [float(x) for x in sys.argv[2].split(',')]; rh = len(sys.argv) > 3 and sys.
     argv[3] == 'rh'
    for d in ds:
        t0 = time.time(); val, w, it = solve(d, m, rh)
        print('m=%d d=%.2f %s C1(grid)=%.5f worst violation=%.1e it=%d  %.0fs' % (m, d, 'constraint at beta
     =1/2 only' if rh else 'all beta', val, w, it, time.time()-t0), flush=True)
```

## opt_line.py: the second-order cone programme of one explicit line

```
# Search program (not a certificate): optimise one explicit line |N(T)-main(T)| <= A1 log T + A3 for a target
      height L* = log T*.
# Kernel U = sum c_iota G_iota (basis R1..R_mR, I2..I_mI), d fixed. Convex programme (second-order cone,
     Clarabel through cvxpy):
#   minimise  (c_R1/2pi) L* + C_zeta(c) + z
#   s.t.      f(b,t) + eta/(1+t^2) <= U(b,t)      on a (b,t) grid, refined by cutting planes
#             2d c_R1 - 2 c_R2 >= eta              (limit of t^2 U at |t| -> oo)
#             z >= F_u(T;c), z >= F_l(T;c)         on a T grid in [1,1e6], and z >= the limit at T = oo
# C_zeta(c) = sum_n Lambda(n) n^{-sigma1} || (a(log n), b(log n)) ||_2  (n <= n0) + quadrature of the tail;
     with PPMAX > 0 the
# primes p <= PPMAX enter through the maximum of f_p over a grid of 96 phases (per-prime phase maximum).
# Usage:  [PPMAX=300] python3 opt_line.py <d> <mR> <mI> <L*> <out.json>
# Output: JSON with the coefficients as exact binary numbers, the input of certify_line_pp.py.
import sys, json, math, time, os, numpy as np, cvxpy as cp, mpmath as mp
PPMAX = int(os.environ.get('PPMAX', '0'))   # per-prime phase maximum in the objective for the primes p <=
     PPMAX (0: none)
sys.path.append(os.path.join(os.path.dirname(os.path.abspath(__file__)), '..', 'hybrid_asymptotic'))   #
     lpcommon.py lives there (or next to this file in the flat archive)
from lpcommon import f, F, tgrid

def basis(mR, mI): return [('R', j) for j in range(1, mR+1)] + [('I', j) for j in range(2, mI+1)]

def kernel_rows(d, B, bb, tt):
    G = np.array([F(d+bb, tt, k, j) + F(d-bb, tt, k, j) for (k, j) in B]).T
    return G, f(bb, tt, d)

def primes(N):
    s = np.ones(N+1, bool); s[:2] = False
    for i in range(2, int(N**0.5)+1):
        if s[i]: s[i*i::i] = False
    return np.nonzero(s)[0]

def czeta_data(d, B, n0):
```

```
    # rows (one per prime power n <= n0, then one per quadrature node of the tail) such that
    # C_zeta ~ sum_rows w * || (rows_a @ c, rows_b @ c + beta0) ||_2;  prime_of[row] = p for the row of n = p
     ^k, 0 for the tail
    s1 = 0.5+d; rows_a = []; rows_b = []; beta0 = []; w = []; prime_of = []
    def add(L, wt, p):
        rows_a.append([-(L**(j-1)/math.factorial(j-1))/math.pi if kk == 'R' else 0 for (kk, j) in B])
        rows_b.append([-(L**(j-1)/math.factorial(j-1))/math.pi if kk == 'I' else 0 for (kk, j) in B])
        beta0.append((math.exp(-d*L)-2)/(L*math.pi)); w.append(wt); prime_of.append(p)
    for p in primes(n0):
        lp = math.log(p); n = p; k = 1
        while n <= n0:
            add(k*lp, lp*math.exp(-s1*k*lp), int(p)); n *= p; k += 1
    # tail n > n0: psi(x) ~ x, sum Lambda(n) g(n) ~ int g(x) dx = int g e^L dL ; Gauss-Legendre in L on [log
     n0, 700]
    xg, wg = np.polynomial.legendre.leggauss(400); a, b = math.log(n0), 700.0
    for x, ww in zip(xg, wg):
        L = (b-a)/2*x+(a+b)/2; add(L, (b-a)/2*ww*math.exp((1-s1)*L), 0)
    return np.array(rows_a), np.array(rows_b), np.array(beta0), np.array(w), np.array(prime_of)


def czeta_pp_data(d, B, PP=300, n0=20000):
    """for primes p <= PP: list of (kvec, wvec, RAk, RBk, b0k) to build max_x f_p(x) on a grid; other n via
     hypot rows."""
    s1 = 0.5+d; groups = []
    for p in primes(PP):
        lp = math.log(p); n = p; k = 1; ks = []; ra = []; rb = []; bz = []
        while n <= n0:
            L = k*lp; w = lp*math.exp(-s1*L)
            ra.append([-w*(L**(j-1)/math.factorial(j-1))/math.pi if kk == 'R' else 0 for (kk, j) in B])
            rb.append([-w*(L**(j-1)/math.factorial(j-1))/math.pi if kk == 'I' else 0 for (kk, j) in B])
            bz.append(w*(math.exp(-d*L)-2)/(L*math.pi)); ks.append(k); n *= p; k += 1
        groups.append((p, np.array(ks), np.array(ra), np.array(rb), np.array(bz)))
    return groups

def elem_data(d, B, Tgrid):
    mp.mp.dps = 25
    s1r = mp.mpf(1)/2+d; s2r = mp.mpf(1)/2+2*d
    base = []; Phi = []
    for T in Tgrid:
        T = mp.mpf(T); s1 = s1r+1j*T; s2 = s2r+1j*T
        Ael = lambda s: mp.im(mp.log(s)+mp.log(s-1))-T/2*mp.log(mp.pi)+mp.im(mp.loggamma(s/2))
        base.append(float((2*Ael(s1)-Ael(s2))/mp.pi-T/(2*mp.pi)*mp.log(T/(2*mp.pi*mp.e))))
        row = []
        for (k, j) in B:
            if j == 1: D = 1/s1+1/(s1-1)-mp.log(mp.pi)/2+mp.psi(0, s1/2)/2
            else: D = s1**(-j)+(s1-1)**(-j)+(-1)**(j-1)/mp.factorial(j-1)/2**j*mp.psi(j-1, s1/2)
            row.append(float(mp.re(D)) if k == 'R' else float(-mp.im(D)))
        Phi.append(row)
    return np.array(base), np.array(Phi)

def optimize(d, mR, mI, Lstar, eta=2e-4, n0=20000, maxit=25, verbose=True):
    B = basis(mR, mI); nB = len(B); i1 = B.index(('R', 1)); i2 = B.index(('R', 2))
    isR = np.array([k == 'R' for k, _ in B])
    bb, tt = [a.ravel() for a in np.meshgrid(np.linspace(0, 0.5, 26), tgrid(300.), indexing='ij')]
    G, Fv = kernel_rows(d, B, bb, tt); Fv = Fv+eta/(1+tt**2)
    RA, RB, b0, w, prime_of = czeta_data(d, B, n0)
    if PPMAX:
        GR = czeta_pp_data(d, B, PPMAX, n0)
        maskPP = np.array([not (p and p <= PPMAX) for p in prime_of])     # rows of the prime powers p^k with
      p <= PPMAX are replaced by the phase maximum
        assert (~maskPP).sum() == sum(len(g[1]) for g in GR)
    Tg = np.unique(np.concatenate([np.geomspace(1, 1e6, 260), [1.0]]))
    base, Phi = elem_data(d, B, Tg); logT = np.log(Tg)
    sc = np.array([(d-0.5)**j for (_, j) in B])        # scaling for conditioning
```

```
    bfine = np.linspace(0, 0.5, 201); tp = np.geomspace(1e-8, 3000., 12000); tfine = np.concatenate([-tp
     [::-1], tp])
    for it in range(maxit):
        y = cp.Variable(nB); z = cp.Variable(); c = cp.multiply(sc, y)
        cons = [G@c >= Fv, 2*d*c[i1]-2*c[i2] >= eta]
        sign = np.where(isR, 1.0, -1.0)
        Fu = base+(Phi@c)/math.pi-c[i1]/(2*math.pi)*logT
        Fl = -base+(Phi@cp.multiply(sign, c))/math.pi-c[i1]/(2*math.pi)*logT
        cons += [z >= Fu, z >= Fl, z >= 7/8-c[i1]/(2*math.pi)*math.log(2*math.pi)]
        if PPMAX:
            xg = np.linspace(0, 2*np.pi, 96, endpoint=False); tp_ = cp.Variable(len(GR))
            for gi, (p, ks, ra, rb, bz) in enumerate(GR):
                Cm = np.cos(np.outer(xg, ks)); Sm = np.sin(np.outer(xg, ks))
                cons.append(tp_[gi] >= Cm@(ra@c)+Sm@(rb@c+bz))
            Cz = cp.sum(tp_)+cp.sum(cp.multiply(w[maskPP], cp.norm(cp.vstack([RA[maskPP]@c, RB[maskPP]@c+b0[
     maskPP]]), 2, axis=0)))
        else:
            Cz = cp.sum(cp.multiply(w, cp.norm(cp.vstack([RA@c, RB@c+b0]), 2, axis=0)))
        obj = c[i1]/(2*math.pi)*Lstar+Cz+z
        prob = cp.Problem(cp.Minimize(obj), cons); prob.solve(solver=cp.CLARABEL)
        assert prob.status in (cp.OPTIMAL, cp.OPTIMAL_INACCURATE), 'solver status: '+str(prob.status)
        cv = sc*y.value
        nb = []; nt = []; worst = -1
        for b in bfine:
            Gr, Fr = kernel_rows(d, B, np.full_like(tfine, b), tfine); v = Fr+eta/(1+tfine**2)-Gr@cv; worst =
      max(worst, v.max())
            idx = np.where((v > 0) & (v >= np.roll(v, 1)) & (v >= np.roll(v, -1)))[0]; nb += [b]*len(idx); nt
      += list(tfine[idx])
        if verbose: print('  it %d obj=%.5f C1=%.5f Cz~%.4f z=%.4f worst=%.2e' % (it, prob.value, cv[i1]/(2*
     math.pi), Cz.value, z.value, worst), flush=True)
        if not nb: break
        G2, F2 = kernel_rows(d, B, np.array(nb), np.array(nt)); G = np.vstack([G, G2]); Fv = np.concatenate([
     Fv, F2+eta/(1+np.array(nt)**2)])
    return dict(d=d, B=B, c=list(map(float, cv)), C1=float(cv[i1]/(2*math.pi)), Cz_est=float(Cz.value), z_est
     =float(z.value),
                A3_est=float(Cz.value+z.value), Lstar=Lstar, eta=eta, PPMAX=PPMAX, discs=[], kappa=[], m=max(
     j for _, j in B))

if __name__ == '__main__':
    d = float(sys.argv[1]); mR = int(sys.argv[2]); mI = int(sys.argv[3]); Ls = float(sys.argv[4]); out = sys.
     argv[5]
    t0 = time.time(); r = optimize(d, mR, mI, Ls)
    print('RESULT d=%.3f mR=%d mI=%d L*=%g  A1=%.5f A3~%.4f  value@L*~%.4f  (%.0fs)' % (d, mR, mI, Ls, r['C1
     '], r['A3_est'], r['C1']*Ls+r['A3_est'], time.time()-t0), flush=True)
    json.dump(r, open(out, 'w'))
```

## Appendix C. Outputs of the certification programs

Each output begins with the SHA-256 digests of its input file and of the programs that produced it. Only the plain decimal numbers printed after `<` or `<=` (without a radius) are produced by directed decimal rounding (`up_str` of `certutil.py`) and are asserted as bounds; every display of the form `[m +/- r]` is a ball printed with the standard `str` of `python-flint`, which rounds the midpoint to nearest, and is diagnostic only (for the final value of $C_{\mathrm{H}}$ the certificate also prints its radius, as a ball enclosing the radius, and a twenty-digit enclosure obtained by directed rounding). The running times printed at the end of the lines are informative only and depend on the machine; they are not part of the certified data. There are $1 + 6 \cdot 4 = 25$ certificates: one for the hybrid inequality and four (kernel inequality, $C_\zeta$, elementary part in modes N and S) for each of the six lines.

**Hybrid: `certify_hybrid.py hyb_m10_P.json 40`**

```
hyb_m10_P.json sha256 2b94bf5d689eee6e5688699be67d4725255aa97453d9115dcf2010b6b0e6a9fc
certify_hybrid.py sha256 f4977c8401a8c2c35b1761b9de4d39307f6fe8a65697799b6009b1784724f55e
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
profile_published.py sha256 06a554e573fabe71b6049e704349df53cad774712423042241436500acadb4bc
python 3.13.18, python-flint 0.9.0, precision 256 bits
d = 9/10  m = 10  discs used: 90  profile e_P (profile_published.py)
C = c_R1/(2pi) + sum kappa_k J_k/pi = [0.074859614271090718755016 45 +/- 3.78e-27]  (ball, midpoint and radius
    )
   radius of the ball: [2.94e-76 +/- 2.74e-79]   enclosure of C with directed rounding:
    [0.074859614271090718755, 0.074859614271090718756]
kernel part c_R1/(2pi) = [0.065582182784544440363 +/- 1.95e-22]
limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 = [0.001408437209 +/- 3.24e-13]  positive: True
t in (0,T1]: certified, boxes 2028149
t in [-T1,0): certified, boxes 2294869
|t|>=T1: certified, boxes 277098
all certified;  C < 0.0748596142711   time 838s
```

**Line $\ell_5$: kernel inequality**

```
sol.json sha256 febfa49528729b67e8af3b65920a6682545be70e6ec46b9a5262448e9e946574
certify_hybrid.py sha256 f4977c8401a8c2c35b1761b9de4d39307f6fe8a65697799b6009b1784724f55e
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
d = 1  m = 5  discs used: 0
C = c_R1/(2pi) + sum kappa_k J_k/pi = [0.201756550426914177038288 6 +/- 4.65e-26]  (ball, midpoint and radius)
   radius of the ball: [3.27e-78 +/- 1.70e-81]   enclosure of C with directed rounding:
    [0.20175655042691417703, 0.20175655042691417704]
kernel part c_R1/(2pi) = [0.20175655042691417704 +/- 1.72e-21]
limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 = [1.133488684 +/- 3.74e-10]  positive: True
t in (0,T1]: certified, boxes 10255
t in [-T1,0): certified, boxes 4207
|t|>=T1: certified, boxes 2
all certified;  C < 0.201756550427   time 1s
```

**Line $\ell_5$: $C_\zeta$**

```
sol.json sha256 febfa49528729b67e8af3b65920a6682545be70e6ec46b9a5262448e9e946574
cz_gen_pp.py sha256 a16e89acdcb472cf121ae6e212f9399c774ac2fefdf731a69e919545d763ee02
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
per-prime phase maximum for p <= 1000 : gain [0.13567339 +/- 3.39e-9]
primes 5761455  partial sum (n<=100000000) = [0.700266525818397 +/- 2.62e-16]  psi(N0) = [99998242.7966268
    +/- 1.77e-8]  51s
g monotone on [2000, oo): crude bound [+/- 4.40e-3] < sigma1
tail <= [0.003333488646 +/- 2.02e-13]  (int_{N0}^{e^LB} G = [0.003174276158 +/- 3.32e-14] , int_{e^L1}^{e^LB}
     g = [+/- 3.23e-163] , far tail <= [2.6758e-167 +/- 3.15e-172] )
C_zeta < 0.703600014465    time 52s
```

### Line $\ell_5$: elementary part, mode N

```
sol.json sha256 febfa49528729b67e8af3b65920a6682545be70e6ec46b9a5262448e9e946574
elem_gen.py sha256 25530a892cba6b7ef5c5c229da418d28a7e86622f11552a8f7abb963a6093670
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
intervals 9215  sup F_u on [1,T0] <= 0.6653643978   sup F_l <= -0.7495433723
T>=T0: F_u <= 0.5045914644  F_l <= -1.245408535  (K = [3.95201 +/- 3.63e-6] )
SUP over T>=1 of max(F_u,F_l) <= 0.665364397774    time 60s
```

### Line $\ell_5$: elementary part, mode S

```
sol.json sha256 febfa49528729b67e8af3b65920a6682545be70e6ec46b9a5262448e9e946574
elem_gen.py sha256 25530a892cba6b7ef5c5c229da418d28a7e86622f11552a8f7abb963a6093670
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
intervals 9215  sup F_u on [1,T0] <= -0.2236696656   sup F_l <= 0.1394908934
T>=T0: F_u <= -0.3703919570  F_l <= -0.3703919570  (K = [4.11780 +/- 2.69e-8] )
SUP over T>=1 of max(F_u,F_l) <= 0.139490893363    time 61s
```

### Line $\ell_{20}$: kernel inequality

```
sol.json sha256 a95801710718f4fdbfb80f2f66ece1d071a440f5650c2a2ae2700fc13032143f
certify_hybrid.py sha256 f4977c8401a8c2c35b1761b9de4d39307f6fe8a65697799b6009b1784724f55e
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
d = 6/5  m = 5  discs used: 0
C = c_R1/(2pi) + sum kappa_k J_k/pi = [0.10822193548420830107702 96 +/- 1.10e-26]  (ball, midpoint and radius)
   radius of the ball: [1.67e-78 +/- 4.53e-81]   enclosure of C with directed rounding:
     [0.10822193548420830107, 0.10822193548420830108]
kernel part c_R1/(2pi) = [0.10822193548420830108 +/- 2.98e-21]
limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 = [0.02372706155 +/- 2.76e-12]  positive: True
t in (0,T1]: certified, boxes 473575
t in [-T1,0): certified, boxes 279773
|t|>=T1: certified, boxes 1022
all certified;  C < 0.108221935485   time 46s
```

### Line $\ell_{20}$: $C_\zeta$

```
sol.json sha256 a95801710718f4fdbfb80f2f66ece1d071a440f5650c2a2ae2700fc13032143f
cz_gen_pp.py sha256 a16e89acdcb472cf121ae6e212f9399c774ac2fefdf731a69e919545d763ee02
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
per-prime phase maximum for p <= 1000 : gain [0.21171880 +/- 3.25e-9]
primes 5761455  partial sum (n<=100000000) = [1.46785649757579 +/- 4.11e-15]  psi(N0) = [99998242.7966268 +/-
     1.77e-8]  53s
g monotone on [2000, oo): crude bound [+/- 0.0145] < sigma1
tail <= [0.0002188390453 +/- 1.36e-15]  (int_{N0}^{e^LB} G = [0.0002060161219 +/- 1.10e-14] , int_{e^L1}^{e^
    LB} g = [+/- 7.27e-234] , far tail <= [7.2489e-238 +/- 1.12e-244] )
C_zeta < 1.46807533663    time 53s
```

### Line $\ell_{20}$: elementary part, mode N

```
sol.json sha256 a95801710718f4fdbfb80f2f66ece1d071a440f5650c2a2ae2700fc13032143f
elem_gen.py sha256 25530a892cba6b7ef5c5c229da418d28a7e86622f11552a8f7abb963a6093670
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
intervals 9215  sup F_u on [1,T0] <= 0.7020707094   sup F_l <= -1.072389149
T>=T0: F_u <= 0.6766231016  F_l <= -1.073376898  (K = [5.21715 +/- 1.54e-6] )
SUP over T>=1 of max(F_u,F_l) <= 0.702070709399    time 61s
```

### Line $\ell_{20}$: elementary part, mode S

```
sol.json sha256 a95801710718f4fdbfb80f2f66ece1d071a440f5650c2a2ae2700fc13032143f
elem_gen.py sha256 25530a892cba6b7ef5c5c229da418d28a7e86622f11552a8f7abb963a6093670
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
intervals 9215  sup F_u on [1,T0] <= -0.1827864943   sup F_l <= -0.1965944846
T>=T0: F_u <= -0.1983603198  F_l <= -0.1983603198  (K = [5.38293 +/- 4.87e-6] )
SUP over T>=1 of max(F_u,F_l) <= -0.182786494342    time 62s
```

### Line $\ell_{60}$: kernel inequality

```
sol.json sha256 79d3366fe78fe865de0b9dbd01bab13c6d5380a91913b78c2f4da095fb389acf
certify_hybrid.py sha256 f4977c8401a8c2c35b1761b9de4d39307f6fe8a65697799b6009b1784724f55e
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
d = 1  m = 5  discs used: 0
C = c_R1/(2pi) + sum kappa_k J_k/pi = [0.094721574236353990058 50854 +/- 1.28e-27]  (ball, midpoint and radius
    )
   radius of the ball: [1.60e-78 +/- 3.05e-82]   enclosure of C with directed rounding:
    [0.094721574236353990058, 0.094721574236353990059]
kernel part c_R1/(2pi) = [0.094721574236353990059 +/- 4.92e-22]
limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 = [0.008043907939 +/- 4.87e-13]  positive: True
t in (0,T1]: certified, boxes 1441685
t in [-T1,0): certified, boxes 483889
|t|>=T1: certified, boxes 4434
all certified;  C < 0.0947215742364   time 109s
```

### Line $\ell_{60}$: $C_\zeta$

```
sol.json sha256 79d3366fe78fe865de0b9dbd01bab13c6d5380a91913b78c2f4da095fb389acf
cz_gen_pp.py sha256 a16e89acdcb472cf121ae6e212f9399c774ac2fefdf731a69e919545d763ee02
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
per-prime phase maximum for p <= 1000 : gain [0.24105429 +/- 2.19e-10]
primes 5761455  partial sum (n<=100000000) = [2.09743955289942 +/- 3.84e-15]  psi(N0) = [99998242.7966268 +/-
     1.77e-8]  52s
g monotone on [2000, oo): crude bound [+/- 0.0149] < sigma1
tail <= [0.006829934351 +/- 6.03e-14]  (int_{N0}^{e^LB} G = [0.006476069071 +/- 3.90e-13] , int_{e^L1}^{e^LB}
     g = [+/- 1.23e-164] , far tail <= [1.7143e-168 +/- 4.48e-173] )
C_zeta < 2.10426948726    time 53s
```

### Line $\ell_{60}$: elementary part, mode N

```
sol.json sha256 79d3366fe78fe865de0b9dbd01bab13c6d5380a91913b78c2f4da095fb389acf
elem_gen.py sha256 25530a892cba6b7ef5c5c229da418d28a7e86622f11552a8f7abb963a6093670
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
intervals 9215  sup F_u on [1,T0] <= 0.7289415578   sup F_l <= -1.047579458
T>=T0: F_u <= 0.7013136727  F_l <= -1.048686327  (K = [4.00282 +/- 3.70e-6] )
SUP over T>=1 of max(F_u,F_l) <= 0.728941557753    time 61s
```

### Line $\ell_{60}$: elementary part, mode S

```
sol.json sha256 79d3366fe78fe865de0b9dbd01bab13c6d5380a91913b78c2f4da095fb389acf
elem_gen.py sha256 25530a892cba6b7ef5c5c229da418d28a7e86622f11552a8f7abb963a6093670
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
intervals 9215  sup F_u on [1,T0] <= -0.1584757589   sup F_l <= -0.1717847946
T>=T0: F_u <= -0.1736697487  F_l <= -0.1736697487  (K = [4.16860 +/- 2.71e-6] )
SUP over T>=1 of max(F_u,F_l) <= -0.158475758991    time 62s
```

### Line $\ell_{150}$: kernel inequality

```
sol.json sha256 88654e2339e4f68925dd36bb344e3ed3ea22a373bdc8fa02dec09407e7c066ea
certify_hybrid.py sha256 f4977c8401a8c2c35b1761b9de4d39307f6fe8a65697799b6009b1784724f55e
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
d = 17/20  m = 5  discs used: 0
C = c_R1/(2pi) + sum kappa_k J_k/pi = [0.093480628436037879597194 71 +/- 2.87e-27]  (ball, midpoint and radius
    )
   radius of the ball: [1.59e-78 +/- 3.49e-81]   enclosure of C with directed rounding:
    [0.093480628436037879597, 0.093480628436037879598]
kernel part c_R1/(2pi) = [0.093480628436037879597 +/- 1.95e-22]
limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 = [0.007450671016 +/- 2.69e-13]  positive: True
t in (0,T1]: certified, boxes 1658655
t in [-T1,0): certified, boxes 353921
|t|>=T1: certified, boxes 3198
all certified;  C < 0.0934806284361   time 121s
```

### Line $\ell_{150}$: $C_\zeta$

```
sol.json sha256 88654e2339e4f68925dd36bb344e3ed3ea22a373bdc8fa02dec09407e7c066ea
cz_gen_pp.py sha256 a16e89acdcb472cf121ae6e212f9399c774ac2fefdf731a69e919545d763ee02
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
per-prime phase maximum for p <= 1000 : gain [0.25085637 +/- 3.02e-9]
primes 5761455  partial sum (n<=100000000) = [2.21455819652827 +/- 4.70e-15]  psi(N0) = [99998242.7966268 +/-
     1.77e-8]  53s
g monotone on [2000, oo): crude bound [+/- 7.01e-3] < sigma1
tail <= [0.04122525901 +/- 2.13e-12]  (int_{N0}^{e^LB} G = [0.03953310421 +/- 1.61e-12] , int_{e^L1}^{e^LB} g
     = [+/- 2.53e-111] , far tail <= [3.6826e-115 +/- 3.45e-120] )
C_zeta < 2.25578345555    time 53s
```

### Line $\ell_{150}$: elementary part, mode N

```
sol.json sha256 88654e2339e4f68925dd36bb344e3ed3ea22a373bdc8fa02dec09407e7c066ea
elem_gen.py sha256 25530a892cba6b7ef5c5c229da418d28a7e86622f11552a8f7abb963a6093670
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
intervals 9215  sup F_u on [1,T0] <= 0.7303168470   sup F_l <= -1.045307483
T>=T0: F_u <= 0.7035135754  F_l <= -1.046486424  (K = [3.19479 +/- 4.84e-6] )
SUP over T>=1 of max(F_u,F_l) <= 0.730316846936    time 61s
```

### Line $\ell_{150}$: elementary part, mode S

```
sol.json sha256 88654e2339e4f68925dd36bb344e3ed3ea22a373bdc8fa02dec09407e7c066ea
elem_gen.py sha256 25530a892cba6b7ef5c5c229da418d28a7e86622f11552a8f7abb963a6093670
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
intervals 9215  sup F_u on [1,T0] <= -0.1575763483   sup F_l <= -0.1695128192
T>=T0: F_u <= -0.1714698460  F_l <= -0.1714698460  (K = [3.36057 +/- 1.57e-6] )
SUP over T>=1 of max(F_u,F_l) <= -0.157576348369    time 62s
```

### Line $\ell_{400}$: kernel inequality

```
sol.json sha256 961b299dafd10eca996a4a542d3f9f6b3afa729cd7bbde4fd11d5759c1ab42c8
certify_hybrid.py sha256 f4977c8401a8c2c35b1761b9de4d39307f6fe8a65697799b6009b1784724f55e
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
d = 17/20  m = 10  discs used: 0
C = c_R1/(2pi) + sum kappa_k J_k/pi = [0.083986018406625982801154 79 +/- 4.65e-27]  (ball, midpoint and radius
    )
   radius of the ball: [1.54e-78 +/- 1.28e-81]   enclosure of C with directed rounding:
    [0.083986018406625982801, 0.083986018406625982802]
kernel part c_R1/(2pi) = [0.083986018406625982801 +/- 1.55e-22]
limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 = [0.0002002739955 +/- 2.69e-14]  positive: True
```

```
t in (0,T1]: certified, boxes 11835193
t in [-T1,0): certified, boxes 9801985
|t|>=T1: certified, boxes 4194302
all certified;  C < 0.0839860184067   time 2319s
```

### Line $\ell_{400}$: $C_\zeta$

```
sol.json sha256 961b299dafd10eca996a4a542d3f9f6b3afa729cd7bbde4fd11d5759c1ab42c8
cz_gen_pp.py sha256 a16e89acdcb472cf121ae6e212f9399c774ac2fefdf731a69e919545d763ee02
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
per-prime phase maximum for p <= 1000 : gain [0.28179774 +/- 3.28e-9]
primes 5761455  partial sum (n<=100000000) = [4.01127229118254 +/- 1.94e-15]  psi(N0) = [99998242.7966268 +/-
      1.77e-8]  80s
g monotone on [5000, oo): crude bound [+/- 0.0114] < sigma1
tail <= [0.4524693305 +/- 1.73e-11]  (int_{N0}^{e^LB} G = [0.4325160730 +/- 4.01e-11] , int_{e^L1}^{e^LB} g =
      [+/- 6.02e-102] , far tail <= [2.8606e-111 +/- 3.11e-116] )
C_zeta < 4.46374162170    time 81s
```

### Line $\ell_{400}$: elementary part, mode N

```
sol.json sha256 961b299dafd10eca996a4a542d3f9f6b3afa729cd7bbde4fd11d5759c1ab42c8
elem_gen.py sha256 25530a892cba6b7ef5c5c229da418d28a7e86622f11552a8f7abb963a6093670
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
intervals 9215  sup F_u on [1,T0] <= 0.7375655061   sup F_l <= -1.027807128
T>=T0: F_u <= 0.7209635502  F_l <= -1.029036449  (K = [3.19527 +/- 2.94e-6] )
SUP over T>=1 of max(F_u,F_l) <= 0.737565506071    time 79s
```

### Line $\ell_{400}$: elementary part, mode S

```
sol.json sha256 961b299dafd10eca996a4a542d3f9f6b3afa729cd7bbde4fd11d5759c1ab42c8
elem_gen.py sha256 25530a892cba6b7ef5c5c229da418d28a7e86622f11552a8f7abb963a6093670
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
intervals 9215  sup F_u on [1,T0] <= -0.1458267316   sup F_l <= -0.1520124644
T>=T0: F_u <= -0.1540198711  F_l <= -0.1540198711  (K = [3.36106 +/- 6.67e-7] )
SUP over T>=1 of max(F_u,F_l) <= -0.145826731639    time 81s
```

### Line $\ell_{6000}$: kernel inequality

```
sol.json sha256 bc61062c39431f9498808273a0673ca1381c0a16313f88d89d2684e53da8431e
certify_hybrid.py sha256 f4977c8401a8c2c35b1761b9de4d39307f6fe8a65697799b6009b1784724f55e
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
d = 17/20  m = 10  discs used: 0
C = c_R1/(2pi) + sum kappa_k J_k/pi = [0.083814170602092884236629767 +/- 2.14e-27]  (ball, midpoint and radius
    )
   radius of the ball: [1.54e-78 +/- 3.35e-82]   enclosure of C with directed rounding:
     [0.083814170602092884236, 0.083814170602092884237]
kernel part c_R1/(2pi) = [0.083814170602092884236 +/- 2.98e-22]
limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 = [0.0002027460239 +/- 3.01e-14]  positive: True
t in (0,T1]: certified, boxes 14760921
t in [-T1,0): certified, boxes 15866017
|t|>=T1: certified, boxes 4194302
all certified;  C < 0.0838141706021   time 3123s
```

### Line $\ell_{6000}$: $C_\zeta$

```
sol.json sha256 bc61062c39431f9498808273a0673ca1381c0a16313f88d89d2684e53da8431e
cz_gen_pp.py sha256 a16e89acdcb472cf121ae6e212f9399c774ac2fefdf731a69e919545d763ee02
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
per-prime phase maximum for p <= 1000 : gain [0.28466228 +/- 4.66e-10]
primes 5761455  partial sum (n<=100000000) = [4.10087443545128 +/- 2.45e-15]  psi(N0) = [99998242.7966268 +/-
      1.77e-8]  80s
g monotone on [5000, oo): crude bound [+/- 0.0108] < sigma1
tail <= [0.4886359120 +/- 3.99e-12]  (int_{N0}^{e^LB} G = [0.4671146906 +/- 3.15e-11] , int_{e^L1}^{e^LB} g =
      [+/- 6.36e-102] , far tail <= [2.9139e-111 +/- 4.83e-116] )
C_zeta < 4.58951034745    time 80s
```

### Line $\ell_{6000}$: elementary part, mode N

```
sol.json sha256 bc61062c39431f9498808273a0673ca1381c0a16313f88d89d2684e53da8431e
elem_gen.py sha256 25530a892cba6b7ef5c5c229da418d28a7e86622f11552a8f7abb963a6093670
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
intervals 9215  sup F_u on [1,T0] <= 0.7357375257   sup F_l <= -1.027508246
T>=T0: F_u <= 0.7212794332  F_l <= -1.028720566  (K = [3.19575 +/- 1.13e-6] )
SUP over T>=1 of max(F_u,F_l) <= 0.735737525685    time 80s
```

### Line $\ell_{6000}$: elementary part, mode S

```
sol.json sha256 bc61062c39431f9498808273a0673ca1381c0a16313f88d89d2684e53da8431e
elem_gen.py sha256 25530a892cba6b7ef5c5c229da418d28a7e86622f11552a8f7abb963a6093670
certutil.py sha256 47322401572e98fab93924be05e7172d9d88a80685ee2818973a1081430e3220
python 3.13.18, python-flint 0.9.0, precision 256 bits
intervals 9215  sup F_u on [1,T0] <= -0.1464103660   sup F_l <= -0.1517135821
T>=T0: F_u <= -0.1537039882  F_l <= -0.1537039882  (K = [3.36154 +/- 2.48e-6] )
SUP over T>=1 of max(F_u,F_l) <= -0.146410366017    time 80s
```

### Grid optima of `kernel_only.py` (Remark 2.5; not certified)

Lines marked `NOT CONVERGED` are runs whose cutting-plane iteration stopped with a residual violation $\geq 10^{-5}$ on the fine grid; they are not quoted in the text. The marks are added here; the file `kernel_only.txt` is unchanged.

```
m=4 d=0.70 all beta C1(grid)=0.08831 worst violation=1.8e-08 it=59  120s
m=4 d=0.80 all beta C1(grid)=0.08894 worst violation=3.0e-13 it=9  20s
m=4 d=0.90 all beta C1(grid)=0.09061 worst violation=1.5e-10 it=3  8s
m=10 d=0.70 all beta C1(grid)=0.08160 worst violation=1.4e-13 it=5  33s
m=10 d=0.80 all beta C1(grid)=0.08071 worst violation=7.4e-08 it=59  233s
m=10 d=0.90 all beta C1(grid)=0.08100 worst violation=8.6e-09 it=59  173s
m=20 d=0.70 all beta C1(grid)=0.07994 worst violation=3.1e-09 it=59  373s
m=20 d=0.80 all beta C1(grid)=0.07983 worst violation=1.6e-09 it=59  378s
m=20 d=0.90 all beta C1(grid)=0.07996 worst violation=1.9e-06 it=59  382s
m=40 d=0.70 all beta C1(grid)=0.07978 worst violation=2.1e-02 it=59  2057s   # NOT CONVERGED (not quoted)
m=4 d=0.70 constraint at beta=1/2 only C1(grid)=0.04375 worst violation=3.3e-16 it=3  0s
m=4 d=0.80 constraint at beta=1/2 only C1(grid)=0.05000 worst violation=1.6e-12 it=2  0s
m=4 d=0.90 constraint at beta=1/2 only C1(grid)=0.05625 worst violation=2.4e-12 it=2  0s
m=10 d=0.70 constraint at beta=1/2 only C1(grid)=0.01747 worst violation=3.3e-04 it=59  2s   # NOT CONVERGED
     (not quoted)
m=10 d=0.80 constraint at beta=1/2 only C1(grid)=0.02002 worst violation=3.5e-14 it=3  0s
m=10 d=0.90 constraint at beta=1/2 only C1(grid)=0.02250 worst violation=1.8e-06 it=59  2s
m=20 d=0.70 constraint at beta=1/2 only C1(grid)=0.01070 worst violation=5.4e-01 it=59  3s   # NOT CONVERGED
     (not quoted)
m=20 d=0.80 constraint at beta=1/2 only C1(grid)=0.01257 worst violation=7.6e-02 it=59  4s   # NOT CONVERGED
     (not quoted)
m=20 d=0.90 constraint at beta=1/2 only C1(grid)=0.01330 worst violation=1.3e-02 it=59  4s   # NOT CONVERGED
     (not quoted)
```

## Appendix D. Input data of the certificates

The following files hold the coefficients as decimal strings that round-trip to the binary floating-point numbers produced by the optimiser (the certifiers convert them to exact rationals), the disc data before rounding to rationals (the certificate rounds $c_k, R_k, v_k$ to denominators $\le 10^4$ and then replaces $R_k$ by $\min(R_k, c_k - \frac{1}{2})$, as in `certify_hybrid.py`), and the parameters of the search. The fields `J_grid` and `C_grid_objective` are the floating-point values of the search programme and are not used by the certificates, which recompute $J_k$ and $C_{\mathrm{H}}$ exactly; they are kept so that the file is the unaltered output of the search (its digest is part of the provenance lines of the certificate). The rational disc data actually certified are those of Table 4, obtained from the decimal values below by the two lines of `certify_hybrid.py` quoted in Appendix B.

**`hyb_m10_P.json` (hybrid)**


```
{
 "m": 10,
 "d": 0.9,
 "prof": "P",
 "eta": 0.001,
 "B": [
  ["R", 1],
  ["R", 2],
  ["R", 3],
  ["R", 4],
  ["R", 5],
  ["R", 6],
  ["R", 7],
  ["R", 8],
  ["R", 9],
  ["R", 10],
  ["I", 2],
  ["I", 3],
  ["I", 4],
  ["I", 5],
  ["I", 6],
  ["I", 7],
  ["I", 8],
  ["I", 9],
  ["I", 10]
 ],
 "c": [0.41206500728461565, 0.37015428795149224, 0.2844015811499438, 0.15861857578668956,
  0.06954223006530842, -0.06157764993877496, -0.040570135781007474, 0.07526718292596676,
  -0.03235073058726206, 0.005632605237216712, 0.7334607489350747, -0.13613005947680004,
  0.15136050241561017, -0.11692974368156732, -0.03656894918659249, 0.10601427043647423,
  -0.06227163099672864, 0.016690927091753422, -0.0017941465528435766],
 "discs": [
  [1.03, 0.18000000000000002, -0.5],
  [1.03, 0.18000000000000002, 0.5],
  [1.03, 0.23000000000000004, -0.65],
  [1.03, 0.23000000000000004, 0.65],
  [1.03, 0.28, -0.65],
  [1.03, 0.28, 0.65],
  [1.03, 0.33, -0.65],
  [1.03, 0.33, 0.65],
  [1.03, 0.38, -0.65],
  [1.03, 0.38, 0.65],
  [1.03, 0.43000000000000005, 1.0],
  [1.03, 0.48000000000000004, -1.0],
  [1.03, 0.48000000000000004, 1.0],
  [1.03, 0.53, -1.6],
  [1.03, 0.53, -1.3],
  [1.03, 0.53, -1.0],
  [1.03, 0.53, 1.0],
```

```
[1.03, 0.53, 1.3],
[1.06, 0.16000000000000006, -0.5],
[1.06, 0.16000000000000006, 0.5],
[1.06, 0.21000000000000005, -0.5],
[1.06, 0.21000000000000005, 0.5],
[1.06, 0.26000000000000006, -0.5],
[1.06, 0.26000000000000006, 0.5],
[1.06, 0.26000000000000006, 0.65],
[1.06, 0.41000000000000003, -0.8],
[1.06, 0.41000000000000003, -0.65],
[1.06, 0.41000000000000003, 0.65],
[1.06, 0.41000000000000003, 0.8],
[1.06, 0.4600000000000001, 0.8],
[1.06, 0.56, -1.6],
[1.06, 0.56, -1.3],
[1.06, 0.56, 1.3],
[1.1, 0.15000000000000008, -0.35],
[1.1, 0.2500000000000001, -0.5],
[1.1, 0.2500000000000001, 0.5],
[1.1, 0.4000000000000001, 0.8],
[1.1, 0.45000000000000007, -0.8],
[1.1, 0.45000000000000007, 0.8],
[1.1, 0.5000000000000001, -0.8],
[1.1, 0.5000000000000001, 0.8],
[1.1, 0.6000000000000001, -1.0],
[1.1, 0.6000000000000001, 1.0],
[1.1, 0.6000000000000001, 1.3],
[1.0986122886681098, 0.1986122886681098, -0.5],
[1.0986122886681098, 0.1986122886681098, 0.5],
[1.0986122886681098, 0.24861228866810978, -0.5],
[1.0986122886681098, 0.24861228866810978, 0.5],
[1.0986122886681098, 0.2986122886681098, -0.5],
[1.0986122886681098, 0.2986122886681098, 0.5],
[1.0986122886681098, 0.44861228866810976, -0.8],
[1.0986122886681098, 0.44861228866810976, 0.8],
[1.0986122886681098, 0.4986122886681098, -0.8],
[1.0986122886681098, 0.4986122886681098, 0.8],
[1.0986122886681098, 0.5986122886681098, -1.3],
[1.15, 0.1999999999999999, -0.5],
[1.15, 0.1999999999999999, -0.35],
[1.15, 0.1999999999999999, 0.35],
[1.15, 0.4499999999999999, -0.8],
[1.15, 0.4999999999999999, -0.8],
[1.15, 0.5499999999999999, -0.8],
[1.15, 0.5499999999999999, 0.8],
[1.15, 0.6499999999999999, -1.3],
[1.15, 0.6499999999999999, -1.0],
[1.15, 0.6499999999999999, 1.0],
[1.15, 0.6499999999999999, 1.3],
[1.2, 0.24999999999999994, -0.5],
[1.2, 0.24999999999999994, -0.35],
[1.2, 0.24999999999999994, 0.35],
[1.2, 0.24999999999999994, 0.5],
[1.2, 0.7, -1.3],
[1.2, 0.7, -1.0],
[1.2, 0.7, 1.0],
[1.2, 0.7, 1.3],
[1.25, 0.3, -0.35],
[1.25, 0.3, 0.35],
[1.25, 0.75, -1.3],
[1.25, 0.75, -1.0],
[1.25, 0.75, 1.0],
[1.25, 0.75, 1.3],
[1.3, 0.8, -1.0],
[1.3, 0.8, 1.0],
```

```
  [1.4, 0.4499999999999999, -0.35],
  [1.4, 0.4499999999999999, 0.35],
  [1.4, 0.8999999999999999, -1.3],
  [1.4, 0.8999999999999999, 1.3],
  [1.5, 0.55, -0.35],
  [1.5, 0.55, 0.35],
  [1.6094379124341003, 0.6594379124341003, -0.35],
  [1.6094379124341003, 0.6594379124341003, 0.35]
 ],
 "kappa": [0.04645216860607228, 0.04300701194678176, 0.047063043236809085,
  0.14669081272175175, 0.025979954843667673, 0.10379166553442853, 0.0238180047300216,
  0.1041736978433328, 0.014539531796755864, 0.0648721619755471, 0.01585298753435496,
  0.0033617963250462696, 0.006488928258029474, 5.4144875650491944e-05,
  0.0003018716070398441, 0.0009151843658643538, 0.004457480242890296,
  0.0014660000151806882, 0.047525841386079126, 0.15907294401238437, 0.025389681493291277,
  0.2223325253406065, 0.08631620266291855, 0.1538221037360442, 0.011498324806702449,
  0.010728457418492028, 0.024094460474547018, 0.08996010049122434, 0.018772230071276146,
  0.013598324363960737, 9.669233392673498e-05, 4.8027806973018017e-05,
  0.0001824982987809 1896, 0.05671657488344152, 0.051472319257400555, 0.18833803873166002,
  0.07545736055574893, 0.0016274867554092252, 0.009838802010945192, 0.0012561800451027275,
  0.005913769696587884, 0.0006183338065604295, 0.002725978105029675,
  0.00021124645907630063, 0.05954976707208953, 0.25468461930865344, 0.016700547594809507,
  0.07440766182674931, 0.01266957077706443, 0.18918149312114735, 0.0014777755579662957,
  0.006728141614374093, 0.0014793988972288392, 0.0056601805478273185,
  5.5623684826215334e-05, 0.05424339889920384, 0.03475948981696701, 0.21438371806294945,
  0.00631637069925039, 0.0033326420626991556, 0.006954420904420063, 0.028592802039103745,
  5.25882403645087e-05, 0.0004219252908076343, 0.0018326737860931267,
  0.00020626507835385952, 0.03706538770583855, 0.030599996800513854, 0.06841653547695875,
  0.332325423847939, 4.57943912298433e-05, 0.0003653681965739939, 0.0015721607201088414,
  0.0001785005555963715, 0.05227534137638917, 0.2040492822717576, 6.871033611742962e-05,
  0.0003184030759719035, 0.001362286750857658, 0.0002746990132641945,
  0.0008212433177690405, 0.003961667358838006, 0.3861919018728296, 0.8551654872594248,
  0.00014937245314941035, 0.0014320991096890576, 0.09799147070095049, 0.33882316551798763,
  0.16714072019417897, 0.5061513200725676],
 "J_grid": [0.006913876985130073, 0.006913876985130073, 0.010786963484310835,
  0.010786963484310835, 0.015048973174851672, 0.015048973174851672, 0.019571517019666976,
  0.019571517019666976, 0.024737702602379327, 0.024737702602379327, 0.030311093588762966,
  0.03614382786661558, 0.03614382786661558, 0.042238079000552084, 0.042238079000552084,
  0.042238079000552084, 0.042238079000552084, 0.042238079000552084, 0.0032593310610211064,
  0.0032593310610211064, 0.006361647634553729, 0.006361647634553729, 0.010091729993481463,
  0.010091729993481463, 0.010091729993481463, 0.02372706874438562, 0.02372706874438562,
  0.02372706874438562, 0.02372706874438562, 0.029208025469220297, 0.04097579484229039,
  0.04097579484229039, 0.04097579484229039, 0.0007855777381792925, 0.00579739181011531,
  0.00579739181011531, 0.01762364909587594, 0.022555747215725654, 0.022555747215725654,
  0.027910450281615512, 0.027910450281615512, 0.039458155882717894, 0.039458155882717894,
  0.039458155882717894, 0.0029087213249446332, 0.0029087213249446332, 0.005814504320743654,
  0.005814504320743654, 0.009368782031364846, 0.009368782031364846, 0.02259349298288061,
  0.02259349298288061, 0.027952579183055815, 0.027952579183055815, 0.03950798254330579,
  0.0006778485850694062, 0.0006778485850694062, 0.0006778485850694062, 0.01654368154193419,
  0.021311247069162823, 0.026509942596292888, 0.026509942596292888, 0.037780805766164105,
  0.037780805766164105, 0.037780805766164105, 0.037780805766164105, 0.0006049812018696226,
  0.0006049812018696226, 0.0006049812018696226, 0.0006049812018696226, 0.03630213022245224,
  0.03630213022245224, 0.03630213022245224, 0.03630213022245224, 0.0005514853690123469,
  0.0005514853690123469, 0.034985510865448044, 0.034985510865448044, 0.034985510865448044,
  0.034985510865448044, 0.033803191348382326, 0.033803191348382326, 0.00044923232543557626,
  0.00044923232543557626, 0.031760036845278866, 0.031760036845278866, 0.000406003160664041,
  0.000406003160664041, 0.0003707553808856433 06, 0.0003707553808856433 06],
 "C_grid_objective": 0.07485961386282106
}
```

### `hyb_m10_P_discs_rational.json` (derived)

The rational disc data $(c_k, R_k, v_k)$ and the exact weights $\kappa_k$ used by the certificate, as pairs (numerator, denominator), obtained from `hyb_m10_P.json` by the conversions of `certify_hybrid.py`; the file is generated by `gen_appendix.py` and is not an input of any certificate. Its first lines are

```
{
 "source": "hyb_m10_P.json",
 "d": [
  9,
  10
 ],
 "discs": [
  {
   "k": 1,
   "c": [
    103,
    100
...
```

### `line_L5/sol.json` (line $\ell_5$)


```
{
 "d": 1.0,
 "B": [
  ["R", 1],
  ["R", 2],
  ["R", 3],
  ["R", 4],
  ["R", 5],
  ["I", 2],
  ["I", 3],
  ["I", 4],
  ["I", 5]
 ],
 "c": [1.2676737932696245, 0.7009294514561635, -0.4946330486621655, 0.09751555052498845,
  -0.0005893962931625271, -0.3784073335097401, 0.13473062893139318, -0.01708382778797021,
  0.0001003433684765934],
 "C1": 0.20175655042691418,
 "Cz_est": 0.8391011111542401,
 "z_est": 0.665362725928392,
 "A3_est": 1.504463837082632,
 "Lstar": 5.0,
 "eta": 0.0002,
 "discs": [],
 "kappa": [],
 "m": 5,
 "val": 2.5132465892172027,
 "PPMAX": 0
}
```


### `line_L20/sol.json` (line $\ell_{20}$)


```
{
 "d": 1.2,
 "B": [
  ["R", 1],
  ["R", 2],
  ["R", 3],
  ["R", 4],
  ["R", 5],
  ["I", 2],
  ["I", 3],
  ["I", 4],
```

```
  ["I", 5]
 ],
 "c": [0.6799784749489147, 0.8041106391650762, 0.7774270907421565, -0.008448613496901295,
  2.955972158446658e-05, 1.1843502097112268, -0.576537489831307, 0.00610293149500534,
  -2.1076986306361527e-05],
 "C1": 0.1082219354842083,
 "Cz_est": 1.6797641910534995,
 "z_est": 0.7020688851693405,
 "A3_est": 2.38183307622284,
 "Lstar": 20.0,
 "eta": 0.0002,
 "discs": [],
 "kappa": [],
 "m": 5,
 "val": 4.546271785907006,
 "PPMAX": 0
}
```


### `line_L60/sol.json` (line $\ell_{60}$)


```
{
 "d": 1.0,
 "B": [
  ["R", 1],
  ["R", 2],
  ["R", 3],
  ["R", 4],
  ["R", 5],
  ["I", 2],
  ["I", 3],
  ["I", 4],
  ["I", 5]
 ],
 "c": [0.5951532035147798, 0.5911312495450366, 0.3889216428262093, -0.004325167636924481,
  1.5378902475155497e-05, 0.8079466147098463, -0.3400480079099986, 0.003649167179891613,
  -1.2750467413103112e-05],
 "C1": 0.094721574236354,
 "Cz_est": 2.344884047970574,
 "z_est": 0.7289397780667036,
 "A3_est": 3.0738238260372777,
 "Lstar": 60.0,
 "eta": 0.0002,
 "discs": [],
 "kappa": [],
 "m": 5,
 "val": 8.757118280218517,
 "PPMAX": 0
}
```


### `line_L150/sol.json` (line $\ell_{150}$)


```
{
 "d": 0.85,
 "B": [
  ["R", 1],
  ["R", 2],
  ["R", 3],
  ["R", 4],
  ["R", 5],
  ["I", 2],
  ["I", 3],
  ["I", 4],
  ["I", 5]
 ],
```

```
 "c": [0.5873561110952275, 0.4955273589230774, 0.17191023951649326, -0.04515429156322006,
  0.00027174675540190917, 0.5292986070563637, -0.26974254048270235, 0.03950670966367516,
  -0.00023459073769482711],
 "C1": 0.09348062843603788,
 "Cz_est": 2.5047342757708364,
 "z_est": 0.7303153300893256,
 "A3_est": 3.235049605860162,
 "Lstar": 150.0,
 "eta": 0.0002,
 "discs": [],
 "kappa": [],
 "m": 5,
 "val": 17.257143871265846,
 "PPMAX": 0
}
```


### `line_C400/sol.json` (line $\ell_{400}$)


```
{
 "d": 0.85,
 "B": [
  ["R", 1],
  ["R", 2],
  ["R", 3],
  ["R", 4],
  ["R", 5],
  ["R", 6],
  ["R", 7],
  ["R", 8],
  ["R", 9],
  ["R", 10],
  ["I", 2],
  ["I", 3],
  ["I", 4],
  ["I", 5],
  ["I", 6],
  ["I", 7],
  ["I", 8],
  ["I", 9],
  ["I", 10]
 ],
 "c": [0.5276997168610267, 0.44844462233410926, 0.20734915440049123, -0.05170881362706953,
  0.04277439204584691, -0.0021686787194867547, 5.225366348519004e-05,
  -7.194711787780742e-07, 5.548232880956561e-09, -1.890562024872484e-11, 0.57950165380938,
  -0.24304154954646912, 0.09598460587810165, -0.022316290439150017, 0.0010166745274645633,
  -2.326790970420105e-05, 3.096452357377663e-07, -2.3274894920069496e-09,
  7.769753796000765e-12],
 "C1": 0.08398601840662599,
 "Cz_est": 4.723835786462193,
 "z_est": 0.7375635792618767,
 "A3_est": 5.46139936572407,
 "Lstar": 400.0,
 "eta": 0.0002,
 "discs": [],
 "kappa": [],
 "m": 10,
 "val": 39.05580672837446,
 "PPMAX": 0
}
```


### `line_D6000/sol.json` (line $\ell_{6000}$)


```
{
 "d": 0.85,
```

```
 "B": [
  ["R", 1],
  ["R", 2],
  ["R", 3],
  ["R", 4],
  ["R", 5],
  ["R", 6],
  ["R", 7],
  ["R", 8],
  ["R", 9],
  ["R", 10],
  ["I", 2],
  ["I", 3],
  ["I", 4],
  ["I", 5],
  ["I", 6],
  ["I", 7],
  ["I", 8],
  ["I", 9],
  ["I", 10]
 ],
 "c": [0.5266199652605132, 0.44752559745950127, 0.21763335933091524, -0.06632610210813339,
  0.049224751946469816, -0.002454597973796714, 5.8415182530844107e-05,
  -7.96150671490144e-07, 6.08645994127779e-09, -2.0583832928014207e-11, 0.5834246015810649,
  -0.23667212703027896, 0.09567840750457196, -0.01858966229544025, 0.0008313848476117577,
  -1.89384002058713e-05, 2.517567194510178e-07, -1.8927048877024136e-09,
  6.322548809318692e-12],
 "C1": 0.08381417060209288,
 "Cz_est": 4.566001466369842,
 "z_est": 0.7357358598268879,
 "A3_est": 5.30173732619673,
 "Lstar": 6000.0,
 "eta": 0.0002,
 "discs": [],
 "kappa": [],
 "m": 10,
 "val": 508.18676093875405,
 "PPMAX": 300
}
```


Universidad Nacional de Educación a Distancia, Madrid, Spain
*Email address*: vcasana1@alumno.uned.es